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MODULO 03 ELECTRICAL FUNDAMENTALS



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Applicability Table

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Construction and basic chemical action of primary cells, secondary cells, lead acid cells, nickel cadmium cells, other alkaline cells; Cells connected in series and parallel; Internal resistance and its effect on a battery; Construction, materials, and operation of thermocouples; Operation of photocells.	•	•
3.6 DC Circuits	B1	B2
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3.7 Resistance/Resistor	B1	B2
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3.15 Transformers	B1	B2
Transformer construction principles and operation; Transformer losses and methods for overcoming them; Transformer action under load and no-load conditions; Power transfer, efficiency, polarity markings; Line and phase voltages and currents; Power in a three-phase system; Primary and Secondary current, voltage, turn ratio, power, efficiency; Auto-transformers.	•	•
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3.17 AC Generators	B1	B2
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Construction, principles of operation and characteristics of: ACsynchronous and induction motors both single-phase and polyphase; Methods of speed control and direction of rotation; Methods of producing a rotating field: capacitor, inductor, shaded or split pole.	•	•
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3.1 Electron Theory

Structure and distribution of electrical charges within: atoms, molecules, ions, compounds

Atomic Theory

The word atom derives from the Greek word “*atomos*” and it means invisible. Greeks were the first to hypothesize the existence of this elementary bricks which constitute the matter, this hypothesis which allowed them, albeit on a more philosophical than scientific basis, to explain its discontinuous nature. Actually, it is only thanks to science’s strides in the last two centuries that the presence of atoms has been supported by scientific experiments and results. It’s only in the last century that it has been possible to understand the actual nature of matter, atom and particle (sub-atomic particle) that constitute it.

An atom is made up of 3 elementary particles:

		MASS [kg]	ELECTRIC CHARGE [C]
<i>Electron</i>	Negative elementary electric charge Its mass is 1836 times smaller than protons and neutrons	$9,1 \times 10^{-31}$	$- 1,6 \times 10^{-19}$
<i>Proton</i>	Positive elementary electric charge	$1,6 \times 10^{-27}$	$+ 1,6 \times 10^{-19}$
<i>Neutron</i>	Neutral elementary electric charge	$1,6 \times 10^{-27}$	0

Protons and neutrons gather in the central part which takes the name of *nucleus*. Around it, electrons move at a very high speed. Most of the volume that the atom occupies remains completely empty.

Each chemical element is characterized by:

- *Atomic number* (represented by letter **Z**) – it indicates the number of protons of an element.
In a neutral atom, Z is also equal to the number of electrons, as they perfectly balance the protons.
- *Mass number* (represented by letter **A**) – it’s equal to the number of particles in the atom’s nucleus, i.e., the sum of protons and neutrons.
Electron is not considered because its mass, in respect to the other two particles, is negligible.
To obtain the number of neutrons of an atom it is sufficient to make the subtraction A minus Z.

As the number of protons, electrons and neutrons vary, the atom's characteristics change.

- The number of protons in the nucleus is what distinguishes the atom of an element from another. Consequently, all the atoms characterized by the same atomic number are considered as a **chemical element** which has been given a name and a symbol that represents it.

Considering atoms with a different number of protons, we are simply talking about **different elements**.

For example, hydrogen has only one proton, while the successive element, helium, has two protons.

The atom of an element is the smallest quantity of it that remains unchanged in every transition or in every chemical reaction. In other words, for example, the smallest quantity of lithium/carbon/oxygen that behaves, binds, and reacts (chemically) like lithium/carbon/oxygen.

Chemically, the atom results unalterable, indivisible and indestructible but, acting through physical transformations (for example by making atoms collide at very high speed), it can be disintegrated, transformed in other atoms, fixed, fused, etc.

*Example: atoms with $Z = 6$ are called **Carbon**, identified with the Capital letter **C**.*

- All the atoms of the same element have the same number of protons, but the number of neutrons can vary. When this happens, we talk about **isotopes**. Isotopes characterized by an excess of neutrons present a form of weak radioactivity caused by the natural tendency to get rid of this excess.

An example of isotope is Carbon 14. Carbon, naturally (99%), occurs on its isotope ^{12}C (Carbon 12), with 6 protons and 6 neutrons. However, it's possible to locate a small quantity ($< 10^{-12}\%$) of ^{14}C with 6 protons and 8 neutrons. The weak radioactivity of this isotope can be used to date organic finds dating back up to 50'000 years earlier.

- When the number of electrons of an atom is equal to the number of protons, the atom is said to be **electrically neutral**, because the positive charges are balanced by the negative ones (no. protons = no. electrons). Considering the neutral atom of a specific element (therefore characterized by a fixed number of protons) it's possible, within certain limits, to modify the number of electrons, obtaining:

- Positive ion** or **cation**: one or more electrons are removed. Positive charges prevail.
- Negative ion** or **anion**: one or more electrons are added. Negative charges prevail.

The electric charge of the ion is expressed in unit of charge of the electron, and it's indicated with a sign at the top right of the symbol of the atom or groups of atoms.

Examples: Na^+ , Cl^- , NH_4^+ .

Summarizing:

Two atoms with different number of protons → these are two different elements. Atomic number Z and mass number A vary.

Two atoms of a same element with different number of neutrons → these are the isotopes of that element. Mass number A varies.

Two atoms of a same element with different number of electrons → these are the ions of that element.

Experiment and Rutherford atomic model

We have seen that the mass of the atom is concentrated in its nucleus. The rest of its structure can be considered as “empty”.

The first evidence of it was obtained through an experiment. By striking a very thin sheet of gold with subatomic particles, it was noticed that most of them passed beyond the sheet undisturbed. This suggested that the particles had not encountered a solid structure on their way through the foil and that therefore the matter was largely empty.

The outcome of this experiment led Rutherford to formulate the hypothesis of an atomic model with the electrons rotating around the nucleus in circular or elliptical orbits at a certain distance and with a speed such as to ensure that the centrifugal force balances the force of electrostatic attraction at any moment between protons and electrons. In other words, he imagined that an atom was a small planetary system.

The hypothesis made by Rutherford, although widely believed, is wrong due to two logical flaws:

- Moving electrons emit an electric field. This field would dissipate the energy of electrons themselves, making them lose speed and then centrifugal force, and consequently causing a fall into the nucleus.
- Although electrons are attracted to the nucleus, atoms with a larger number of electrons would be extremely unstable due to the repulsion between them.

Atomic orbitals

With the emerging of quantum mechanics, and, in particular, thanks to great scientists like Bohr, Somerfield, Schrödinger e Heisenberg, it was possible to understand how in reality the electron did not move around the nucleus on a closed trajectory.

An important concept is expressed through the *Heisenberg uncertainty Principle*, according to which, experimentally, it is impossible to simultaneously determine the position and velocity of an electron around an atom.

Instead, it's possible to determine the region of the space around the nucleus where there is, at least, 90% probability of finding the electron. This region of space is called *orbital*. Orbitals can be visualized as “clouds” where the electron is found 90% of the time.

Orbitals around the nucleus are not all similar, but they are distinguished by quantum (or energetic) level and by their form. The orbital form is identified with the letters s, p, d, f, g, h etc. and the energetic level is identified with an increasing number from number 1.

For example, type ‘s’ orbitals have spheric shape and type ‘p’ have a two-lobed shape.

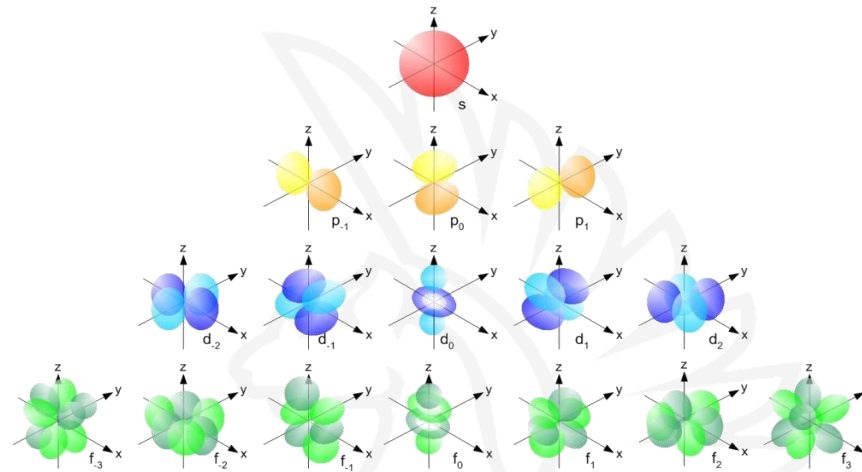


Figure 1 – Scheme of the shape and arrangement of the orbitals

Electrons arrange themselves in orbitals around the nucleus according to well-defined rules:

- **Pauli exclusion principle:** There can be no more than two electrons in each orbital. This couple is called **electronic doublet**.
- If in an orbital there are two electrons, they must have opposite spin (spin indicates the direction of rotation around its own axis).
- The electron always occupies the available orbitals at the lowest energy level.
- **Maximum multiplicity rule of Hund:** having two orbitals at the same energy level, with one of them already occupied by another electron and the other empty, the electron will always occupy the empty orbital assuming a spin parallel to the electron in the other orbital.

It is defined as a **total electronic configuration** the arrangement of all electrons in the various orbitals.

It is defined as an **external electronic configuration** the arrangement of electrons in **s** and **p** orbitals at the outermost energy level. This electronic configuration is fundamental: atoms tend to interact by involving only the electrons found in the outermost orbitals, denominated **valence electrons**, which therefore determine the chemical and physical properties of the element.

Interactions between atoms occur because atoms try to achieve the greatest possible energetic stability, by completing the s and p orbitals of the outermost energy level. This completion is called **octet configuration** and corresponds to the presence of 8 electrons (or 2 in the case of hydrogen only) in the outermost orbitals of the atoms, with two electrons in the s-orbital and six in the p-orbitals (s^2p^6).

Light emissions

By providing energy, typically in the form of heat, to an electron, this is able to pass to an higher energy level (following the rules previously defined), as long as its agitation is such as to make it jump to a higher energy level.

The electron will tend to regress to its initial energy level, in which it had greater stability, thus releasing excess energy in the form of light.

Basically, when an electron passes to a lower energy level, it emits light. Different atoms, having different electronic configurations, emit a spectrum of lights of different colors.

For example, lithium, sodium and potassium emit yellow, green and blue light respectively.



The Periodic Table of the Elements

1

IA

1A

1

H

Hydrogen

1.008

-259.1

2

IIA

2A

3

Li

Lithium

6.941

180.5

4

Be

Beryllium

9.012

1287

11

Na

Sodium

22.990

97.8

12

Mg

Magnesium

24.305

650

19

K

Potassium

39.098

63.5

20

Ca

Calcium

40.078

842

37

Rb

Rubidium

85.468

39.3

38

Sr

Strontium

87.62

777

55

Cs

Cesium

132.905

28.44

87

Fr

Francium

223.020

27

2

He

Helium

4.003

-272.24 (2.17 K)

10

Ne

Neon

20.180

-248.609 TP (4.1 K)

18

Ar

Argon

39.948

-189.36 TP (83.8 K)

36

Kr

Krypton

83.80

-157.36 TP (115.8 K)

54

Xe

Xenon

131.29

-117.74 TP (161.3 K)

86

Rn

Radon

222.018

-71

13

B

Boron

10.811

2075

14

C

Carbon

12.011

graphite 4489 TP (10.3 MPa)
diamond 3550

15

N

Nitrogen

14.007

-210

16

O

Oxygen

15.999

-218.79

17

F

Fluorine

18.998

-219.67 TP

31

Al

Aluminum

26.982

933.47

32

Si

Silicon

28.086

1414

33

P

Phosphorus

30.974

441.5 (white)

34

S

Sulfur

32.06

217.9 (grey)

35

Cl

Chlorine

35.453

-101.5

36

Ar

Argon

39.948

-189.36 TP (83.8 K)

49

In

Indium

114.818

156.6

50

Sn

Tin

118.71

231.93

51

Sb

Antimony

121.760

630.63

52

Te

Tellurium

127.6

449.51

53

I

Iodine

126.905

113.7

54

Xe

Xenon

131.29

-117.74 TP (161.3 K)

81

Tl

Thallium

204.383

304

82

Pb

Lead

207.2

327.46

83

Bi

Bismuth

208.980

271.4

84

Po

Polonium

[209]

254

85

At

Astatine

209.987

302

86

Rn

Radon

222.018

-71

113

Uut

Ununtrium

unknown

114

Fl

Flerovium

unknown

115

Uup

Ununpentium

unknown

116

Lv

Livermorium

unknown

117

Uus

Ununseptium

unknown

118

Uuo

Ununoctium

unknown

13

III A

3A

14

IV A

4A

15

V A

5A

16

VIA

6A

17

VII A

7A

18

VIIIA

8A

8

VIII

8

9

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397

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Figure 2- Periodic Table of the Elements

In nature it is possible to meet about a hundred elements, while in laboratory, it is possible to create other elements. All the elements find their place in the *Periodic Table of Elements*, on which they are ordered with a crescent atomic number Z . The resulting system is made up of rows called *periods* and columns called *groups*.

The apparently irregular shape of the table depends not only on the atomic number but also on the arrangement of the electrons in the atomic orbitals on the various quantum levels. Through the Pauli principle and the Hund rule it is possible to proceed with the progressive filling of the atomic orbitals (we are talking about the *Aufbau Criterion*) and to determine the levels occupied by the electrons of each atom.

It is thus possible to verify the close correlation between the periodicity of the chemical properties and the periodicity of the atomic structure.

In particular, we can see some examples:

- *Period 1: hydrogen (Z=1) and helium (Z=2) complete with their electrons the first quantum level, consisting of a single s-type orbital.*
- *Period 2: with lithium (Z = 3) the filling of the second level begins and ends with neon (Z = 10). Elements included between these two all belong to the second period.*
- *Period 3: the atoms of the elements from sodium (Z = 11) to argon (Z = 18) are characterized by the filling of the third quantum level.*
- *Period 4: it begins with potassium (Z = 19) and calcium (Z = 20). At this point, the nuclear charge is such as to make subsequent electrons occupy the orbitals belonging to the lower quantum level. The filling of the fourth level is therefore interrupted with the insertion of 10 electrons in the 3d orbitals from scandium (Z = 21) to zinc (Z = 30). After this, there is gallium (Z = 31), with which the filling of the fourth level ending with the crypto (Z = 36). In fact, the fourth period of the periodic system is characterized by 18 elements, for the insertion of the first series of 10 transition elements.*
- *Period 5: from rubidium (Z = 37) to xenon (Z = 54) has a trend similar to the previous one, with the insertion of the elements belonging to the second transition series, from yttrium (Z = 39) to cadmium (Z = 48), which correspond to the filling of the 4d orbitals.*
- *Period 6: it begins with cesium (Z = 55) and presents a further insertion of 7 4f orbitals, that is, belonging to an innermost quantum level. Thus, we have the appearance of a new series of 14 elements, called rare earths (or lanthanides), ranging from cerium (Z = 58) to lutetium (Z = 71); this period, which is made up of 32 elements, 14 more than the previous one, ends with radon (Z = 86).*
- *Period 7: it's the last level and it is incomplete and, starting with uranium (Z = 92), it is made up of unstable elements (transuranic elements) as they are radioactive.*

By comparing the atoms of the elements of the same group, it results that they have the same external electronic configuration, which determines their main chemical characteristic, first of all the valence: thus, for example calcium and strontium, both bivalent and belonging to group 2, have $3s^2$ and $4s^2$ configuration respectively.

Consequence of the progressive filling of a level over a period is the variation of the metallic properties which, marked in the elements of group 1 (**alkali metals**), gradually decrease until reaching the group of **halogens**, typically non-metallic, and that of **noble gases**, whose complete external quantum level characterizes its poor reactivity.

Furthermore, from the electronic structure it results that the elements of the same group have the external electrons placed on the atomic orbitals progressively further away from the nucleus: consequently, descending along a group, external electrons are less and less bound and this justifies the increase in metallic properties.

It then happens that the same group, for example that of nitrogen, is formed by typically non-metallic light elements (nitrogen and phosphorus) and heavier elements having metallic properties.

It turns out, therefore, that the stable element with the most marked metallic properties is cesium, which is found in the lower left corner of the periodic system, while the element with the greatest electronic affinity (see the following pages) is fluorine, which is found in the upper right.

Finally, the fact that the **transition elements** differ from each other only for the progressive filling of the d and f orbitals of the quantum levels lower than the main external one, accounts for the greater gradualness with which the chemical properties for these elements vary, so that, for example, the rare earth elements have a remarkable chemical similarity to each other.

Metals

Metals are on the left part of the periodic table.

METALS										SEMIMETALS					NON-METALS						
H																	He				
Li	Be															B	C	N	O	F	Ne
Na	Mg															Al	Si	P	S	Cl	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr				
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe				
Cs	Ba	La-Lu	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn				
Fr	Ra	Ac-Lr																			
			La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu				
			Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr				

Figure 3 - Metallic characteristic of the elements on the Periodic Table

A metal is a chemical element characterized by these properties:

- Often high ductility (ability to be reduced into thin sheets)
- High reflective power
- Opacity to light (it is not transparent)
- Good thermal conductivity
- Good electrical conductivity, although it decreases with increasing temperature
- Ability to supply positively charged ions when in solution
- Combined with oxygen, it forms basic oxides.

At room temperature, except for mercury, metals are crystalline solids.

Their electronic structure is characterized by the fact that the valence orbitals contain altogether a small number of electrons compared to that corresponding to the saturation of their energy level.

Some classifications:

- In relation to their density
We speak of **heavy metals** if the relative density is higher than 4 (examples: iron, lead, copper, etc.) and of **light metals** if relative density is lower than 4.
- Metals that are difficult to oxidize and which, for this reason, can also be found in nature in their metallic state (so not at their oxide state) are called **noble metals**, and are for example platinum and gold.
- **Alloying (or addition) metals**
They are those metals typically used as components for obtaining metal alloys (example: iron combined with carbon generates steel).
It is defined as **white metal** a particular group of anti-friction alloys based on tin, lead and antimony.
- Hard metal
They are very hard alloys, mostly obtained by sintering, and containing tungsten carbides, molybdenum etc.

In other words, the transition from metals to nonmetals, and therefore the relative change of properties, occurs gradually, without a clear discontinuity. For example, the basic character of oxides gradually decreases over the course of a period, from left to right, passing from metallic oxides (basic) to amphoteric oxides and, finally, to nonmetallic oxides (acidic oxides or anhydrides).

Non-metals and semimetals

The elements that appear on the right on the periodic table are defined as non-metals.
At room temperature half of these are gaseous, bromine is liquid, while all the others are solid.

The classical definition of metals and non-metals is based on the fact that they have opposite characteristics. Actually, however, it is not possible to make a clear distinction between the two classes of elements since, although a line is drawn in the periodic system of elements that allows a division between the two above mentioned categories, the elements that come to be close to the line of separation have properties that cannot, in a definite way, be qualified as either metallic or non-metallic. These elements fall within the more modern definition of **semimetals**.

In other words, the transition from metals to non-metals, and therefore the relative change of properties, takes place gradually, with no net discontinuity. For example, the basic character of the oxides gradually decreases over a period, from left to right, passing from metallic oxides (basic) to amphoteric ones and, finally, to non-metallic ones (acid oxides or anhydrides).

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Chemical bonds

Atoms, in nature, rarely are found isolated one from each other. A molecule of a substance represents the smallest aggregation of atoms that preserves the chemical characteristics of that substance. A molecule can be formed by equal atoms (for example, gaseous oxygen O₂) or different atoms.

The formation of chemical bonds between atoms can be understood by knowing some important quantities:

- ***Ionization energy*** of an atom or molecule
It is the minimum energy required to move an electron away from it to an infinite distance. If the starting atom is neutral, this process will transform it in a positive ion (first ionization energy). If the starting particle already is a positive ion, this process will increase its ionic electric charge by one (n-ionization energy, with n = second, third, fourth, etc.).
- ***Electronic affinity***
It is the amount of released or absorbed energy when an electron is added to an isolated neutral atom to form an ion with a charge of -1.
- ***Electronegativity***
It is the tendency that an atom has to attract electrons shared with other atoms, creating a bond. The opposite characteristic to the electronegativity is the ***electro-positivity***. The elements of the second-to-last group of the periodic table, i.e., halogens, are characterized by the lack of a single electron to reach the octet. So, they have a high electronegativity, and the electron will be strongly attracted to create the bond.
Considering, instead, the elements of the first group (alkaline metals), electronegativity is the lowest in relation to the other elements, because they have a single extra electron compared to the octet. Therefore, forming the bond, they will attract electrons very weakly.

Chemical bonds can be ***strong/primary*** or ***weak/secondary***.

Primary bonds develop between atoms and between ions during the formation of molecules and need high energy to break. They can be covalent, ionic and metallic bonds.

Secondary bonds usually form between molecules and are characterized by lower energies.

Covalent bond

The covalent bond is a chemical bond in which two atoms share couples of electrons, so that both can reach the higher possible stability, i.e., octet configuration. The shared couple consists of an electron of the first atom and an electron of the second one.

The external orbitals of the two atoms, containing the unpaired electrons, overlap, forming ***molecular orbitals*** (and so a molecule). The electrons belong to both atoms at the same time.

Some particularly reactive elements, for example oxygen and chlorine, if they do not have other preferable elements to which to bind, they tend to try to stabilize by binding together. For this reason, in nature, it is very common to find gases in diatomic form rather than monatomic form, for example: O₂, Cl₂.

A specific type of covalent bond is the *coordination bond*. In this case, electrons are made available by an atom only. The other involved atom simply “accepts” the electrons, that, also in this case, will belong to both atoms at the same time.

Ionic bond

Ionic bond is the strongest bond that can be established between two atoms, and, unlike the others, its *nature* is *electrostatic*. This bond forms between two elements with a great difference of electronegativity, like sodium and chloride, forming NaCl (the cooking salt).

Electrons, in this case, are not shared but are permanently transferred from an atom to the other.

For example, atoms of Na and Cl ionize each other becoming respectively Na^+ and Cl^- ions. At this point, due to the electrostatic force, the two ions attract each other, forming the ionic bond.

Given a certain number of ions, these will tend to aggregate in an orderly way forming a crystal lattice with characteristics that depends on the ion's charge and their dimension.

Metallic bond

Metallic bond is what gives to the metals their characteristics. Therefore, it's a fundamental factor for distinguish a metallic element from a non-metallic one.

The atoms of metallic elements have few valence electrons. These electrons can “separate” from the atom, which becomes a positive ion, and acquire a certain freedom of movement. Basically, the electrons no longer belong to the individual atoms but are free to move among all the positive ions involved, forming a sort of “electronegative glue” which is called an *electronic cloud*.

Positive ions form an orderly structure and electrons will arrange among them. The metallic bond is nothing more than the electrostatic attraction between the two types of particles.

The freedom of movement of the electrons is the characteristic which determines the high electric and thermal conductivity of metals.

Secondary chemical bonds

The secondary chemical bonds are characterized by lower bond energies, and they establish between molecules. We can distinguish dipole-dipole bond, ion-dipole bond, Van der Waals forces and hydrogen bond (which forms, for example, between water molecules).

Chemical reactions

A homogeneous substance composed by atoms of different elements and representable with a chemical formula, is called **compound**. For example, the water H_2O , composed by hydrogen and oxygen, is a compound. Compounds are obtained through a chemical reaction, which is a transformation that involves the break and the formation of chemical bonds. To these transformations it's possible to link a certain energy exchange.

In a chemical reaction we can distinguish the **reagents** (the substances from which the reaction starts) and the **products**. During the reaction, atoms of the reagents recombine with each other, absorbing or emitting energy in accordance with how the chemical bonds are modified:

- **Endothermic reactions**

When a certain amount of energy is required to create new bonds, the reaction products absorb energy from the external environment in the form of heat or light, or they can absorb the energy released from the break of other bonds.

A certain number of chemical reactions, even very common ones, can be accelerated by increasing the temperature of the environment in which they occur.

Example: the decomposition of water in gaseous hydrogen and oxygen is an endothermic reaction, i.e., for it to occur it's necessary to supply heat.

- **Exothermic reactions**

When during a reaction the bonds break and release energy. If this energy is not reutilized for creating new bonds (within the same reaction), it is released in the environment in form of light and heat.

Often, an exothermic reaction does not start unless a "start" is provided, which is a certain quantity of energy from outside.

Example: combustion is an exothermic reaction because it produces energy in form of heat. In order for it to occur it's necessary to supply a start.

Basic chemical reactions are:

- Metal + oxygen $\rightarrow$ basic oxide (example: the iron oxide Fe_2O_3 which compose rust)
- Non-metal + oxygen $\rightarrow$ acid oxide or anhydride (example: the carbon dioxide CO_2)
- Basic oxide + water $\rightarrow$ base or hydroxide (example: the caustic soda $NaOH$)
- Anhydride + water $\rightarrow$ acid (example: the sulfuric acid H_2SO_4)
- Base + acid $\rightarrow$ salt (example: the cooking salt $NaCl$).



Reversible chemical reactions are those in which, starting from the products, it's possible to re-obtain the initial reagents.

Acids and bases

The most commonly used classification of acids and bases is built on Arrhenius theory which, although outdated and limited, is sufficient for some uses of chemistry.

- **Arrhenius' acid**
 A compound which, when dissolved in water, dissociates, releasing H^+ ions. There are strong acids and weak acids depending on the relative amount of *hydrogen ions* they release.
 For example, hydrochloric acid HCl , which in ambient conditions is in gaseous state, is as a strong acid because, when dissolved in water, all its molecules dissociate into H^+ and Cl^- . The acetic acid CH_3COOH , instead, is a weak acid, as in water only a part of its molecules dissociates.
- **Arrhenius' base**
 A compound which, when dissolved in water, dissociates, releasing OH^- groups. Similarly to acids, strong bases and weak bases can be distinguished depending on how much they dissociate and, therefore, on the relative quantity of *hydroxide ions* they release.
- **Neutral substances**
 They are compounds which are neither acid nor basic.

Basing on this theory it has been developed a measurement scale to indicate the degree of acidity (or basicity) of a solution, by measuring the concentration of hydrogen ions. It is applied the specific formula to find values from 0 to 14, on a scale called pH:

- **$pH < 7$ acidic solution**
 The concentration of hydrogen ions is greater than the concentration of hydroxide ions. The value obtained is lower than 7 the higher the concentration of H^+ is.
- **$pH = 7$ neutral solution**
 The concentration of hydrogen ions is equal to that of hydroxide ions. The value obtained is equal to 7 (example: pure water).
- **$pH > 7$ basic (or alkaline) solution**
 The concentration of hydrogen ions is lower than that of hydroxide ions. The value obtained is higher than 7 the lower the concentration of H^+ is.

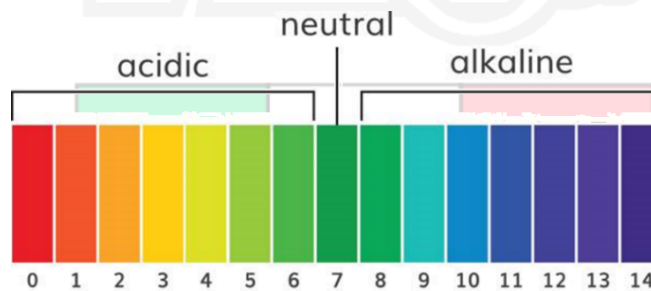


Figure 4 - pH scale

To know the pH of a solution there are suitably calibrated potentiometers which take the name of **pH-meters**.

Alternatively, a coarser measure can be obtained with **litmus papers**, that colors based on the pH of the substance being examined (as in the previous figure).

A more recent theory is *Lewis's theory* which defines acids and bases more precisely. A Lewis's acid is a substance able to create a coordination bond accepting a couple of electrons. A Lewis's base, on the contrary, is a substance able to create a coordination bond giving a couple of electrons.

Thanks to this definition, completer and more exhaustive, we can free ourselves from the needs of putting a substance in solution, managing to explain the behavior of other various substances.

A consequence, for example, is that according to Lewis's theory, hydrochloric acid is not acidic itself, but the released ion H^+ is acidic, due to the reaction with water.



Molecular structure of conductors, semiconductors and insulators

Remember that the chemical properties of an element depend primarily on the atomic number Z , which corresponds to its total number of electrons.

The electrons in the outermost orbit (*valence band/orbit*), called valence electrons, are those least influenced by the attraction of the nucleus.

In conductive materials, in defined conditions and thanks to the supply of external energy, the outermost electrons can leave the valence band and acquire a certain freedom of movement within the lattice of the atoms. The movement of the *free electrons* occurs in the so said *conduction band*. For “band” we always mean a certain energetic level.

The electrons in motion hit the electrons of the atoms nearby, giving them enough energy to also move from the valence band to the conduction band.

This cascade phenomenon is what generates the electric current, which is nothing more than a flow of electrons caused by the application of a potential difference (which translates into a supply of energy from the outside).

Generally, we could say that every material is conductive, because applying a very high difference of potential we are always able to produce an electron flow.

Metals are excellent conductors because they have the valence and the conduction bands very close to each other (closeness meant as energy levels), for which the electrons don't need so much energy to pass from a level to the other.

Viceversa, materials such as wood, plastic, porcelain, rubber, glass, etc., are bad conductors (or *insulators*), precisely because they have the two energy bands distant from each other. This gap between the bands, called blackout zone or band gap, is very large. Even providing a great amount of energy, the electron flow is very small, and currents are practically zero.

A third class of materials is that of *semiconductors* (silicon, germanium, selenium), that have intermediate electrical properties between conductors and insulators. These elements have 4 electrons in the valence orbital, therefore each atom is placed so that it is supported by 4 other electrons of a nearby atom (covalent bond). These materials are at the base of the construction of components used in the electronic field.



3.2 Static Electricity and Conduction

Based on the state of the electric charges, there are two types of phenomena:

- *Static electricity*, related to stationary electrical charges.
- *Electric current*, related to electric charges in motion.

Static electricity and distribution of electrostatic charges

When electrical charges are accumulated on a conductor, we talk about static electricity.

The laws of physics that govern electrostatic determine that:

- The charges accumulate only on the surface of the conductor and not in the internal volume.
- The charges tend to be distributed uniformly over the entire surface available to them due to mutual repulsion, even if the "more pointed" areas of the conductor tend to concentrate electrons compared to the flat areas.

The consequence of the fact that the charges always tend to "spread out" over the entire surface is that when a body charged with electrons meets a neutral body, the electrons will use also the surface of the neutral conductor to spread out as much as possible. This is what happens when a cable is connected to an electric generator.

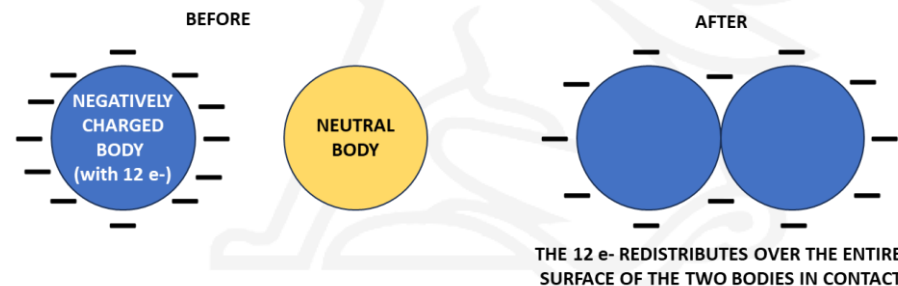


Figure 5 - Distribution of static charges

To obtain static electricity, it is necessary to create an accumulation of negative electrical charges and hinder their natural tendency to disperse. This phenomenon is typical, in fact, of insulating materials. In general, until between the two surfaces there isn't a potential difference, the accumulated electrons cannot discharge.

Aircraft are subject to a continuous accumulation of charges, due to the friction with the atmosphere, and it's essential that no potential differences develop between the parts that make them up. If this were the case, the accumulated charges could discharge, creating discharges of current from the panels with higher potential to those with lower one. These discharges, even if very small, potentially could cause a fire.

Two solutions are applied:

- **Electrical bonding**: is a link between more metallic parts with a conductive wire, which ensures that every surface of the structure is at the same electrical potential, therefore avoiding that electrical discharges occur.
- **Static wick / static discharger**: are devices that take the advantage of the “tip effect”, used to remove the accumulation of charges from the aircraft’s surfaces during flight. They are positioned on the surfaces’ trailing edge, for example of wings or of rotor’s blades. When a big quantity of charges accumulates, these elements allow them to disperse into the atmosphere, thus avoiding that a possible discharge occurs at the most thin/sharp parts of the structure itself, where charges tend to accumulate mostly.



Figure 6 - Bonding



Figure 7 - Static Wicks

Some examples of static electricity:

Getting out of a car after driving on a dry day, it can happen to experience the discharge of static current that has accumulated on the bodywork. The dust suspended in the air, not lubricated by the presence of humidity, flows on the bodywork giving it a certain number of electrons. Charges are not able to discharge from the bodywork because, despite being metallic and therefore a good conductor, it's insulated from the ground due to the rubber that makes up the tires. Although it is a slow process, driving for a long time, many electrons accumulate. When getting off the vehicle, when we touch the bodywork, the electrons use the human body as a conductor to discharge to the ground. It is for this reason that it happens to get a little shock. The most obvious solution to avoid the unpleasant phenomenon is to discharge the charges into the atmosphere or into the ground by means of a conductor or to wear insulating shoes, so that our body does not constitute a way for the charges.

During a storm, the agitation of the air creates an accumulation of charges in the clouds in a similar way to the automobile. The number of electrons accumulated reaches an amount that overcomes the immense resistance of the kilometers of air that separates that certain altitude from the ground, generating a lightning bolt.

Electrostatic laws of attraction and repulsion

An object with a negative charge is simply a body with an excess of electrons. Viceversa, an object that has lost some electrons has a positive charge. These two bodies, due to their charges, attract or repulse each other, respectively if they have opposite or same charge.

NOTE: even if this phenomenon has some similarities with magnets, in facts it has nothing to do with them.

Each electric charge generates an **electric field** around itself (measured in Volt/meter), that is space within which the charge exerts a force on other electric charges. Greater charges will produce stronger fields and therefore will exert more intense forces. Around an electric charge it is possible to draw lines of force that indicate the direction of the electric field at that point. These electric field lines of force have the following properties:

- They exit from the positive charges and enter into negative charges
- They always are **perpendicular** to the surface of a conductor
- They can never intersect with each other.

As trivial as they may be for an isolated charge, the lines of force can be very complicated in the presence of several sources, following in every case the same three rules.

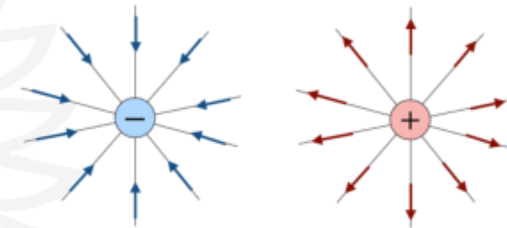


Figure 8 - Electric field lines of an isolated (negative or positive) charge

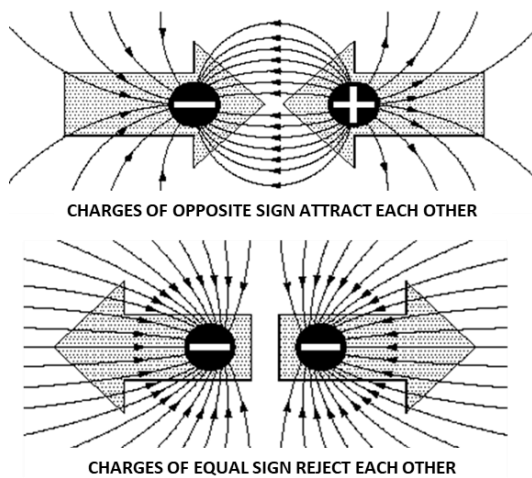


Figure 9 – Electric field lines of two charges

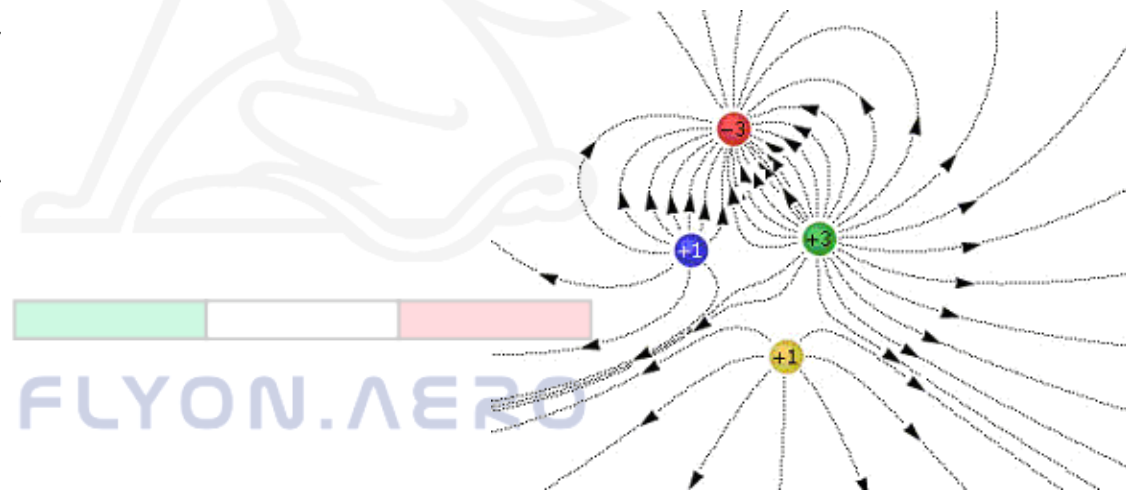


Figure 10 - Electric field lines of a set of charges

Units of charge, Coulomb's Law

The **electric charge** Q , measured in **Coulomb** (C), is defined by the International System as “the charge that flows in one second through the section of a conductor in which a stationary current with an intensity of 1 Ampere passes”.

The **elementary charge**, that is the smallest existing in nature, is the charge q of the electron (in absolute value), equal to $1.6 \cdot 10^{-19}$ Coulomb.

The charge Q of any generic body is simply a integer multiple (defined by number n) of this elementary charge:

$$Q = n \cdot q$$

Coulomb's law

The French physicist Jean Coulomb was the first to experimentally verify the link that exists between electric charges and the force of attraction or repulsion generated between them. His studies defined that:

“The force acting on two electric charges is directly proportional to their product and inversely proportional to the square of their distance and depends on a coefficient of the mean in which the charges are.”

$$F = \frac{k \cdot q_1 \cdot q_2}{d^2} \quad \text{with} \quad k = \frac{1}{4\pi \cdot \epsilon_0}$$

Where d is the distance between the charges; q_1 and q_2 are the respective amounts of charge of the two particles (in Coulomb) and k is the **Coulomb constant**. Coulomb force F , as all the other forces, is measured in Newton (N). In particular, we distinguish two cases:

- $F < 0$ it means that the charges have opposite sign $\rightarrow$ Attractive force.
- $F > 0$ it means that the charges have same sign $\rightarrow$ Repulsive force.

Regarding Coulomb constant, that in vacuum is about $8.988 \cdot 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2$, we can say that it is obtained starting from the **dielectric constant of the vacuum** ϵ_0 .

In general, dielectric constant, is a physical quantity that represents the propension of a mean to oppose to the flow of electric current through it.

The dielectric constant of air is approximal to that of vacuum.

If we apply Coulomb law to a mean different than vacuum (or than air), it's necessary to know the **absolute dielectric constant** (ϵ) of that mean, which will be in the formula in place of ϵ_0 . The constant can also be expressed in relative terms. The **relative dielectric constant** (ϵ_r) of a mean is the (non-dimensional) ratio between its absolute constant and the dielectric constant of vacuum (or of air) ϵ_0 :

$$\epsilon_r = \frac{\epsilon}{\epsilon_0} \quad \text{with} \quad \epsilon_0 = 8,85 \cdot 10^{-12} \text{ F/m}$$

Conduction of electricity in solids, liquids, gases and a vacuum

In solids (metals) conduction is electronic in nature, i.e., due to the motion of electrons and is unidirectional. In liquids and gases, instead, conduction occurs thanks to a displacement of ions and is therefore bidirectional.

Conduction in solids

Metals, as seen previously, has an orderly structure (crystal lattice), composed by positive ions surrounded by electrons free to move in the conduction band. In isolated conditions, these electrons move between the various atoms, not acquiring a specific motion.

If a potential difference is applied to the two ends of a metallic body through a generator, the electrons acquire an ordered motion. Conduction in metals is **unidirectional** because current is carried by electrons only, which are repelled by the negative pole and attracted by the positive one and are therefore forced to move in the direction imposed by the generator. The electrons are reconstituted by the generator (the number of electrons in the metallic body remains constant, while in the generator it varies) and a continuous flow of electrons is therefore determined, i.e., a **conduction current**.

Conduction in liquids

NOTE: in a chemically pure liquid conduction cannot occur because there are not free charges. Consequently, pure (distilled) water **is not** a conductor.

When a substance is dissolved in pure liquids, an electrolytic solution is obtained: the substance dissociates into pairs of positive and negative ions. By applying a potential difference, the positive ions move towards the negative pole while the negative ones move towards the positive pole. The current is **bidirectional** because, in fact, the two types of carrier particles move in two opposite directions.

The resulting electric current (see Chapter 3.4 for a more complete explanation) is called **ionic current** or **convection current** and a transport of matter also is associated with it.

Conduction in gases

In gases also we speak of ionic conduction. Under normal conditions, a gas contains no charge carriers, so it can be considered a good insulator.

The gas can be ionized by subjecting its atoms to a "bombardment" of Alpha or Beta particles or by other types of subatomic particles or neutrons (all generated by radioactive decay), exploiting the transfer of energy due to collisions. Another way to ionize the gas is to use electromagnetic radiation such as X or Gamma rays, in this case exploiting the photoelectric effect.

In both cases, the energy supplied allows the outermost electrons of the atoms to separate, thus generating positive and negative ions. By applying a potential difference at the two ends of, for example, a tube filled with gas, a discharge of current or a glow can be obtained. Also in this case, as in that of liquids, the positive and negative carriers move in two opposite directions.

An example of conduction in gases is lightning: atmospheric air is ionized by solar or cosmic rays and, when there is a great potential difference between the clouds and the ground, the discharge is generated.

Conduction in vacuum

There are no particles present in a vacuum, so the vacuum itself cannot generate current carriers and is an excellent insulator.

However, by applying two electrodes to the ends of a tube in which a vacuum has been created, if the negative electrode is heated or illuminated, it will emit electrons. The electrons will then be attracted to the positive electrode at the other end, constituting a passage of current.

This process is used for example in cathode ray tube monitors.



3.3 Electrical Terminology

The following terms, their units and factors affecting them: potential difference, electromotive force, voltage, current, resistance, conductance, charge, conventional current flow, electron flow.

Water analogy

One of the classic methods for understanding how electric current works is to use the analogy of water.

Consider a water tank placed at a certain altitude and a pipe connected to it, that goes downwards. Water will flow from the tank downwards through the pipe, with a flow rate that depends on the height of the tank above the ground and the diameter of the pipe.

The water analogy associates the electric voltage (potential difference) with the difference in height, the current with the water flow and the resistance of the conductor with the pipe's diameter.

{ tank higher → greater water flow rate (water flow is faster)
{ higher potential difference → greater current

{ bigger pipe section → greater water flow rate
{ lower conductor resistance → greater current

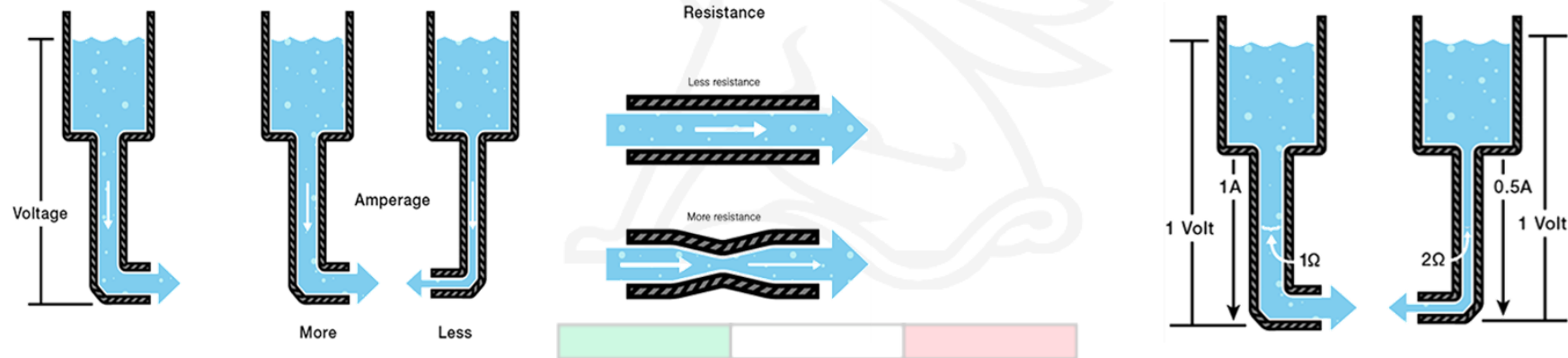


Figure 11 - Water analogy

Potential difference, electromotive force, voltage

Electric potential, measured in **Volt (V)**, is a quantity that derive from the electric field. It can be said that it measures the energy needed to move an electric charge. In fact, if we want to generate a flow of charges, we need a potential difference between two points in space, which individuate the direction of the displacement of charges.

For example, considering a circuit made of conductive material, the electrical potential of one of its points can be increased, thus establishing a **potential difference** between that point and the rest of the circuit, with a lower electrical potential. A flow of electrons will be established, which tends to balance the difference in level imposed on the circuit. The potential difference is also indicated with the term **electrical voltage** (indicated with the capital letter **V**).

In an electrical circuit, the voltage is kept constant by the generator, which allows a continuous flow of current. To better understand this concept, let's go back to the water analogy. Suppose we have two tanks A and B connected to each other via a pipe circuit and with A placed higher than B.

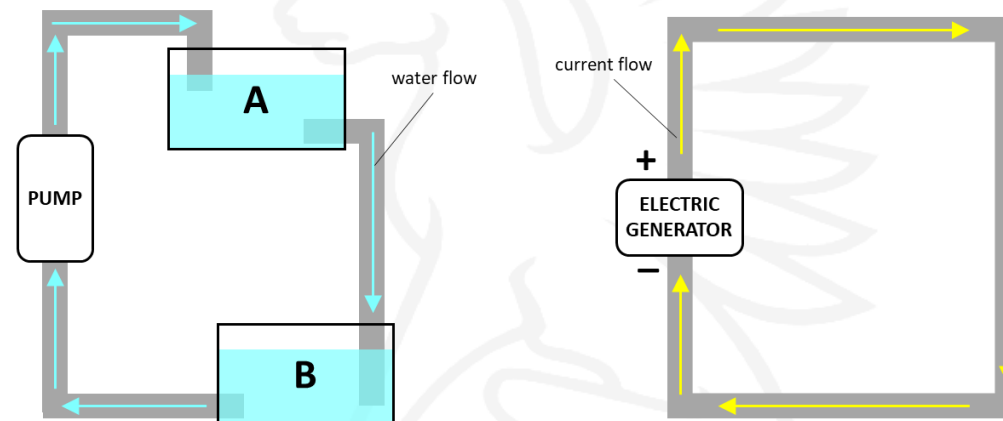


Figure 12- Water analogy – electromotive force

Water naturally flows from A to B. However, to keep the circuit active, it's necessary to supply energy to the water, so that it can return from tank B to tank A and continue the cycle. At the same way, charges naturally flow from higher to lower potential, but this potential difference must be kept.

In the hydraulic circuit, energy is supplied by the pump. The higher is the difference in level between A and B, the higher will be the energy that the pump must supply. In the electrical circuit, the pump corresponds to the generator. The generator must ensure that the electrons are never able to balance the difference of potential (otherwise the current flow would cease), forcing them to pass towards the negative pole and therefore always maintaining an excess of positivity at the other pole. The energy spent for it increases as the potential difference increases (associated with the difference in level A and B) and is linked to a quantity called electromotive force.

Electromotive force (emf) is the maximum (**ideal**) voltage that a generator can deliver. We distinguish it from the voltage at the generator terminals (or difference of potential) because in the latter the dissipation due to the internal resistance of the generator itself is also considered.

Electric generators are nothing more than machines designed to convert a form of energy (mechanical, chemical, etc.) into electromotive force.

NOTE: despite its name, electromotive force is **not** a force, and therefore should never be indicated in Newtons. Like difference of potential, in fact, it is measured in Volts.

Current, charge, conventional current flow, electron flow

Electrical current, indicated by the capital letter *I* and measured in **Amperes (A)** is a measure of the flow of charges going through a conductor in the time unit. It depends on the potential difference and on the characteristics of the considered conductor.

The electrical charge of a generic body can be calculated as an integer multiple of the elementary charge, that is the charge (in absolute value) of the electron.

As we know thanks to the study of the atomic theory, electrons flow from the negative to the positive pole. But the definition of the direction of the electric current is however an historic derivation. Benjamin Franklin believed that the “electric fluid” traveled from a higher level named “positive” to a lower level named “negative” and called this flow “current”. Consequently, by convention, we still indicate today the current as if it flows from the positive to the negative pole, as if it was due to a migration of positive charges.

From a conceptual point of view, the two reference systems are perfectly interchangeable because the difference consists only in the direction. It’s only necessary to remember to execute all the calculations without modify the reference system.

Unless otherwise specified, we will always use the conventional direction of the current: from positive to negative.

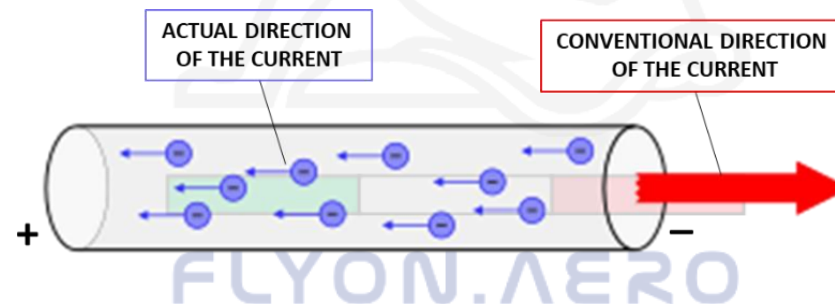


Figure 13 - Direction of the electric current

Current can be calculated as:

$$I = \frac{Q}{t}$$

Therefore we understand that the Ampere is equal to Coulomb per seconds.

Resistance

It can be noticed that current flows more easily in some conductors than others. *Electrical resistance* is a quantity that measures the tendency of a body to oppose the passage of current. It is indicated with the capital letter **R** and is measured in *Ohms* (Ω). Its value depends on the geometry of the conductor: it is directly proportional to the length and inversely proportional to the section. Consequently, a thinner and longer conductor will have greater resistance.

The value of the resistance obviously also depends on the material of which a conductor is made. Each material can be associated with a *resistivity* or *specific resistance* (indicated with the Greek letter 'rho' ρ), which is the characteristic resistance of the specific material with unit length and section. It is indicated in Ohm meters (Ωm) and also depends on the temperature at which the body is located.

As the resistivity of the material increases, the resistance of the conductor increases.

Example: a copper cable ($\rho = 1.68 \cdot 10^{-8}$) has a lower resistance than a glass cable ($\rho = 10^{10} \div 10^{14}$) of the same size, and is therefore more "prone" to being crossed by current.

In circuits, the total resistance is **always considered**, i.e., both that of the load (for example a light bulb) and that of the cable that makes up the circuit itself.

Conductance

Electrical conductance is the inverse quantity of resistance, that is, it measures the tendency of a body to let current pass through it. It is indicated with the capital letter **G** and is measured in *Siemens* (S). Conductance also depends on the geometry of the conductor.

Similarly, the inverse of resistivity is *conductivity/conductivity* or *specific conductance*, denoted by the Greek letter 'sigma' σ and measured in Siemens per meter (S/m). As with resistivity, conductivity depends on the material and temperature.

Example: a copper cable ($\sigma = 5.96 \cdot 10^7$) has a higher conductivity than a glass cable ($\sigma = 10^{-14} \div 10^{-10}$) of the same size.

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3.4 Generation of Electricity

Production of electricity by the following methods: light, heat, friction, pressure, chemical action, magnetism and motion

The generation of electric current, basically, is based on systems that use various forms of energy (mechanical, chemical, etc.) to free the electrons of a metal and use their flow.

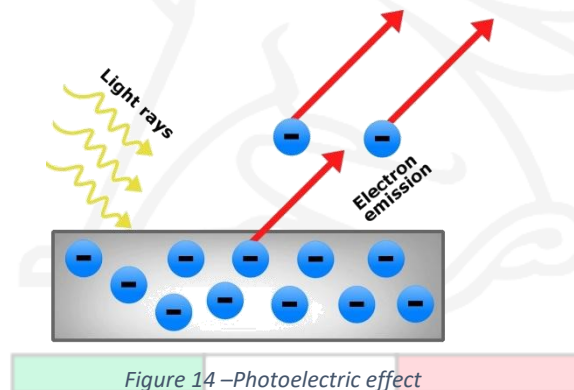
Among the various possible current generation methods that we will see, only the action of moving magnetic fields allows the development of powers suitable for large-scale use. The other methods manage to develop lower powers or are the basis for the construction of some sensors.

Production of electricity by light – photoelectric effect

Consider a metallic surface, hit by an electromagnetic radiation (light). If the frequency of this radiation overcome a certain threshold value, the material will start to emit electrons. This phenomenon is called *photoelectric effect*.

This *threshold value* is a different frequency for each metal (or semimetal), below which the photoelectric effect does not trigger.

This effect can be explained using the quantum theory, according to which the light is simultaneously a wave and a particle. When a particle of light (photon) hits an electron of the metal, it gives it enough energy to allow it to separate from it, passing from the valence band to the outside.



The photoelectric effect is exploited using two electrodes. The first electrode is hit by the light radiation, while the second attracts to itself the electrons just released by the first, which with their movement constitute an electric current.

Another effect that links light to electricity is the photovoltaic effect. It is an effect similar to the photoelectric one, but in this case the electron simply passes from the valence band to the conduction band, thus constituting a current within the material.

Production of electricity by heat – thermoelectric effect

All physical phenomena that convert heat into electricity and vice versa are called *thermoelectric effects*.

An important phenomenon is that which occurs due to the *Seebeck effect*, concerning two different metals placed in contact with each other. If the two metals are at different temperatures, a potential difference and therefore a current is generated between them. This effect is exploited in *thermocouples*, which are, in fact, electrical circuits based on two metals joined together.

The opposite effect to this is called the Peltier effect. Let's consider two different metals in contact and impose a current flow. Depending on its direction, the junction emits or absorbs heat, generating a heating or cooling device (Peltier cell).

A similar phenomenon is the Thomson effect, which however considers a single homogeneous metallic conductor with two ends at different temperatures. If we impose a flow of current along it, the conductor cools down if the current goes towards the hotter end and heats up if it goes towards the colder one.

The *Joule effect* also can be considered a thermoelectric effect. It concerns the dissipation of electrical energy in the form of heat in any current-carrying conductor. Unlike the Peltier effect, in the case of the Joule effect the heating occurs independently of the direction of the current.

Production of electricity by friction – triboelectric effect

When two different materials, of which at least one is an insulator, are rubbed together, a voltage and a transfer of electrical charges are generated. The effect is called *triboelectricity* and is nothing more than an accumulation of static electricity.

For example, when we walk on a nylon carpet with rubber shoes, the contact between the two materials causes the carpet to become positively charged and the shoes negatively charged. Being insulating materials, the charges remain accumulated until the human body encounters another material, which allows this accumulation to be discharged by generating a small spark.

Production of electricity by pressure – piezoelectric effect

Some crystalline materials, such as quartz, when subjected to mechanical deformation generate an electric potential difference, proportional to the deformation. This happens because the crystalline structure (and therefore the electric dipoles) deforms, resulting in a negatively and positively charged face of the crystal. When the stress is no longer applied, the crystal returns to its normal shape and the potential difference disappears.

This phenomenon is called *piezoelectricity* and can also occur in reverse, i.e., when a mechanical deformation is obtained by applying a potential difference to crystalline materials.

Production of electricity by chemical action – electrochemical effect

Electrochemistry is based on the transfer of electrons generated by chemical reactions. This effect is the basis of devices such as batteries and accumulators.

A typical electrochemical system consists of two solid conductors separated by an ionic conductor (hence an electrolyte solution). Solid conductors are called electrodes and can be made of either metals or semiconductors. The electrolyte solution is usually divided into two parts by a porous wall. An electrode is immersed in each of the two parts. The device described is called a *galvanic cell* or *battery*. We can identify:

- **Anode** – is the negative electrode. Here the oxidation reaction occurs. Electrons are released.
- **Cathode** – is the positive electrode. Here the reduction reaction occurs. Electrons released by the anode are attracted.

NOTE: the definition of anode and cathode as negative and positive is **not a univocal definition**, but it's suitable only for electrochemistry. In electronics and electric engineering, cathode is connected to the negative pole while anode is connected to the positive one.

A classic example of a device that exploits spontaneous chemical reactions for the transformation of chemical energy into electrical energy is the Daniell battery. This galvanic cell is made up of three fundamental elements:

- Anodic compartment (–), consisting of a zinc bar immersed in a solution of zinc sulphate (ZnSO_4).
- Cathodic compartment (+), consisting of a copper rod in a solution of copper sulphate (CuSO_4).
- Salt bridge, consisting of a tube filled with a saturated solution of a salt (e.g.: potassium chloride KCl) with two caps at the ends permeable to ions and water. The salt bridge connects the two compartments.

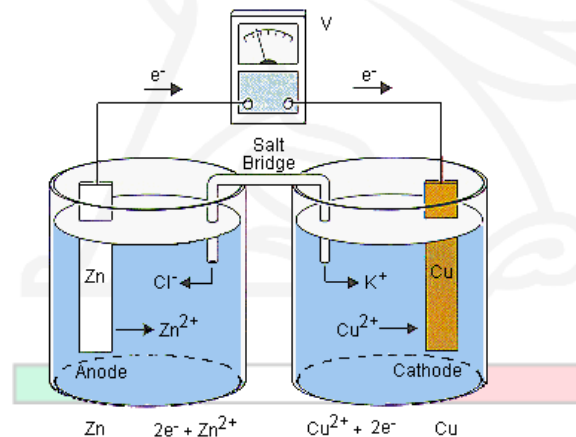


Figure 15 – Pile of Daniell

When the two electrodes are connected by a conducting wire, chemical reactions are activated:

- Oxidation occurs in the anodic compartment: the metallic zinc detaches from the foil and is added to the solution in the form of a positive ion (Zn^{2+}).

- Reduction occurs in the cathodic compartment: the positive ions (Cu^{2+}) in solution disappear, depositing as metal on the copper foil.

The consequence of these reactions is an excess of positive charges in the anode compartment and an excess of negative charges in the cathode one. This constitutes a potential difference that the electrons released by the reaction at the anode tend to compensate, moving towards the cathode. An electric current was generated. Remembering what was said about its direction: the electrons go from the anode to the cathode, but the conventional current goes from the cathode to the anode.

To ensure the continuous operation of the battery it is necessary to maintain the potential difference. This is the task of the salt bridge, which releases negative ions towards the anode and positive ions towards the cathode, re-establishing the electroneutrality of the solutions and allowing the chemical reactions to continue naturally.

An observable consequence of these reactions is that the zinc foil tends to thin while the copper foil tends to thicken.

The electromotive force that a Daniell battery can deliver is 1.10 V.

NOTE: the anode and cathode compartments are also called "semi-cells".

Production of electricity by magnetism and motion

According to *Maxwell laws*, which describe the fundamental relationships of the electromagnetism, variable electrical fields generate magnetic fields and variable magnetic fields generate electrical fields.

To generate electricity, it is therefore sufficient to have a conductive element influenced by a magnetic field that varies over time. This variation can be obtained thanks to the relative movement between the conductor and the field, as occurs in generators. In fact, one of the most classic methods is to create a coil of cables around a group of rotating magnets. The same effect is achieved by rotating the cables and holding the magnets still. The topic will be explored in depth in the next chapters.

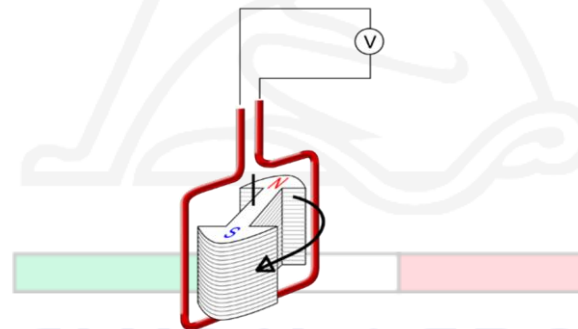


Figure 16 - Scheme of a simple generator

3.5 Sources of DC Electricity

Construction and basic chemical action of: primary cells, secondary cells, lead acid cells, nickel cadmium cells, other alkaline cells

Direct Current (DC) is a current characterized by a flow of charge with a constant direction over time.

We have seen that the galvanic cell is the elementary device which converts chemical energy into electrical energy. It's important to make a technical distinction:

- **Cell** – it's the single element.
- **Battery** – it's made up of multiple cells, organized in series or in parallel.

Cells can be distinguished based on the type of electrolyte. **Wet cell** batteries have a liquid electrolyte, while **dry cell** batteries use a slightly moistened paste. The electrolyte can also be acidic or basic, defining batteries with non-alkaline cells and those with **alkaline cells** respectively.

Considering their operation, galvanic cells are divided into:

- **Primary cells** – devices that, once discharged, cannot be recharged.
- **Secondary cells** – devices whose charge can be fully restored by applying a voltage.

By joining together multiple secondary cells, an **accumulator** or **secondary battery** is obtained.

Parameters of a battery

The main parameters that determine the performance of a battery, primary and/or secondary, are:

- **Nominal voltage** [V – Volt] – is the value indicated by the manufacturer on the battery plate. It is the potential difference between the positive and negative poles, without a load connected. It depends on the electrochemical system used and the number of cells that make up the battery.
- Working voltage [V] – is the voltage actually measurable between the two battery terminals when a load is inserted.
- End of charge voltage [V] – is the voltage measurable between the battery terminals in a closed circuit, at the end of charging.
- End of discharge voltage [V] – is the voltage at which the battery can no longer provide energy, measured in a closed circuit. It is indicated by the manufacturer.
- **Capacity** [Ah – Ampere hour] – is the amount of current that a fully charged battery can supply before running out of power. It is indicated by the manufacturer.
- Discharge time [h – hours] – is the useful life of a battery under specific conditions. It depends on the rated capacity and the current absorbed by the specific load.

- Power [W – Watt] – represents the speed of energy transfer. It is a value that we can calculate by multiplying the value of the nominal voltage (on the plate) with that of the current supplied (measurable).
- *Internal electrical resistance* [Ω – Ohm] – causes a voltage drop compared to the nominal one. It depends on the construction characteristics of the battery. The value increases as the battery ages.
- Efficiency in quantity of charge [dimensionless] – it is the ratio between the quantity of charge that can be used in the discharge phase and that used for charging. The typical value is around 90%.
- Energy efficiency [dimensionless] – it is the ratio between the amount of energy that can be delivered during the discharge phase and that used for charging. It usually has a value around 70-80%.
- Energy efficiency measured in cycles – is the ability of a battery to withstand successive charging and discharging cycles. It depends on the conditions of use of the battery. Usually this is judged at the end of its operational life when it has a capacity equal to 80% of the original value.
- *Shelf-life in storage* – is the storage time beyond which the battery is no longer able to guarantee the specified performance. All batteries, over time, suffer a decay of the accumulated charge. The expiry date is indicated by the manufacturer and refers to correct use in certain environmental conditions.

Maintenance and storage of primary batteries and accumulators

Speaking of maintenance, transport and storage of batteries, some precautions must be taken:

- Spread the terminals, nuts, connectors between cells and contacts with Vaseline or light oil, so that they do not oxidize in contact with air.
- Do not turn the battery upside down or tilt it. It is advisable to apply clearly visible “TOP” and “BOTTOM” labels.
- Keep the battery away from heat sources.
- The storage room must be clean, dry, adequately ventilated, not subject to vibrations, with a watertight electrical system and there must be signs explicitly prohibiting smoking and the use of open flames. The temperature of the storage room must be "non-critical", between +5°C and +40°C, even if the optimal temperature is 20÷25°C.

The latest generation sealed batteries, such as lead or nickel-cadmium batteries, do not require maintenance and do not present problems if they are turned upside down. However, older batteries, even if fully functional, require proper handling and charging (**always read the instructions of the manufacturer**).

It should also be noted that primary batteries and accumulators are classified as "hazardous urban waste" as they contain heavy metals, including mercury, lead and cadmium, which are particularly toxic to humans and the environment. Their *disposal* in the appropriate separate waste collection is mandatory.

Primary batteries

Some examples of primary batteries follow.

Type	Nominal voltage	Description	Application	Image
Zinc-carbon battery	1.5 V per cell	• Very cheap and low performing • It is dry cell • Typical construction: the anode is a cylindrical zinc container that encloses the electrolyte (a gelatinous paste of manganese dioxide, ammonium chloride and carbon). The cathode is a central graphite electrode, the end of which protrudes from the zinc container and is covered by a metal cap, separated from the container itself by a sealing plastic cap that avoids short circuits	Suitable for low power devices, they are the classic AA/AAA batteries on the market	
Alkaline manganese battery	1.5 V per cell	• It is the most common type of battery • Responds very well to high stress and continuous use	Like the previous ones	
Lithium metal battery	1.5 V per cell	• High performance • Lightweight but expensive	Suitable for small appliances such as meters and burglar alarms	
Silver oxide battery	1.5 V per cell	• Button cell with medium-high charging capacity • It was used in the aeronautical industry but was supplanted by less expensive batteries	Used for example in wristwatches or calculators	

Secondary batteries

Accumulators (or secondary/rechargeable batteries) are more expensive than primaries but, with appropriate processes, they can be recharged up to a thousand times.

One of the most important aspects to consider is the type of charging. Different batteries have different electrolyte solutions and produce different charging fumes. These fumes are dangerous in themselves and become even more dangerous when they encounter different fumes in poorly ventilated environments. For this reason, checking and recharging operations must be carried out in suitable spaces, equipped with fume and vapor extractors, and using acid-resistant gloves.

Generally, battery maintenance includes at least these activities:

- Check and restore the characteristics of the electrolyte solution
- Visual inspection of connectors between cells
- Inspection for any damage to the container or individual cells.

Currently the most common batteries on the market are lead-acid and nickel-cadmium accumulators.

Lead-Acid batteries

Lead-acid batteries are those used on most vehicles.

CONSTRUCTION AND CHARACTERISTICS

Usually, the accumulator is made of 6 cells, which include:

- Anode (–): electrode of spongy lead (Pb)
- Cathode (+): electrode of lead dioxide (PbO₂)
- Electrolyte: sulfuric acid solution (H₂SO₄). The electrolytic solution usually has a density around $1.25 \div 1.27 \text{ g/cm}^3$, variable with the charge.

Each cell produces a voltage of approximately 2 V (varies between 1.8 V and 2.2 V) and is conventionally considered discharged below 1.8 V.

NOTE: the voltage of a battery represents the sum of the no-load voltages of the individual cells. Consequently, if the battery does not supply current, its state of charge cannot be measured using a voltmeter.

To minimize the internal resistance of the battery, the half-cells must be very close together. To prevent short circuits from occurring due to contact between cells of opposite polarity, the cells (also called plates) are separated by layers of porous plastic.

DISCHARGE PHASE

Let's consider a single cell and assume it is charged: by connecting a circuit with a user to the electrodes, the same processes seen in the Daniell battery will happen.

The oxidation reaction takes place in the anodic half-cell, which releases electrons. The electrons will go towards the cathode half-cell, where the reduction reaction takes place. In this case the migration of electrons is balanced by the ionization of sulfuric acid.

When the cell is completely discharged, the electrodes are covered in lead salt and the electrolyte consists of a minimal concentration of acid in water.

RECHARGE PHASE

By providing energy from the outside and applying an adequate potential difference, the cells can be recharged by exploiting the reverse chemical reactions compared to those occurred during the discharge phase. The electrons are forced to return to the negative pole of the cell, restoring the electrodes to their original state (spongy lead and lead dioxide). The concentration of the electrolyte solution is reestablished, bringing the amount of acid in the water back to the maximum.

During charging, the **voltage** supplied externally must be kept **constant**. For this, a specific charger equipped with a voltage stabilizer is needed.

It is important that the battery charging process is not prolonged beyond the necessary time. In fact, if we continue to supply electricity, once the ions to be recombined with the lead salt have run out, this energy is used to trigger an electrolysis reaction of the water, which generates gaseous hydrogen and oxygen. Furthermore, due to the Joule effect, the energy supplied heats the circuit.

Oxygen and hydrogen are extremely flammable gases, and an overheating situation can easily lead to a battery explosion.

Theoretically, the discharge and charge cycles can occur infinite times, but some loss due to evaporation of the electrolyte water is inevitable. It is therefore necessary to intervene regularly to restore the density to the defined value.

TESTS

Lead-acid batteries are regularly subjected to some checks:

- **Momentary load test**
The voltage on individual cells is measured, by applying a very high current load for a short period of time.
Any voltage drops on individual elements indicate connection problems between the cells.
- **Capacity test**
The battery is completely discharged to measure its capacity.
The measured value, compared with the nominal capacity given by the manufacturer, gives information on the remaining life of the battery.
- **Control of the density of the electrolyte**
A sample of electrolyte solution is taken from the battery and an instrument called a hydrometer is used to measure the density of the acid in the water.

Nickel-Cadmium batteries

Nickel-cadmium batteries have a high charge capacity and are used in starting aircraft engines and in electric cars. However, cadmium is a highly harmful metal for the environment so the use of this type of battery is very limited.

CONSTRUCTION AND CHARACTERISTICS

Ni-Cd batteries can have up to 12 cells. Each one made up of:

- Anode (–): cadmium (Cd) electrode
- Cathode (+): nickel oxide hydroxide (NiO(OH)) electrode
- Electrolyte: potassium hydroxide (KOH) solution. It usually has a density around $1.24 \div 1.30 \text{ g/cm}^3$, variable with the charge.

Modern electrodes are not solid sheets but are porous, as they are obtained by sintering. They are then plated with a layer of cadmium for the anode and nickel for the cathode. This production process allows the contact surface to be substantially amplified, increasing the performance of the battery and reducing its internal resistance. To prevent the electrodes from meeting, porous separators made of nylon or cellophane are used.

Each cell produces a voltage of approximately 1.25 V (varies between 1.1 V and 1.5 V) and is conventionally considered discharged below 1.1 V.

DISCHARGE PHASE

Connecting the charged battery to a user, the reactions start. The potassium hydroxide of the electrolytic solution dissociates in positive K^+ ions and negative OH^- ions. The latter react with the cadmium at the anode (oxidation reaction), forming cadmium hydroxide and releasing the electrons that move towards the cathode. At the cathode the reduction reaction occurs between the nickel oxyhydroxide, the water of the solution and the electrons from the anode. The result is the formation of nickel hydroxide, but there's also a release of other OH^- ions.

Basically, in the solution only OH^- and K^+ (said above) ions remain. For this reason, actually, the potassium hydroxide KOH is not consumed in the reaction, but it can be said that it just gives a path for the OH^- charges. What varies in concentration is water.

RECHARGE PHASE

If energy is supplied from the outside, the reverse reactions of the previous ones take place, allowing the water in the solution to reconstitute. The nickel hydroxide must oxidize, reconstituting the initial oxyhydroxide of the cathode. The cadmium hydroxide at the anode must reduce back to metallic cadmium. The first reaction is faster than the second.

This means that, during the time it takes for the anode to recharge, the cathode becomes overloaded, forming oxygen gas, which binds to the cadmium and further slows the anode's charging. To prevent this oxygen from escaping, posing the risk of an explosion, batteries are produced with excess cadmium (so that all the oxygen can react) and the cells are sealed.

Nickel-cadmium batteries are charged at *constant current*; therefore, they are placed in parallel with an appropriately regulated resistance.

The optimal charging current is 1/10 of the maximum current supplied by the battery. If you want fast charging, you can increase this value up to 1/4 of the maximum current, reaching 1/2 of the rated current for batteries with sintered electrodes.

Slow charging of the batteries is certainly preferable. Recharging is necessary to reconstitute the electrolyte, through a chemical reaction that requires precise times to occur. Increasing the current to try to reduce the reconstitution time significantly lowers the efficiency of the process, because a lot of electrical energy is not transformed into chemical energy but is dispersed in the form of heat.

By themselves, Ni-Cd batteries have very low internal resistance. It may happen that some cells have a lower resistance than others, and in this case the current will concentrate in them causing a localized increase in temperature. This heating, in turn, further reduces the internal resistance, and as a result the current increases again, and so on. This cyclic phenomenon is called "thermal run-away" and can lead to battery explosion.

TESTS

These checks are performed on nickel-cadmium batteries:

- **Polarization test**
The polarization of a cell is a non-equilibrium condition that generates overvoltage and therefore overheating. To check for its presence, the (charged) battery is kept without users for approximately 1 hour and then the no-load voltage of each individual cell is measured and compared with the nominal one.
- **Insulation test**
It is used to check that the battery case is effectively insulated, so that there are no current leaks, which cause the battery to discharge too quickly and constitute a fire risk. The test is carried out by applying a potential difference of 250 V (or greater) between the electrodes and the container. The minimum acceptable resistance value for Ni-Cd batteries is 10 Ω .
- **Temperature measurement**
Typically, high-power Ni-Cd batteries are equipped with an internal precision thermostat. This sensor can only be checked if the battery is removed from its location.

Other accumulators

Other types of accumulators are:



- Nickel – metal hydride: they have a voltage of 1.2 V per cell. They are used in the same applications as Ni-Cd but are less harmful to the environment.
- Lithium ion/polymer: they have a voltage of 3.7 V per cell. They have high charge capacity and energy density. Used for example in smartphones.
- Zinc – air: they have a voltage of 1.4 V per cell. They are high performance batteries. Used almost exclusively in hearing aids.

Cells connected in series and parallel

In a battery, galvanic cells can be connected in series or parallel.

In a **series connection**, the positive terminal of one cell is connected to the negative terminal of the next, which is connected to the positive of the next, and so on. This connection allows for higher voltages with the same current supplied.

The total current at both ends of the battery is equal to that flowing through each individual cell, which always has the same value.

The total capacity of the battery is the same as that of a single cell (remembering that they all have equal capacity).

The total battery voltage is the sum of the individual cell voltages.

The total internal resistance of the battery is equal to the sum of the internal resistances of each individual cell.

In a **parallel connection**, all positive terminals are connected to each other, as are all negative terminals. This allows an increase in current.

In fact, the current between the two ends of the battery is equal to the sum of the currents supplied by each cell.

The total battery capacity is the sum of the capacities of the individual cells.

The total voltage of the battery is equal to that of a single cell, which has the same value for all cells.

The total internal resistance of the battery is equal to the resistance of one cell divided by the number of cells connected (all have equal resistance).

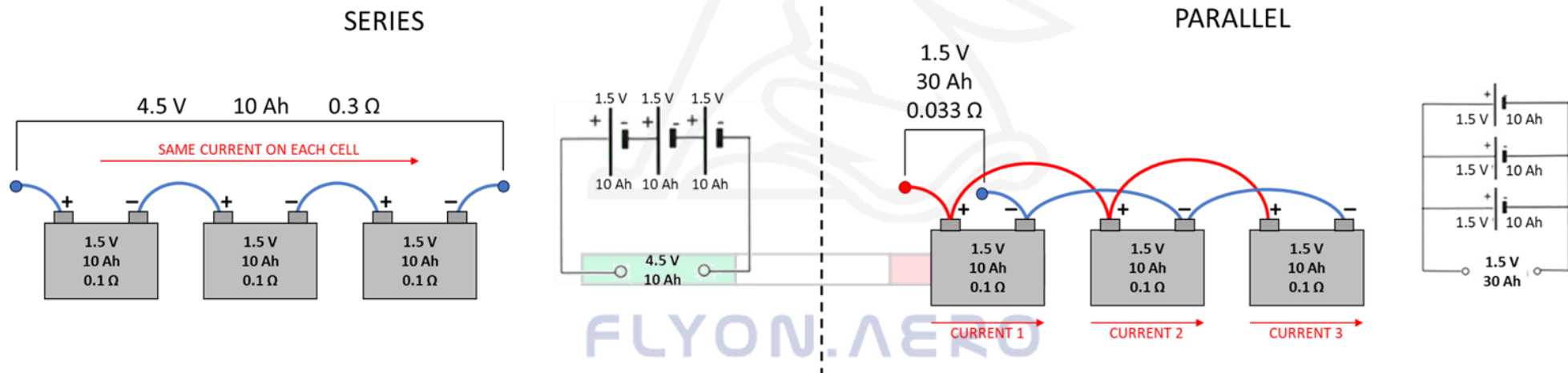


Figure 17 - Cells connected in series and parallel

Internal resistance and its effects on a battery

Internal electrical resistance is one of the most important parameters when choosing a battery. In general, resistance is that which a conductor opposes to the passage of current through itself.

As the battery is used, the materials that compose it and its structure deteriorate. This determines an increase in internal resistance (and therefore an increase in heat generated during use) and a decrease in capacity → *aging of the battery*.

The increase in internal resistance of a battery can have various causes:

- Corrosion, even slight, of the internal circuit
- Loss of active electrode material
- Deformation of the plates
- Drying of the electrolyte.

The internal resistance of a cell increases slowly in the first period of use but, when a capacity value of approximately 80% of the original capacity is reached, the degradation rate increases significantly and, therefore, its replacement is recommended.

The internal resistance, when the cells are in these conditions, is approximately 25-50% higher than the value of new cells.

Typically, manufacturers provide information on battery life considering a temperature reference of 25°C and very regular charge/discharge cycles.

The internal resistance of a battery can be reduced by applying two methods:

- Increasing the surface area of the electrodes: not only the size but making the surface porous (for example by sintering)
- Reducing the space between the electrodes.

Example: nickel-cadmium batteries in stylus format, to increase performance with the same space, are produced with the "Jelly-Roll" method, in which a single very large cell is used, wrapped around itself like a spiral.

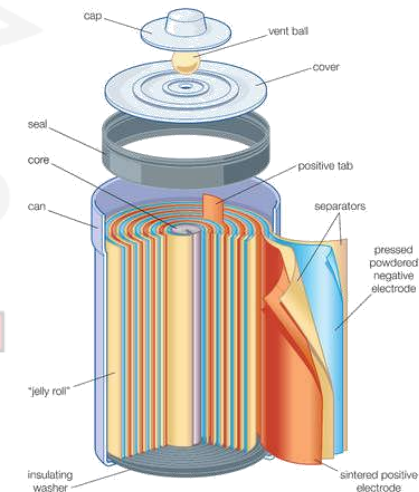


Figure 18 - Ni-Cd cell with "Jelly-Roll architecture"

Construction, materials and operation of thermocouples

Thermocouples are devices that exploit the Seebeck effect, concerning two different metals at different temperatures placed in contact with each other, between which a potential difference is generated. Depending on the materials and how marked the temperature difference is, this effect will be more or less evident.

A thermocouple is therefore a device used to detect a temperature (**temperature transducer**). It is made up of two conductors of known material (leg + and leg -), joined together at a point called **hot junction**. The other two ends of the conductors are joined in a terminal block called **cold junction**, connected to a voltmeter, which measures the voltages.

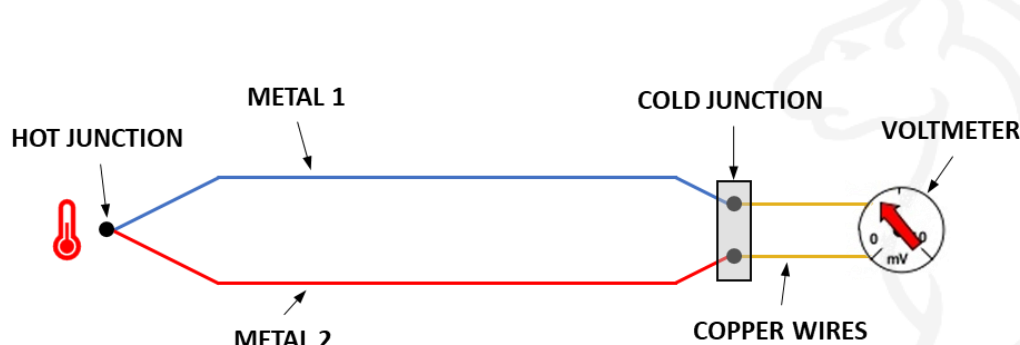


Figure 19 - Schematic of a thermocouple

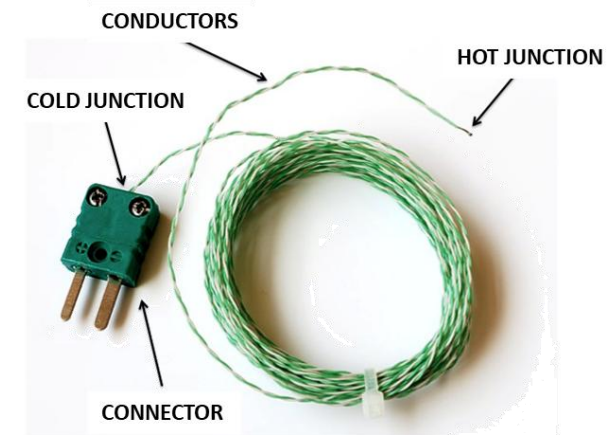


Figure 20 - A thermocouple

The hot junction is positioned in the area whose temperature you want to know. The temperature difference between the hot junction and the cold junction generates a potential difference between the free ends at the cold junction (those that connect to the copper cables), measured by the voltmeter. Once the potential difference is known, the temperature can be calculated.

The cold junction has a reference temperature maintained thanks to the terminal block; however it can be influenced by the ambient temperature.

For this reason, it is necessary to **calibrate the cold junction**. It can be done in two ways:

- Always keeping it at a **known temperature**
 The reference values provided by the sensor manufacturer usually refer to a cold junction held at 0°C. This means, for example, that if the temperature of this junction rises to 10°C and the temperature to be measured is 100°C, the measured temperature actually is 90°C rather than the correct one. Therefore, it must be ensured that the cold junction is always at a temperature of 0°C. However, this method is not practical and involves some energy consumption.

- Using a **temperature compensator**

An electrical circuit consisting of a voltage generator and resistors (one of which is sensitive to temperature) is connected to the cold junction. When the junction temperature is not equal to 0°C, the system is not in equilibrium. A voltage is then supplied which balances the system and ensures that the temperature detected in the cold junction is always equal to 0°C.

The choice of metals used for the construction of thermocouples is mainly a function of the maximum operating temperature, but also of properties such as resistance to oxidation and corrosion.

Example: K thermocouples are very resistant to oxidizing environments and T thermocouples are resistant to corrosion in humid environments.

Thermocouple Type	Composition	Temperature Range
B	Platinum 30% Rhodium (+)	2500-3100 degrees F
	Platinum 6% Rhodium (-)	1370-1700 degrees C
C	W5Re Tungsten 5% Rhenium (+)	3000-4200 degrees F
	W26Re Tungsten 26% Rhenium (-)	1650-2315 degrees C
E	Chromel (+)	200-1650 degrees F
	Constantan (-)	95-900 degrees C
J	Iron (+)	200-1400 degrees F
	Constantan (-)	95-760 degrees C
K	Chromel (+)	200-2300 degrees F
	Alumel (-)	95-1260 degrees C
M	Nickel (+)	32-2250 degrees F
	Nickel (-)	0-1287 degrees C
N	Nicrosil (+)	1200-2300 degrees F
	Nisil (-)	650 -1260 degrees C
R	Platinum 13% Rhodium (+)	1600-2640 degrees F
	Platinum (-)	870-1450 degrees C
S	Platinum 10% Rhodium (+)	1800-2640 degrees F
	Platinum (-)	980-1450 degrees C
T	Copper (+)	negative 330-660 degrees F
	Constantan (-)	negative 200-350 degrees C

Figure 21 – Examples of thermocouple types

The temperature ranges also vary for the same thermocouple, depending on the insulators used and the section of the conductors. For example, a type K (Chromel-Alumen) thermocouple insulated with Teflon measures temperatures from -200°C to +260°C, while a type K thermocouple insulated with ceramic fiber can measure up to 1200°C.

Operation of photocells

A photoelectric sensor, or photocell, is a device based on the photoelectric effect. It generates a current when hit by an electromagnetic wave (light).

A photocell in its essential form consists of a vacuum casing in which two electrodes are located:

- Cathode (**photocathode**) – is the negative electrode: it emits electrons when hit by light.
- Anode – is the positive electrode: it attracts the electrons emitted by the cathode.

NOTE: with this topic we demonstrate the non-uniqueness of the definitions of anode and cathode, as mentioned in Chapter 3.4.

Basically, in the photocell, a circuit is created that only closes when the cathode is illuminated.

For practical applications, a light source (usually an infrared LED) and a remotely located photoelectric receiver are used. As long as the light coming from the source hits the cathode of the receiver, the electrons continue to be transferred to the anode (also it is part of the receiver!). As soon as the cathode is not hit by light, the current ceases. It is the operating principle of burglar alarms or automatic gates.

The opening and closing of automatic gates stop when an obstacle comes between the source and the photocell, because the passage of current is interrupted.

The current generated by the photocells is directly proportional to the intensity of the light, therefore, they can also be used for measuring distances.

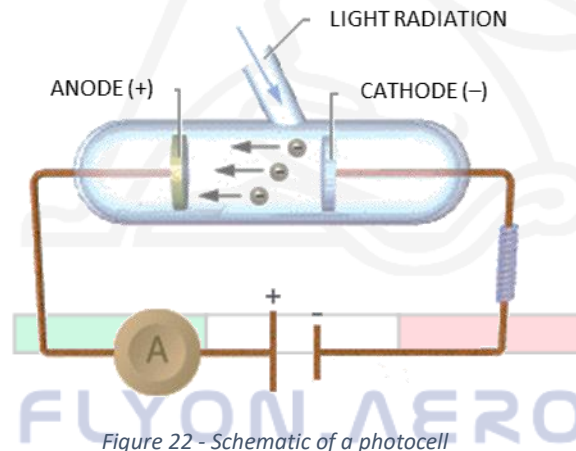


Figure 22 - Schematic of a photocell

Modern devices are no longer based on vacuum lamps but use semiconductors.

3.6 DC Circuits

In a basic electrical circuit, we can identify these elements:

- **Generator** – It has the function of transforming the supplied energy (mechanical, thermal, etc.) into electrical energy.
- **User** – Conceptually is the opposite of a generator. It's any device which transforms electrical energy into other forms of energy.
Example: the light bulb is a user that transforms electrical energy into luminous energy.
- **Connection conductors** – they allow the passage of current (electrons) from the generator to the user.
 - **Node** – is any point in a circuit where three or more conductors converge.
 - **Mesh** – is a closed path (meaning it starts from a node and returns to the same node) of conductors that can be traced in the circuit.
 - **Switch** – opens or closes a circuit, blocking or allowing the passage of current.

The conventions of the direction of the current and voltage within the circuit are:

- For generators: both the current I and the voltage V are indicated starting from the $-$ pole towards the $+$ pole.
- For users: the current I follows the convention already seen (from the $+$ pole to the $-$ pole), while the voltage drop V has the opposite direction.

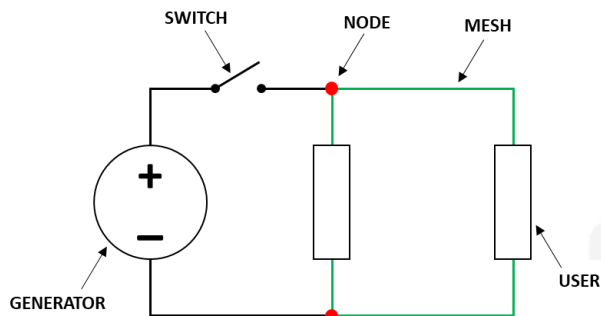


Figure 23 - Elements of a circuit

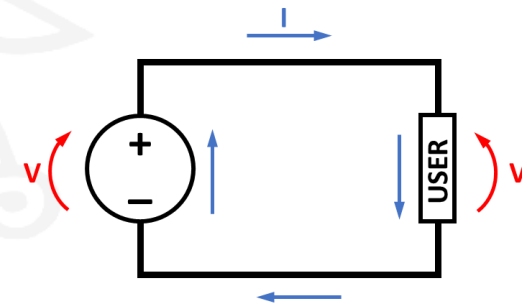


Figure 24 - Convention of users and generators

Two concepts related to circuits:

- **Short-circuit:** it's a direct contact between two branches of the circuit. In this condition, resistance can be considered null. The consequence is an uncontrolled increase of current.
- **Open circuit:** there's no contact between two branches of a circuit. Resistance can be considered infinite. The consequence is that current cannot flow.

Ohm's Law, Kirchhoff's Voltage and Current Laws

Ohm's law and Kirchhoff's laws of currents and voltages allow us to "solve a circuit", that is, calculate all the currents and all the voltages of generators and users.

- *Ohm's law*

The potential difference at the two ends of a conductor is directly proportional to the current flowing through it and its resistance.

The law can be expressed through the formula:

$$V = R \cdot I$$

Where V is the voltage (or potential difference), R is the resistance of the conductor (expressed in Ohms), and I is the current.

- *First law of Kirchhoff*

The algebraic sum of the currents flowing into a node is zero.

In other words, the current flow entering a node is equal to the current flow leaving it.

- *Second law of Kirchhoff*

The algebraic sum of the electromotive forces in a mesh is equal to the algebraic sum of the potential differences at the two ends of the users in the mesh.

In other words, the sum of the voltage drops is equal to the voltage supply of the circuit.

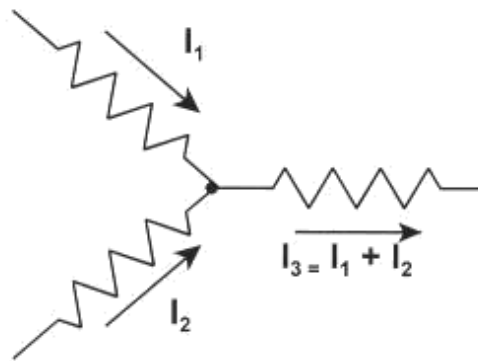


Figure 25 – 1st Kirchhoff's Law

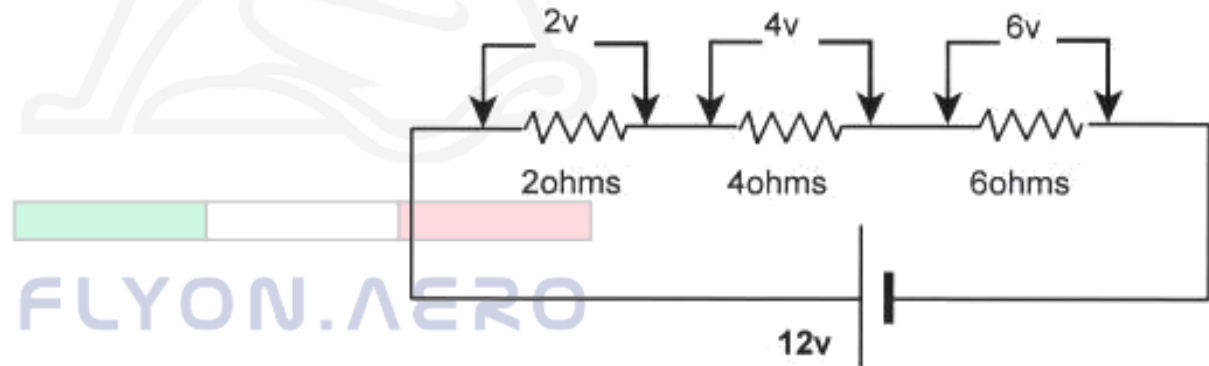
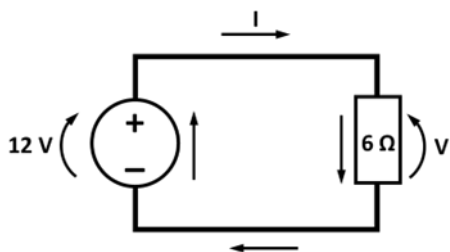


Figure 26 – 2nd Kirchhoff's Law

Calculating using the above laws to find resistance, voltage and current

Some examples of circuit resolution follow.

Example 1 – Calculate current I

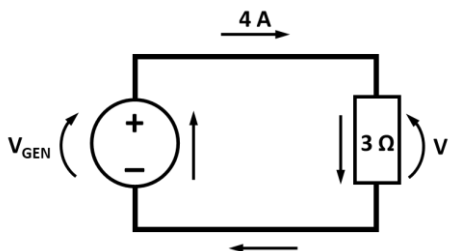


The circuit is made up of a single mesh, so there are no nodes. According to Kirchhoff's laws, the current is the same throughout the circuit and the voltage drop (V) on the user has the same value as the voltage supplied by the generator.

So, we can apply Ohm's law and find the value of the current:

$$V = R \cdot I \rightarrow I = \frac{V}{R} \rightarrow I = \frac{12\text{ V}}{6\ \Omega} = 2\text{ A}$$

Example 2 – Calculate voltage V_{GEN}

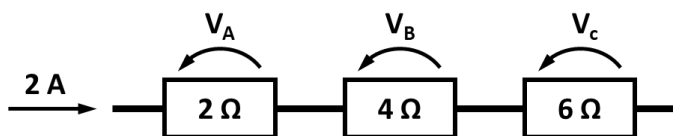


Also in this case the circuit is made of a single mesh, so the voltage V_{GEN} supplied by the generator is certainly equal to the voltage drop V on the user.

We can calculate V using Ohm's law:

$$V = R \cdot I \rightarrow V = 3\ \Omega \cdot 4\text{ A} = 12\text{ V} \rightarrow V_{\text{GEN}} = 12\text{ V}$$

Esempio 3 – Calculate voltages V_A , V_B and V_C



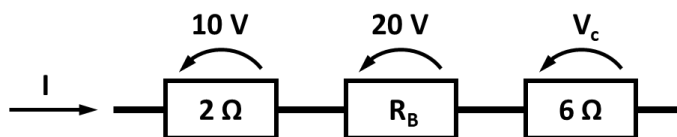
There's only a branch in the circuit, therefore the current is the same on each user. We can apply Ohm's law:

$$V_A = R_A \cdot I \rightarrow V_A = 2\ \Omega \cdot 2\text{ A} = 4\text{ V}$$

$$V_B = R_B \cdot I \rightarrow V_B = 4\ \Omega \cdot 2\text{ A} = 8\text{ V}$$

$$V_C = R_C \cdot I \rightarrow V_C = 6\ \Omega \cdot 2\text{ A} = 12\text{ V}$$

Esempio 4 – Calculate current I , resistance R_B and voltage V_C



As in the previous case, the circuit is made up of only one branch, therefore the current is the same on all three users. We know the voltage drop on the first user and its resistance, therefore, by applying Ohm's law we can find the value of the current:

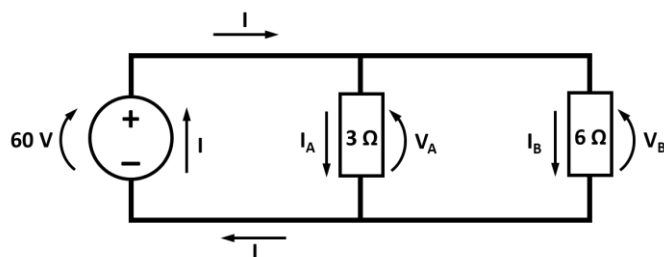
$$I = \frac{V_A}{R_A} \quad \rightarrow \quad I = \frac{10 \text{ V}}{2 \Omega} = 5 \text{ A}$$

At this point, once the current is known, we can calculate the resistance of the second user and the voltage drop on the third, again using Ohm's law:

$$R_B = \frac{V_B}{I} \quad \rightarrow \quad R_B = \frac{20 \text{ V}}{5 \text{ A}} = 4 \Omega$$

$$V_C = R_C \cdot I \quad \rightarrow \quad V_C = 6 \Omega \cdot 5 \text{ A} = 30 \text{ V}$$

Esempio 5 – Calculate currents I , I_A and I_B



There are two meshes. Mesh on the left: according to Kirchhoff's second law the voltage drop V_A is equal to the generator voltage V_{GEN} . Mesh on the right: V_A is equal to the voltage V_B . The larger rectangle that includes the generator and B is also a mesh, so we further prove that the voltage V_B is equal to V_{GEN} .

$$V_A = V_B = V_{GEN} = 60 \text{ V}$$

With Ohm's law we calculate the currents flowing on A and on B:

$$I_A = \frac{V_A}{R_A} \quad \rightarrow \quad I_A = \frac{60 \text{ V}}{3 \Omega} = 20 \text{ A}$$

$$I_B = \frac{V_B}{R_B} \quad \rightarrow \quad I_B = \frac{60 \text{ V}}{6 \Omega} = 10 \text{ A}$$

At this point, according to Kirchhoff's first law, we can calculate the current I entering the central node as the sum of the currents I_A and I_B exiting:

$$I = I_A + I_B \quad \rightarrow \quad I = 20 \text{ A} + 10 \text{ A} = 30 \text{ A}$$

Significance of the internal resistance of a supply

We define an ideal generator as a device capable of generating a constant voltage over time, independent of the load connected to its ends. As already seen, the ideal voltage at the terminals of the capacitor is equal to the electromotive force.

If we could insert an ideal generator into a circuit with very low or even zero resistance, we could generate a current of infinite intensity.

In reality, as we have already mentioned, each generator is characterized by its own internal resistance, due to its construction and the materials it is made of. The internal resistance limits the amount of current that can be supplied by the generator, because (by applying Ohm's law) it defines a maximum value given by the ratio between the voltage that can be supplied and the internal resistance. Consequently, even if the load applied to the circuit were to have zero resistance, the current would still be limited.

We can therefore write that an **ideal generator** delivers a voltage V :

$$V = EMF$$

While a **practical generator** has a voltage drop due to the internal resistance R_{INT} :

$$V = EMF - I \cdot R_{INT}$$

When we indicate the voltage of a generator in a circuit, we are implying the presence of this resistance. The current delivered will therefore not only depend on the load applied, but also on R_{INT} .

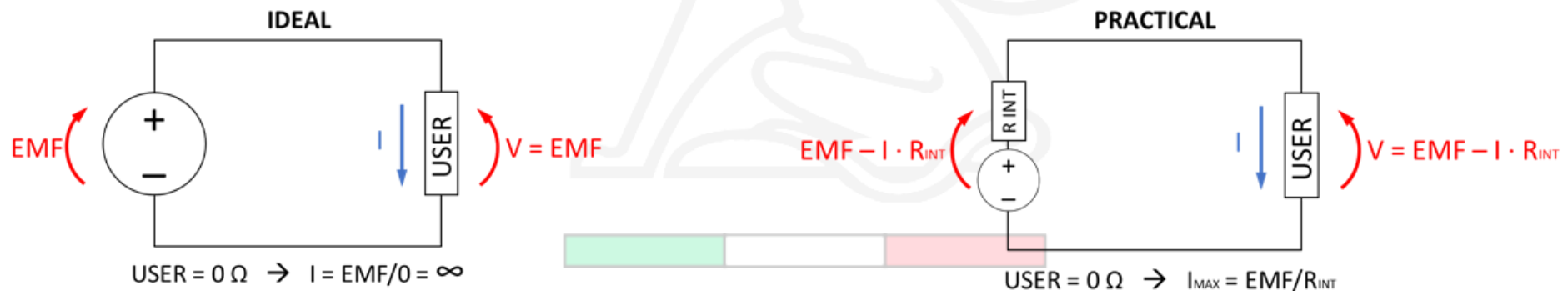


Figure 27 - Comparison between ideal and practical generator

3.7 Resistance/Resistor

SECTION A

Resistance

The elements designed to offer a specific resistance to the flow of current are called *resistors*.

The resistance R of a conductor in general can be calculated as:

$$R = \frac{\rho l}{S}$$

The value of the resistance is directly proportional to the length l of the conductor and inversely proportional to its section S . Consequently, for example, a conductor with double the cross-section of another of the same material will have half the electrical resistance compared to it. A thinner and longer conductor will have greater resistance.

The value of the resistance obviously also depends on the material of which a conductor is made, on the basis of which a parameter called resistivity varies, which has direct proportionality with the resistance.

Specific resistance

Each material can be associated with a *specific resistivity* or *resistance* (indicated with the Greek letter 'rho' ρ), which is the characteristic resistance of the specific material with unit length and section. It is indicated in Ohm meters ($\Omega \cdot m$) and also depends on the temperature of the body.

As the resistivity of the material increases, the resistance of the conductor increases.

Example: a copper cable ($\rho = 1.68 \cdot 10^{-8}$) has a lower resistance than a glass cable ($\rho = 10^{10} \div 10^{14}$) of the same size and is therefore more "prone" to being crossed by current.

We have seen that the inverse of the resistance is called **conductance** (G) and is measured in Siemens (S).

Similarly, the inverse of resistivity is *conductivity* or *specific conductance*, denoted by the Greek letter 'sigma' σ and measured in Siemens per meter (S/m). As with resistivity, conductivity depends on the material and temperature.



Calculation of total resistance using series, parallel and series-parallel combinations

When multiple resistors are present in a circuit, an equivalent resistance can be calculated, that considers all their values. The calculation of the equivalent resistance varies depending on whether the resistors are placed in series or in parallel and is obtained starting from Ohm's and Kirchhoff's laws.

Series

Resistors connected in series carry the same current through them and have different voltage drops. The symbol of fixed resistors in circuits is as follows:

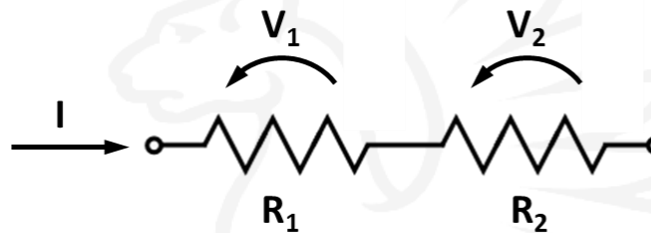


Figure 28 - Connection of two resistors in series

According to Kirchhoff's second law, the sum of the voltage drops V_1 and V_2 equals the voltage supplied to the system, which we call V . It can be said that the series connection “**distributes**” the voltage across the resistors.

$$V = V_1 + V_2 \quad \rightarrow \quad V = R_1 \cdot I + R_2 \cdot I = (R_1 + R_2) \cdot I$$

V and I are the “total” voltage and current values on the circuit. Consequently, we can define a total resistance value, which we call equivalent resistance and is equal to:

$$R_{EQ} = R_1 + R_2$$

In general, the equivalent resistance of multiple resistors in series is the sum of the resistance of each resistor.



Parallel

Resistors connected in parallel carry different currents but have the same voltage drop.

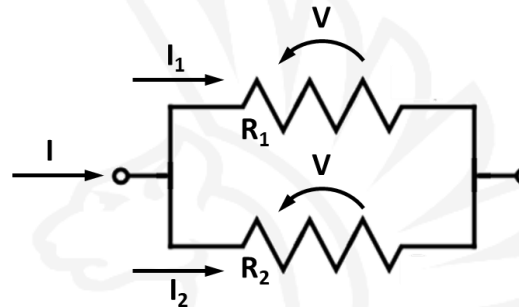


Figure 29 - Connection of two resistors in parallel

By Kirchhoff's first law, the sum of the currents I_1 and I_2 equals the total current of the system, which we call I . It can be said that the parallel connection “**distributes**” the **current** across the resistors.

$$I = I_1 + I_2 \quad \rightarrow \quad I = \frac{V}{R_1} + \frac{V}{R_2} = V \cdot \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

V and I are the “total” voltage and current values on the circuit. In this case, starting from the equation above, we can define the reciprocal of a total resistance value (the equivalent resistance), and consequently obtain the expression for the calculation of the resistance itself:

$$\frac{1}{R_{EQ}} = \frac{1}{R_1} + \frac{1}{R_2} \quad \rightarrow \quad R_{EQ} = \frac{R_1 \cdot R_2}{R_1 + R_2}$$

NOTE: the last formula obtained above is valid only when there are TWO resistors in parallel.

In general, to calculate the equivalent resistance of multiple resistors in parallel, the reciprocal of the equivalent resistance is calculated as the sum of the reciprocals of all the resistances.

Series-Parallel

When multiple resistors appear positioned both in series and in parallel with each other, it is convenient to find the value of the equivalent resistance, to calculate the total voltage or current on the circuit. The calculated values will then allow you to find the individual voltage drops and individual currents on the branches of the circuit.

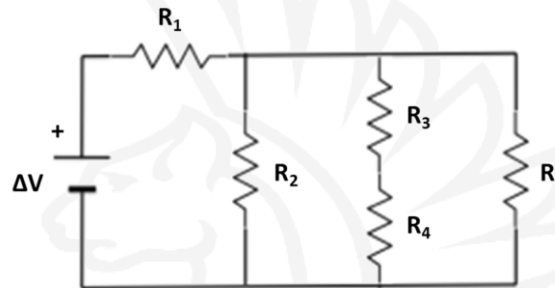


Figure 30 – Series/parallel equivalent resistance calculation example – pt.1

Considering the figure above, to find the equivalent resistance it is convenient to:

1. Find the equivalent resistance R_{34} of the two resistors R_3 and R_4 in series;
2. At this point R_4 is in parallel with R_2 and $R_5 \rightarrow$ Find equivalent resistance R_{2345} ;
3. R_{2345} appears in series with $R_1 \rightarrow$ The resistance R_{EQ} will be their sum.

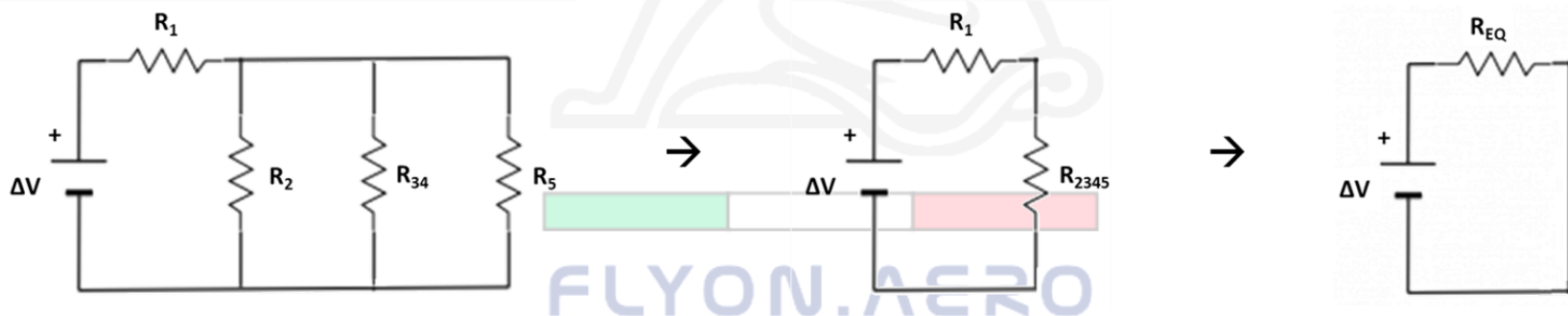
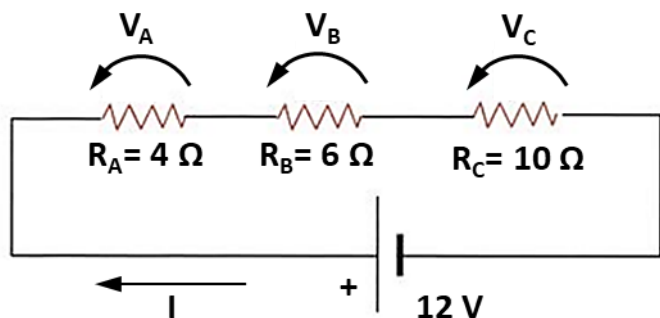


Figure 31 - Series/parallel equivalent resistance calculation example – pt.2

Examples

Example 1 – calculate voltages V_A , V_B , V_C



The resistors are all in series with each other, therefore they are crossed by the same current I , which causes voltage drops proportional to their resistance.

To find the current I we must consider the equivalent resistance:

$$R_{EQ} = R_A + R_B + R_C \rightarrow R_{EQ} = 4\ \Omega + 6\ \Omega + 10\ \Omega = 20\ \Omega$$

At this point, by applying Ohm's law we obtain:

$$I = \frac{V}{R_{EQ}} \rightarrow I = \frac{12\ V}{20\ \Omega} = 0.6\ A$$

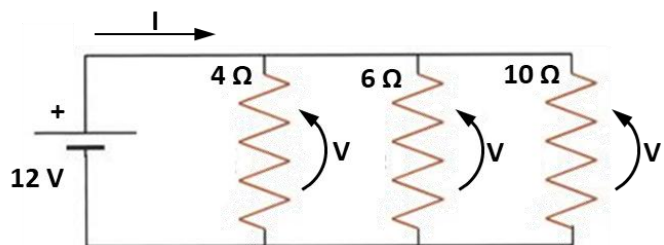
By knowing the value of I and each individual resistance we can find the individual voltage drops:

$$V_A = R_A \cdot I \rightarrow V_A = 4\ \Omega \cdot 0.6\ A = 2.4\ V$$

$$V_B = R_B \cdot I \rightarrow V_B = 6\ \Omega \cdot 0.6\ A = 3.6\ V$$

$$V_C = R_C \cdot I \rightarrow V_C = 10\ \Omega \cdot 0.6\ A = 6.0\ V$$

Example 2 – calculate equivalent resistance R_{EQ}



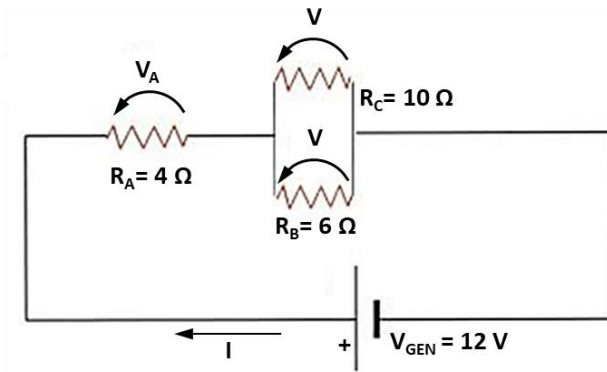
The resistors are all in parallel with each other, so they have voltage drops all equal to the generator voltage. The currents passing through them are proportional to their resistance.

Let's calculate the equivalent resistance:

$$\frac{1}{R_{EQ}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \rightarrow \frac{1}{R_{EQ}} = \frac{1}{4} + \frac{1}{6} + \frac{1}{10} = \frac{15 + 10 + 6}{60} = \frac{31}{60}$$

$$\rightarrow R_{EQ} = \frac{60}{31} = 1.94\ \Omega$$

Example 3 – calculate equivalent resistance R_{EQ}



To find the equivalent resistance we first find the equivalent resistance between R_B and R_C , in parallel with each other:

$$R_{BC} = \frac{R_B \cdot R_C}{R_B + R_C} \rightarrow R_{BC} = \frac{6 \cdot 10}{6 + 10} = 3.75 \Omega$$

The R_{BC} resistor obtained is in series with the R_A resistor. The final equivalent resistance is:

$$R_{EQ} = R_A + R_{BC} \rightarrow R_{EQ} = 4 + 3.75 = 7.75 \Omega$$



Operation and use of potentiometers and rheostats

There are also variable resistors, in which the value of the resistance varies depending on different parameters. Some types of variable resistors are rheostats and potentiometers.

The *rheostat* is a device that connects to the circuit via two terminals. Using a slider, it modifies the resistance, allowing you to *adjust the current* passing through a load placed in series.

If the component has three terminals, it can also be used as a *potentiometer*. It is connected in parallel to the load and "takes" a fraction of the *voltage*, *regulating* its value.

A very trivial use of these components is to combine them with an incandescent lamp to lower the intensity of the light it emits.

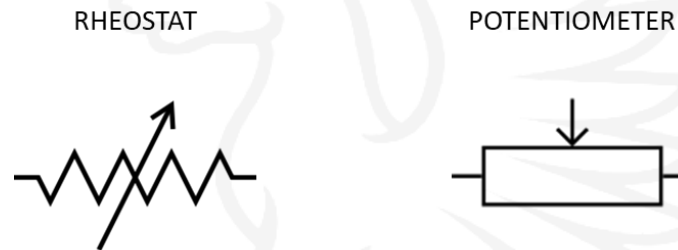


Figure 32 - Symbols of rheostats and potentiometers

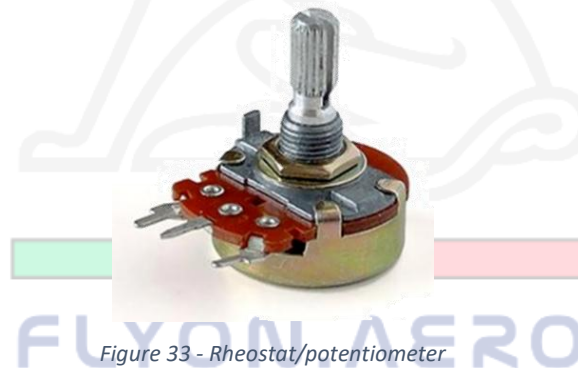


Figure 33 - Rheostat/potentiometer

Operation of Wheatstone Bridge

The Wheatstone bridge, in the figure, is a device made up of two branches of fixed resistors (the top two) and variable resistors (bottom two). When the resistance bridge is in equilibrium, i.e. when the galvanometer in the center detects 0 Ampere of current, it allows us to easily calculate the unknown resistance R_x .

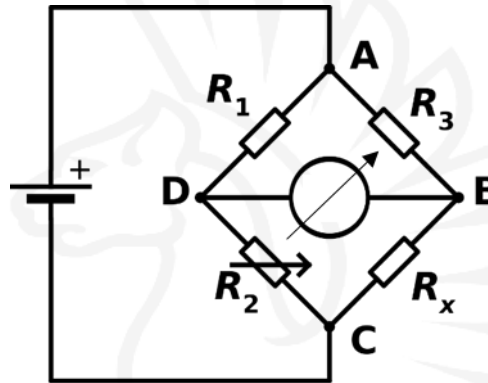


Figure 34 - Wheatstone bridge

The Wheatstone bridge finds application because the resistance of materials can vary as a function of temperature, deformation, and other factors. The conservation of the equilibrium conditions of the bridge and the value of the unknown resistance are closely linked to the variation of the physical parameters considered.

In fact, the bridge can be used as:

- Temperature sensor
Example: to control the windscreen heating system circuit.
- Strain gauge (measurement of compression or traction)
- A base for other sensors (light, quantity of liquid, humidity, pressure, etc.).

Example: in pressure sensors, the R_x resistance is combined with an elastic diaphragm that tightens according to the pressure to which it is subjected. As it lengthens, the section narrows, causing an increase in resistance. The detection of the current generated in the BC bridge due to an imbalance in the resistances can be traced back to the value of the pressure exerted on the diaphragm.

SECTION B

Resistor

Resistors are the devices designed to oppose to current flow. There are different types of them:

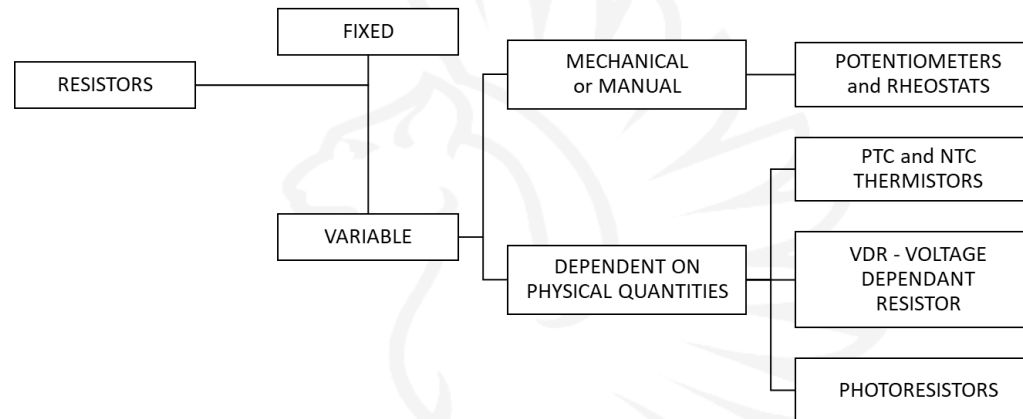


Figure 35 - Classification of resistors

Positive and negative temperature coefficient conductance

We have seen that the conductivity or electrical conductivity of a material is the inverse of the resistivity, and that both depend on temperature.

The resistivity of a material can be calculated by considering the resistivity at a reference temperature and a variation that depends on the temperature, as a function of a value called *thermal coefficient*.

- If an increase in temperature corresponds to an increase in resistivity → the material has a positive coefficient.
- If the resistivity decreases as the temperature increases → the material has a negative coefficient.

In **metals**: resistivity and conductivity vary greatly as a function of temperature. An increase in temperature causes more vibrations in the electrons, which make it harder for them to move in an orderly manner. Therefore, an increase in temperature causes an increase in resistivity and a decrease in conductivity. The coefficient is positive.

Some exceptions are graphite, in which the conductivity increases as the temperature increases, and constantan (Cu-Ni) in which there are no variations.

In **semiconductors** and **insulators**: an increase in temperature causes a decrease in resistivity and an increase in conductivity. The coefficient is negative.

Resistor colour code, values and tolerances, preferred values, wattage ratings

A designer chooses a resistor considering these factors:

- **Resistance value**
- **Tolerance**
Resistors are mass produced, so it is not possible to guarantee that the resistor actually has the indicated value. Tolerance is a parameter that indicates the acceptable limits of variations in the actual value compared to the nominal one.
For example, if we have a $200\ \Omega$ resistor with a tolerance of $\pm 10\%$, we know that it will work over a range of $180\ \Omega$ to $220\ \Omega$.
- **Stability**
The stability of a resistor is the ability to keep its resistance value constant in the circuit for a certain period of time.
- **Maximum power which can be dissipated.**

Wattage rating

Each resistor has a nominal power (in Watts) which indicates the maximum value of power that can be dissipated in heat, without suffering damage due to temperature.

The power that can be dissipated, and therefore the heating, is directly proportional to the current that passes through it. If the current exceeds the maximum value that the resistor can sustain, overheating and subsequent carbonization occurs. Sometimes, high-power resistors have a built-in heat sink.

The maximum value of power is also proportional to the size of the resistor; in fact, a larger resistor will have greater power. Carbon layer resistors (the ones most used in the electronic field), for example, have standardized power values, ranging from $1/8$ of a Watt to 2 Watts and are recognizable by their size.

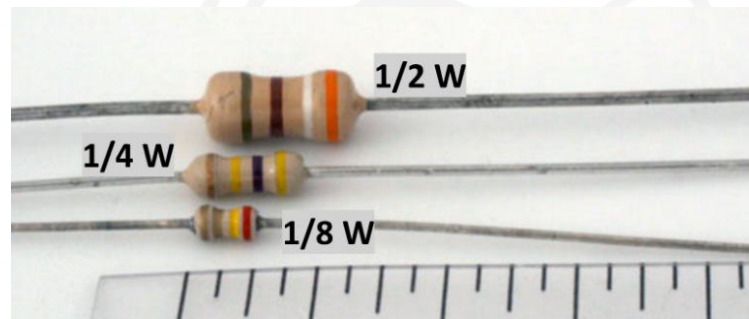


Figure 36 – Examples of wattage rating of some carbon film resistors

Preferred values

The resistors (with fixed values) on the market do not have just any values but have standard values ("*preferred values*"), established on the basis of the IEC 60063 (International Electrotechnical Commission) standard, which establishes tables of resistor values to be used depending on of tolerances.

The first three series have values indicated with only two significant figures, while the last three have three significant figures, being more precise.

STANDARD SERIES	TOLERANCE	SIGNIFICANT FIGURES
E6	20 %	2
E12	10 %	
E24	5 %	
E48	2 %	3
E96	1 %	
E192	0,5% - 0,25% - 0,1%	

For example, the E6 series resistors, which are the least precise, can only have the six numerical values defined by the standard (10, 15, 22, 33, 47, 68).

These numerical values define the **significant figures** of the resistance rating. Therefore, both a $15 \Omega \pm 20\%$ resistor and a $47'000 \Omega \pm 20\%$ resistor belong to the E6 series.

Example: the $1.04 \Omega \pm 0.5\%$ resistor and the $976 \Omega \pm 0.25\%$ resistor belong to the E192 series.



Resistor alphanumeric code

One way to indicate the resistance values of a fixed resistor is to write an alphanumeric code on the body of the element. This method is used for example on wire resistors, ceramic layer resistors and thick film resistors. The code is divided into two parts:

- The first part is made up of a number representing the significant figures of the resistance value and a letter. The letter has the dual function of identifying the position of the comma within the number and defining the unit of measurement:
 - R – Ohm
 - K – Kilo Ohm ($10^3 \Omega$)
 - M – Mega Ohm ($10^6 \Omega$).
- The second part is a letter indicating tolerance, according to this table:

CATEGORY	TOLERANCE
M	20 %
K	10 %
J	5 %
G	2 %
F	1 %

Some examples:

R47 M → $0.47 \Omega \pm 20\%$

3R3 J → $3.3 \Omega \pm 5\%$

68K J → $68 k\Omega \pm 5\%$

4M7 K → $4.7 M\Omega \pm 20\%$

825M F → $825 M\Omega \pm 1\%$



Resistor colour code

Some types of resistors (such as carbon composition resistors, carbon layer resistors, and metal oxide layer resistors) are only a few millimeters long. To write the values on the body of the resistor you would have to use very small characters, which would not be easily readable. To solve this problem, color-based coding was developed. Resistors using this coding can have 4 or 5 colored bands with specific values associated with them.

Most of the resistors on the market have 4 bands, divided into a group of 3 bands close together and the fourth a little further away, in order to define the reading order of the values from left to right:

- **First and second band:** they represent the first and second significant figures of the resistance value.
- **Third band:** it's the multiplier. It indicates the order of magnitude of the resistance value, defining the power of 10 by which the number of 2 digits must be multiplied.
- **Fourth band:** it indicates tolerance. If the fourth band is absent, tolerance is considered to be 20%.

It is useful to remember that the resistors indicated by only two significant figures are part of the E6, E12 and E24 series, with tolerances of 20%, 10% and 5% respectively. Consequently, in four-band resistors, the last band will necessarily be gold or silver (or absent). You will be able to find the other tolerance colors only in five-band resistors, where the resistance value is given by three significant figures.

Example: the resistor in the figure, with green-blue-red-gold bands, has a resistance value of 5600 Ohm and a tolerance of 5% (E24).

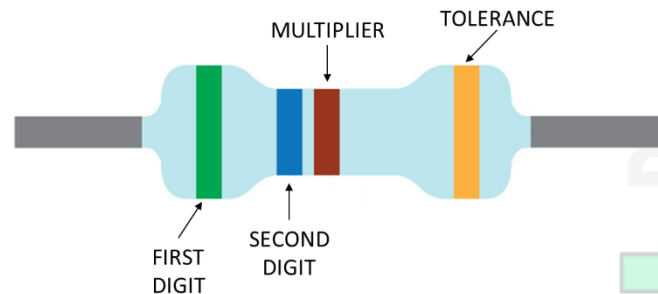


Figure 37 - Resistor colour code

COLOUR	DIGIT	MULTIPLIER	TOLERANCE
Black	0	10^0	
Brown	1	10^1	1 %
Red	2	10^2	2 %
Orange	3	10^3	
Yellow	4	10^4	
Green	5	10^5	0,5 %
Blue	6	10^6	0,25 %
Purple	7	10^7	0,1 %
Grey	8	10^8	
White	9	10^9	
Gold	-	10^{-1}	5 %
Silver	-	10^{-2}	10 %
(no colour)	-	-	20 %

Resistors in series and parallel

We have seen that resistors in series with each other have the same current flowing through them and have different voltage drops. For this reason, when we have a connection of resistors in series, we can talk about a *voltage divider*.

A common example of series connection is that of Christmas lights.

Resistors positioned in parallel, instead, have the same voltage drop but are crossed by different currents, as the current is divided into the different branches of the circuit. In this case we are talking about a *current divider*.

When resistors are placed in parallel, the failure of one has no impact on the others. This type of architecture is best in vehicle electrical systems.

Example: domestic users are connected in parallel



Fixed resistors, stability, tolerance and limitations, methods of construction

Most of the resistors we have seen so far have been fixed value, but we have seen that there are also resistors where the resistance can vary. We can make this distinction:

Fixed resistors, in their most common construction, are made up of a cylindrical element and two terminals called *rheophores*.

The stability of these resistors depends greatly on the temperature and humidity of the environment in which they are used, that can cause the actual resistance value to permanently vary compared to the nominal one. The degree of stability of the resistors can be measured through tests in various environmental conditions, comparing the final resistance with the initial one.

The actual value of the resistance, compared to the nominal one, can also vary during the construction phase, and, in this case, we are talking about tolerance (examined in Section A of this Chapter).

Based on the construction technology we can classify fixed resistors into three categories: wire resistors, mixture resistors and film resistors.

Wire-wound resistors

In wire resistors, a conductive wire is wound onto an insulating support (usually ceramic), and then the whole thing is insulated using vitreous enamel coating, which can withstand temperatures up to 350°C. In fact, these resistors have high thermal **stability**.

The resistance values depend on the diameter of the wires used, made up of different nickel alloys depending on the performance required. The **resistance** can vary from a few Ohms up to hundreds of kilo-Ohms.

The **powers** can reach values around one hundred Watts. For high power applications, enameling the component is avoided to facilitate heat dissipation.

These resistors have **tolerances** between 5% and 20%.

Since they are made of a wound wire, the occurrence of inductive components must be considered during their operation. They are therefore not suitable for high frequency **applications**.

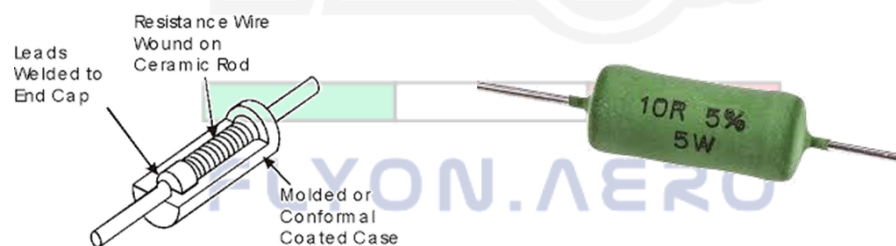


Figure 38 - Construction and example of a wire-wound resistor

Carbon composition resistors

Carbon composition resistors are made of a core of resistive mixture made of carbon, clay and talc, pressed and placed in a resin container, in which the rheophores, which come out laterally, are immersed.

Resistance values range from hundreds of Ohms to hundreds of Mega-Ohms and the **tolerance** can be 5%, 10% or 20%.

The **powers** are very low, reaching maximum values of 2 Watts. The larger the resistor, the greater the power that can be dissipated.

These resistors have small cost and size. They are very resistant from both a mechanical and electrical point of view and do not present the problem of parasitic inductance.

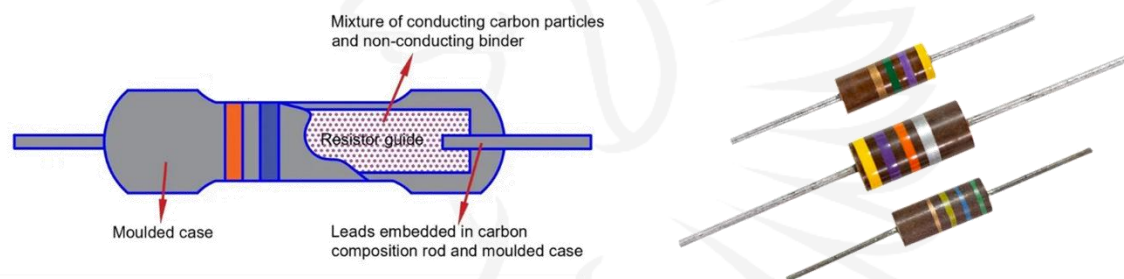


Figure 39 - Construction and example of carbon composition resistors

Film resistors

Film resistors consist of an insulating cylindrical support covered with a layer of resistive material a few millimeters thick. A spiral groove is engraved on this layer, which determines the resistance value. These resistors have high precision and stability and perform well at high frequencies.

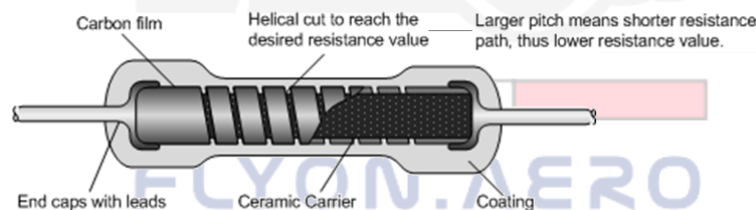


Figure 40 - Construction of film resistors

The outer layer can be made of various materials:

Carbon film resistors	Consisting of a ceramic cylinder covered in graphite	• Not suitable for high temperatures • Typical resistance: between 1 Ω and 20 MΩ • Tolerances: 20%, 10%, 5%, 2% • Dissipable powers: from 1/8 W to 2 W	
Metal oxide film resistors	Consisting of a ceramic cylinder covered with metal oxide	• Suitable for higher temperatures than the previous • Typical resistance: between 1 Ω and 2 MΩ • Tolerances: 5%, 2%, 1% • Dissipable powers: from 1/4 W to 1 W	
Metal film resistors	Consisting of a ceramic cylinder covered with a thin layer of nickel-chromium	• High stability • Typical resistance: between 1 Ω and 10 MΩ • Tolerances of 1% or less (high precision) • Dissipable powers: up to 2 W	
Ceramic film resistors (cermet)	The layer is made up of a mixture of ceramic and metallic material ("cermet")	• Suitable for high voltages, small dimensions, fireproof • Typical resistance: between 10 kΩ and 1 TΩ • Tolerances of 1% • Dissipable powers: up to 2 W	

Variable resistors, thermistors, voltage dependent resistors

Variable resistors can be resistors whose value can be changed manually or automatically by the system in which they are inserted, depending on the desired resistance value.



Figure 41 - Symbol of a variable resistor

Furthermore, resistors can have a value that varies depending on factors such as temperature (thermistors), light (photoresistors) or applied voltage (VDR).

NTC and PTC thermistors

We have seen that the value of the resistance of a conductor also depends on its temperature. In thermistors, this dependence is very significant. Thermistors are generally made of metal oxides, and are divided into:

- **NTC Thermistors** (Negative Temperature Coefficient) – resistance decreases as temperature increases.
 Usually constructed using oxides of chromium, manganese, cobalt, iron, or nickel. These oxides are covered by a ceramic layer, into which noble metal terminals such as silver or tin are inserted.
 Negative temperature coefficient thermistors are used as current limiters and temperature detectors.
- **PTC Thermistors** (Positive Temperature Coefficient) – resistance increases as temperature increases.
 Usually constructed using oxides of barium, strontium, or lead titanate. Also in this case there is a ceramic coating or other materials depending on the temperature of the environment in which they are used.
 Positive temperature coefficient thermistors are used as resettable fuses to avoid overcurrent.



Figure 42 - NTC and PTC resistors – symbols and examples

VDR – Voltage Dependant Resistors

Voltage-dependent resistors, also known as *varistors*, are devices for which Ohm's law does not apply because current and voltage are not linearly related.

The VDRs, in a circuit, are connected in parallel to the mains voltage. When the voltage is below a certain value, the current through the VDR remains low because the resistance is high. When the voltage increases too much, the resistance decreases, causing an exponential increase in current. This increase allows the overvoltage to be discharged into the VDR, which would otherwise go into the circuit and cause damage.

The behavior of the VDRs is symmetrical, in fact they are activated even when the voltage decreases too much.

Varistors, therefore, are particularly used for the suppression of high-energy impulsive disturbances, such as those produced by lightning or those caused by the interruption of inductive loads. These interruptions may be due to the action of a switch, a fuse, or a semiconductor.

VDRs are made of silicon carbide, zinc oxide or titanium oxide.

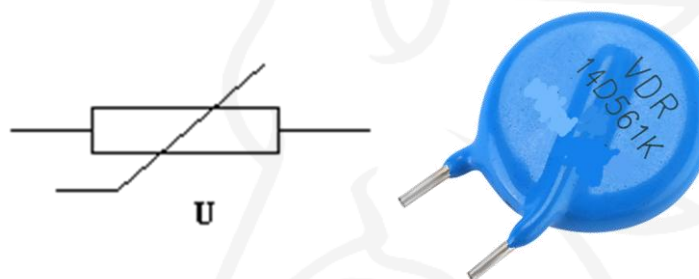


Figure 43 – Symbol and example of a VDR



Construction of potentiometers and rheostats

We have seen that rheostats and potentiometers are devices which, thanks to a cursor, allow you to manually (or mechanically) modify the value of the resistance to be applied to a circuit.

The simplest type of device is the linear one.



Figure 44 - Linear rheostat

The **rheostat** is connected in series, between a generator and a load. Referring to the diagram in the figure above, the branch with the generator is connected to terminal A while the branch with the load is connected to terminal C. By moving the cursor left and right you can determine the portion of the resistor wire that must be crossed by the current arriving from A. The greater the number of windings crossed, the greater the resistance will be and therefore the current arriving at the load will decrease.

Using the third connection (B) of the rheostat, it can be connected in parallel with the circuit, obtaining a voltage regulator (and therefore functioning as a **potentiometer**).

In the case of rotary rheostats/potentiometers, the cable windings are placed in a toroidal shape and the cursor moves around it like the hand of a clock.

The resistive element can also be made up of different types of resistors than the wire winding.



Construction of Wheatstone Bridge

The Wheatstone Bridge is constructed as follows:

- R_1 and R_3 are fixed resistors
- R_2 is a variable resistor
- R_x is a variable resistor of which we do not know the value, i.e., its resistance is an **unknown** that we can measure when the bridge is in equilibrium
- At the center of the structure there is a galvanometer (a device that allows you to measure currents and voltages).

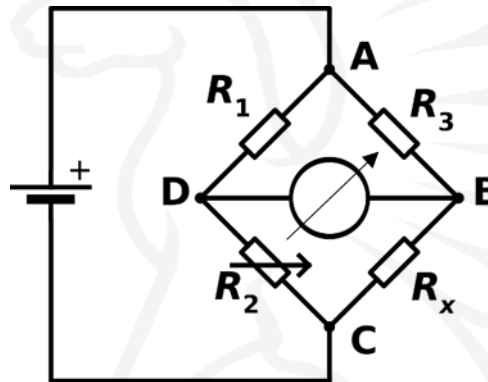


Figure 45 - Wheatstone bridge

When the Wheatstone bridge is connected to a generator, the current reaches point A, where it divides into the two branches ADC and ABC. The galvanometer also measures a flow of current in the central branch DB. To ensure that the bridge is **in equilibrium**, the resistance R_2 must be varied so that the current passing through it is exactly equal to the current passing through R_1 . Consequently, in the DB branch it will not be possible to have a flow of current and the galvanometer will read 0 A.

At this point, using Kirchhoff's and Ohm's laws it is very simple to find the unknown resistance. Let's consider that:

- R_1 and R_2 are in series (being crossed by the same current).
- R_3 and R_x are in series, because no current can flow on the DB branch; therefore, all the current from AB flows to BC.
- R_1 and R_3 are in parallel, because they are part of the ABD mesh where there are no other voltage sources.
- R_2 and R_x are in parallel for the same reason.

By applying Ohm's law, we can find the voltage on R_1 and that on R_3 :

$$V_1 = V_3 \quad \rightarrow \quad R_1 \cdot I_1 = R_3 \cdot I_3$$

In the same way we find the other two tensions. Recall that the current on R_2 is equal to that on R_1 and the current on R_X is equal to that on R_3 :

$$V_2 = V_X \quad \rightarrow \quad R_2 \cdot I_2 = R_X \cdot I_X \quad \rightarrow \quad R_2 \cdot I_1 = R_X \cdot I_3$$

We can combine the two equations found:

$$\frac{R_3 \cdot I_3}{R_1} = \frac{R_X \cdot I_3}{R_2}$$

Simplify I_3 and write as a function of R_X :

$$R_X = \frac{R_2 \cdot R_3}{R_1}$$

We found the unknown resistance. At this point we can use this value for measuring temperatures, pressures, deformations, etc.



3.8 Power

Power, work and energy (kinetic and potential)

Energy is defined as the ability of a body to do work. It is measured in Joule (J).

We can distinguish two concepts:

- **Potential energy (E_p)**: a form of energy that depends on the conditions in which a body is set. Considering its simplest form, i.e. gravitational potential, its value is directly proportional to the weight (mg) of an object and the height at which it is located.
- **Kinetic energy (E_k)**: depends on the speed of a body. It is also directly proportional to the mass of the body.

Let's take a simple example. Let's consider an object placed (still) on a table with a certain height. As long as the object remains in these conditions its speed and therefore its kinetic energy are zero, but its potential energy has a certain value other than zero. If the object is dropped, its speed and kinetic energy increase until it hits the floor. We could say that the floor is at "0 altitude", so the potential energy is zero.

What happens is a transformation of energy: potential energy is gradually transformed into kinetic energy.

Logically: higher table (greater initial E_p) $\rightarrow$ greater fall time $\rightarrow$ greater increase in speed (greater final E_k).

*If we do not consider friction, the sum of the two types of energy (**mechanical energy**) is a constant value.*

Work is the energy exchanged by a body when a force acting on it causes it to move. It is calculated as the product between the force F (Newton) and the displacement s (meters) and, like energy, it is measured in Joules.

$$W = F \cdot s$$

We can consider an object that in situation A falls from a 1-meter-high table, and in situation B falls from a 3-meter-high table. In case B the initial potential energy is greater, as the table is higher.

The gravitational force that attracts the object towards the floor is always the same, but in case B the movement made is greater. Consequently, in case B, the work done to move the object has a higher value than the work done in case A. This shows that greater energy corresponds to greater work.

Power is a concept that provides a measure of how much work is done in a unit of time. In fact, it is calculated as the ratio between work W (J) and time t (seconds). Its SI unit of measurement is the Watt (W), but it can also be measured in Horsepower (1 HP $\approx$ 736 W).

$$P = \frac{W}{t} = \frac{F \cdot s}{t} = F \cdot v$$

Power, work, energy in electric circuits

The concepts just seen can be applied to electrical circuits considering that a movement of electrons occurs in them.

In circuits, current can circulate thanks to the electromotive force of the generator. The potential difference in the circuit can be maintained over time only by supplying energy to the generator.

The **work W (J)** necessary to move a certain quantity of charge Q (C) can be calculated as the voltage across the generator V (V) multiplied by the quantity of charge. Recall that Q is always a multiple of the elementary charge of the electron q .

$$W = V \cdot Q$$

At this point we can calculate the **power P (W)** by applying the same formula seen above:

$$P = \frac{W}{t} = \frac{V \cdot Q}{t}$$

The ratio of charge to time is nothing more than the current I (A), so the power can be expressed more simply as:

$$P = V \cdot I$$

The power absorbed by a load or supplied by a generator is calculated as the product of the applied voltage and the current flowing through it.

The **energy consumed over time** is often calculated in Watt-hours (Wh) or kilowatt-hours (kWh) and depends on the power absorbed by the user in question and the time of use. Wh is not a SI quantity, but it can easily be transformed into Joule considering that 1 hour is equal to 3600 s, therefore: 1 Wh = 3600 J.

The Watt-hour is a more intuitive quantity than the Joule, in fact, it is used in daily life to express the consumption of the devices around us.

For example, a light bulb that absorbs 100 Watts, kept on for 2 hours, consumes 200 Watt-hours (=7200 J). The same energy consumption could be used to keep a low-energy light bulb (20 W) on for 10 hours.

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Dissipation of power by a resistor

Devices that transform energy never have perfect efficiency: there is always a part of energy that is dissipated, in the form of heat.

In electrical circuits, the resistance of the conductor hinders the motion of the electrons (current). At a microscopic level, the collisions that occur between the electrons and the atoms of the conductor generate an exchange (dissipation) of energy, which translates into heating of the conductor. As the resistance of the conductor and the intensity of the current increase, the collisions increase, and consequently the heating.

The heating of the conductor corresponds to the dissipated power because we consider the energy dissipated per unit of time.

The phenomenon explained is called *Joule Effect*:

A current-carrying conductor heats up.

The power, corresponding to the energy dispersed in heat over time, is directly proportional to the current and the resistance of the conductor.

Resistors are devices designed to maximize and exploit this effect: they absorb energy and dissipate it as heat. However, the amount of energy they can absorb has a limit, beyond which they suffer permanent damage. The parameter that expresses this limit is the *Wattage Rating WR (W)*.

Resistors on the market can withstand power values from a few cents to hundreds of Watts.

Examples of devices that exploit the Joule effect:

- *Incandescent bulb. The resistant filament inside is very thin and, when crossed by intense currents, undergoes such an increase in temperature that it becomes incandescent. If the filament were not in an inert atmosphere (therefore without oxygen), it would burn, making the bulb unusable.*
- *Fuse. If a short circuit occurs and the current reaches an out-of-control peak, the fuse melts, preventing further current from flowing and thus protecting the circuit.*
- *Electric heater. They contain "lamps" which when energized greatly increase their temperature, allowing the environment to be heated.*



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Power formula

We have seen that current can be calculated as the product of voltage and current.

If we consider a resistor, which follows Ohm's law ($V=R \cdot I$), then we can rewrite the formula as:

$$P = V \cdot I = R \cdot I^2 = \frac{V^2}{R}$$

The second formulation is what we have expressed as the Joule effect.

Using the inverse formula, if we know the resistance value and Wattage Rating of a resistor, we can calculate the maximum current that can pass through it without damaging it.

Calculations involving power, work and energy

Example 1 – Calculate the maximum current that a resistor with a resistance of 4Ω and a Wattage Rating of 36 W can withstand.

Let's apply the formula for the power dissipated by a resistor:

$$P = R \cdot I^2 \quad \rightarrow \quad I = \sqrt{\frac{P}{R}} \quad \rightarrow \quad I = \sqrt{\frac{36}{4}} = 3 \text{ A}$$

Example 2 – A load is powered at 30 V and has a resistance of 2Ω . Calculate the power absorbed and energy consumption in 12 hours.

Let's apply the formula of the power dissipated by a resistor:

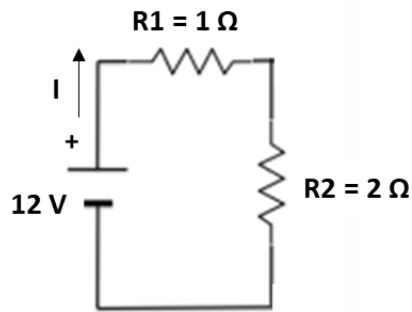
$$P = \frac{V^2}{R} \quad \rightarrow \quad P = \frac{900}{2} = 450 \text{ W}$$

Power is the energy per unit of time, which we can calculate in **Watt-hours** and convert to Joules:

$$P = \frac{E}{t} \quad \rightarrow \quad E = P \cdot t \quad \rightarrow \quad E = 450 \text{ W} \cdot 12 \text{ h} = 5400 \text{ Wh} = 5.4 \text{ kWh}$$

$$E = 5400 \text{ Wh} \cdot 3600 \text{ s/h} = 19440000 \text{ J} = 19.44 \text{ MJ}$$

Example 3 – Calculate the resistance dissipated by each resistor and the power delivered by the generator.



Let's first calculate the equivalent resistance of the two resistors in series and then the current (Ohm's law):

$$R_{EQ} = R_1 + R_2 \rightarrow R_{EQ} = 3 \Omega ; I = \frac{V}{R} \rightarrow I = \frac{12}{3} = 4 A$$

Let's calculate the two dissipated powers:

$$P = R \cdot I^2 \rightarrow P_1 = 1 \cdot 4^2 = 16 W ; P_2 = 2 \cdot 4^2 = 32 W$$

And the power supplied by the generator:

$$P = V \cdot I \rightarrow P = 12 \cdot 4 = 48$$

Example 4 – A 60 W lamp is turned on for 3 hours every evening. Calculate the electricity consumed in a month (30 days).

Let's first calculate the total time:

$$h_{TOT} = 3h \cdot 30days = 90 h$$

Energy in watt-hours is given by:

$$E = P \cdot t \rightarrow E = 60 W \cdot 9 h = 5400 Wh = 5.4 kWh$$

Example 5 – A direct current motor is powered at 200 V by a current of 10 A. Calculate the energy absorbed by the motor in 12 hours of operation (in kWh and J).

Let's calculate the engine power:

$$P = V \cdot I \rightarrow P = 200 \cdot 10 = 2000 W$$

The energy absorbed is:

$$E = P \cdot t \rightarrow E = 2000 W \cdot 12 h = 24000 Wh = 24 kWh$$

Convertiamo in Joule:

$$E = 24000 Wh \cdot 3600 s/h = 86.4 MJ$$

3.9 Capacitance/Capacitor

The term "electrical capacity" represents the ability of a body to store charges which, remaining stored (still), can be defined as electrostatic.

Capacitors are electrical components capable of storing electrostatic energy.

Operation and function of a capacitor

Capacitors are made up of two conductors (*plates* or *armatures*) separated by an insulator (*dielectric*). Each armature has a terminal to be connected to the circuit.

We can find **linear** or **flat capacitors**, with the plates parallel to each other or **cylindrical capacitors**, in which the plates are two coaxial cylinders separated by the dielectric.



Figure 46 - Linear and cylindrical capacitor

When a capacitor is connected to a generator, a potential difference is established between its plates, which generates an electric current. The charges "separate": the negative ones accumulate on the low potential plate and the positive ones, consequently, accumulate on the other plate (basically it is a lack of negative charges). The charges do not disperse in the circuit thanks to the electrostatic attraction between them.

To charge a capacitor it is therefore necessary to supply energy from the outside. Once charged, even if we disconnect the external power supply, the capacitor will keep its accumulated charges. Basically, the capacitor can store *electrical potential energy*. The accumulated charges cannot move because between the two plates there is dielectric, i.e. non-conducting, material.

In direct current (DC) circuits, capacitors are used to store electrical energy that can be used at a later time. They can also be used in alternating current (AC) circuits to filter certain frequencies.

The **total charge Q stored** in the capacitor is the absolute value of the positive or negative charges (the number of positive and negative charges is the same). The maximum charge that can be accumulated is proportional to the potential difference V applied to the plates and to the capacitance C of the capacitor, which depends on its geometry and composition.

$$Q = C \cdot V$$

The **capacitance of the capacitor** is measured in **Farads (F)**.

1 Farad corresponds to the capacity of a capacitor that can accumulate 1 Coulomb of charge if a potential difference of 1 Volt is applied to its plates.

To store a 1 C charge with a voltage of only 1 V would require using a huge capacitor. Capacitors commonly found on the market have capacities ranging from **picoFarad (pF)** to **milliFarad (mF)**.

The **energy stored** in the capacitor is measured in Joules and is calculated as:

$$E = \frac{1}{2} CV^2$$

At most, the potential difference across the capacitor can reach a value equal to the voltage of the generator to which it is connected.

It must be underlined that the dielectric placed inside the charges has limits. If the potential difference applied to the capacitor increases beyond a certain value, which depends on the material the dielectric is made of, then the dielectric **breaks down**. In this situation the material no longer behaves as an insulator, but temporarily behaves like a conductor, allowing the passage of the accumulated charges. This can lead to permanent capacitor damage.



Factors affecting capacitance: area of plates, distance between plates, number of plates, dielectric and dielectric constant, working voltage, voltage rating

The capacity of a capacitor, and consequently the charge and energy accumulated, depend on the shape of its plates, their distance, and the material of which the dielectric between them is made. For simplicity, let's consider the case of the parallel plate capacitor.

The formula of the capacity C (F) is:

$$C = \epsilon \cdot \frac{S}{d}$$

In which ϵ (F/m) is the absolute dielectric constant of the mean (*studied in chapter 2 of this manual*), S is the surface area of the plates (m^2) and d is the distance between them (m).

As the surface area and the dielectric constant increase, capacity increases. Instead, it decreases as distance between the plates increases.

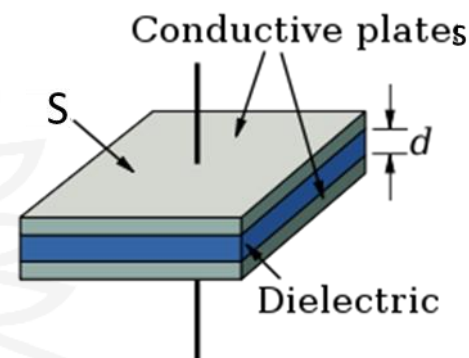


Figure 47 - Schematic of a linear capacitor

A large **surface area** of the armatures determines a high capacity as it can store more charges for the same energy supplied. By connecting multiple *capacitors in parallel*, we obtain the equivalent of a single capacitor with a larger surface area, and therefore a greater equivalent capacity.

If the plates are not **distant** from each other, the capacity is greater, because the charges of opposite sign are very close to each other and attract each other more easily, dispersing less along the circuit. However, for the same reason, by reducing the distance between the reinforcements, the risk of breaking the dielectric increases. To solve this problem, "multiple plates" can be used, by connecting multiple *capacitors in series*. It can be obtained the equivalent of a capacitor with plates further apart from each other, which allows to apply higher voltages than can be supported by a single capacitor. The disadvantage is that the equivalent capacity is reduced.

Capacitors can exploit different types of **dielectrics**, including vacuum, air, paper, etc. Each material is characterized by its own dielectric constant, which can be expressed in an absolute way (ϵ) or, if compared with that of the vacuum ϵ_0 , in a relative way (ϵ_r) (dimensionless). A material with a higher dielectric constant results in a capacitor with a higher capacity.

The ability of an insulator to withstand a potential difference without being crossed by a current (breakdown of the dielectric) is indicated by a quantity called *dielectric strength*. This quantity is commonly measured in kilovolts per millimeter (*kV/mm*) and defines the limit value of the applicable electric field, beyond which a discharge is produced.

Some commonly used dielectrics:

DIELECTRIC MEAN	RELATIVE DIELECTRIC CONSTANT $\epsilon_r = \epsilon/\epsilon_0$	DIELECTRIC STRENGTH [kV/mm]
DRY AIR	1	3
MINERAL OIL	2.2 ÷ 2.5	7.5 ÷ 16
BAKELITE (a type of resin)	5.5 ÷ 8.5	10
GLASS	6 ÷ 8	25 ÷ 100
PURE WATER	81.07	15

Regarding the applicable **voltages**, each capacitor is characterized by an operating voltage and a peak voltage.

The first (*working voltage*) is the maximum voltage (direct or alternating) that can be applied to the capacitor continuously without suffering damage.

The second (*voltage rating*) is the maximum voltage that must never be exceeded, regardless of the application time.

NOTE: pico- (p) 10^{-12} ; nano- (n) 10^{-9} ; micro- (μ) 10^{-6} ; milli- (m) 10^{-3}



Capacitor types, construction and function

There are different types of capacitors, which are mainly divided into **fixed** and **variable**. Fixed capacitors, in turn, can be polarized or non-polarized.

Polarized capacitors: there is a designated direction for the passage of current, which if not respected can lead to dangerous damage to the circuit. Polarity may be indicated differently depending on the manufacturer, for example a shorter terminal may be used to indicate negative, or a colored mark or dash may be used on the capacitor.

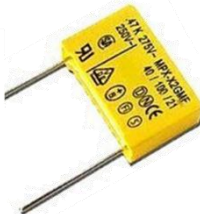
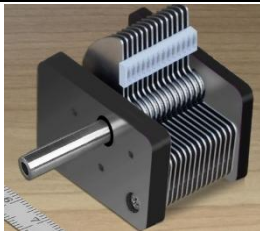
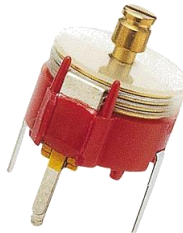
The symbolism is the following:



Figure 48 - Capacitors symbols

Capacitors are defined based on the dielectric and are characterized by the capacitance value, tolerance and maximum operating voltage.

Fixed not polarized	Disc (single layer) or multilayer ceramic capacitors	The dielectric is made of ceramic material, which depending on the processing can have a low or high dielectric constant.	• Capacitance: between 1 pF and 100 nF • If ϵ has a low value: they are very stable and are preferred in high-precision circuits • If ϵ is high: they have high capacities with little dimensions; they are preferred in high frequency applications • Single layer: higher maximum operating voltage (up to 1 kV) • Multilayer: they are the most used. They have higher capabilities	DISK MULTILAYER
	Paper capacitors	The dielectric is a special paper impregnated with a fluid substance. Often there are multiple layers that are wrapped and dipped in oil or resin.	• Capacitance: between 500 pF and 50 nF • Working voltages up to 2 kV • Suitable for high voltage applications • Usually have tolerances of 20% • They self-repair if the dielectric is punctured	

	Polyester capacitors	The dielectric is plastic (polyester). The conductor can be made up of two metal sheets or a film deposited on the dielectric.	• Capacitance: between 1 nF and 1 μF • Working voltages up to 1 kV • Among the most used • Suitable for low frequency uses	 
Fixed polarized	Aluminum or tantalum electrolytic capacitors	Made of an electrode (anode) on which a layer of dielectric oxide with very high ϵ is deposited. The other electrode (cathode) often coincides with the container itself. The two electrodes are connected by an electrolyte.	• Capacitance: the thin dielectric layer allows for very high capacities - Aluminum: between 1 μF and 10 mF - Tantalum: from 1 μF to 70 mF • Working voltages depend on the thickness of the dielectric: up to a few tens of Volts • The shorter connector represents the negative electrode • Tantalum: they can lead to thermal instability, so they are placed in containers that dissipate heat; they are smaller in size	 ALUMINUM  TANTALUM
Variable	Variable air capacitors	They consist of a pack of mobile plates connected to each other and immersed in the air.	• Capacitance: 10 pF to 360 pF • Breakdown voltage up to 100 V • Used for radio frequency applications	
	Capacitive trimmers	They are equipped with a screw which, when rotated, modifies the value of the capacity by moving the armatures further or closer.	• Capacitance in the order of tens of pF • Small dimensions: suitable for mounting on printed circuits	

Capacitor colour coding

The capacitance value of the capacitors, as we have already seen for resistors, can be indicated via a color code, or using an alphanumeric code.

The color code is used for ceramic capacitors, polyester capacitors, and tantalum capacitors. Specific values are used for each type of capacitor.

Polyester capacitors have five colored bands to read from top to bottom:

- First and second bands: represent the first and second significant figures of the capacitance value, always expressed in picofarads (pF).
- Third band: it is the multiplier. It indicates the order of magnitude of the capacity value, defining the power of 10 by which the 2-digit number must be multiplied.
- Fourth band: indicates tolerance.
- Fifth band: indicates the working voltage.

Example: the polyester capacitor in the figure, with purple-red-green-white-yellow bands, has a capacitance value of 72×10^5 pF, a tolerance of 10% and a working voltage of 400 V.

Tantalum polarized capacitors, on the other hand, have a color coding made up of three bands to be read from top to bottom and a circle in the center:

- First and second bands: represent the first and second significant figures of the capacitance value, always expressed in microfarads (μ F).
- Central circle: it is the multiplier.
- Third band: indicates tolerance.

Example: the polarized tantalum capacitor in the figure, with red-purple-yellow bands and a brown circle, has a capacity of 270 μ F and a working voltage of 6.3 V.

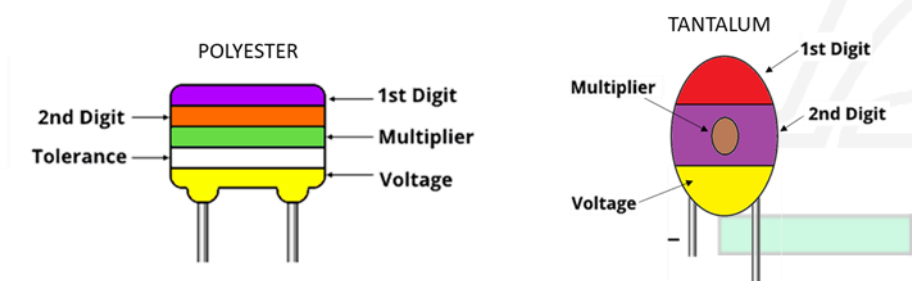


Figure 49 - Polyester capacitor and tantalum capacitor color codes

COLOUR	DIGIT	MULTIPLIER		TOLERANCE	WORKING VOLTAGE	
		POLYESTER	TANTALUM	POLYESTER	POLYESTER	TANTALUM
Black	0	10^0	10^0	20 %	-	10 V
Brown	1	10^1	10^1	1 %	100 V	1.6 V
Red	2	10^2	10^2	2 %	250 V	4 V
Orange	3	10^3	-	-	300 V	35 V
Yellow	4	10^4	-	-	400 V	6.3 V
Green	5	10^5	-	5 %	500 V	16 V
Blue	6	-	-	-	630 V	20 V
Purple	7	-	10^{-3}	-	700 V	50 V
Grey	8	-	10^{-2}	-	800 V	25 V
White	9	-	10^{-1}	10 %	900 V	2.5 V

Alphanumeric coding

On some capacitors, the parameters can also be indicated using an alphanumeric code, which varies based on the type of capacitor but also depending on the manufacturer.

For example, **ceramic capacitors** have various coding. We can find a three-digit one, where the first two indicate the capacity value and the third the number of zeros to add. The unit of measurement in this type of coding is always the picofarad (pF).

If the capacitance value is a number with a decimal point, then it is indicated in nano-farads (nF) and the decimal point is represented by the letter “n” in the corresponding position.

Manufacturers that use microfarads (μF) use a dot instead of a comma and omit the leading zero.

Coding examples of ceramic capacitors:

- 472 $\rightarrow$ capacity of 4700 pF
- 10n $\rightarrow$ capacity of 10 nF = 10'000 pF
- 3n3 $\rightarrow$ capacity of 3.3 nF = 3'300 pF
- .01 $\rightarrow$ capacity of 0.01 μF = 10'000 pF

Polyester capacitors can have the coding just described for ceramic ones. The decimal point is represented by the letter “n” as in the previous case, or by the letter “u” which represents microfarads (μF).

In these capacitors we can also find a letter indicating the tolerance: J for 5%, K for 10% and M for 20%.

Furthermore, there is a last number that indicates the working voltage.

Example of coding of polyester capacitors: .15M50 $\rightarrow$ capacitance of 0.15 μF = 150'000 pF, tolerance of 20%, working voltage of 50 V.

Tantalum electrolytic capacitors have the parameters clearly marked on the body. The value is always in microfarads (μF). The negative lead can be indicated in various ways.



Calculations of capacitance and voltage in series and parallel circuits

Capacitors, like other elements of electrical circuits, can be connected in series or parallel.

- **Series connection:** it is like having a capacitor with plates further apart → the capacity is reduced but the applicable voltage is higher.

The equivalent capacitance of several capacitors in series is calculated as the equivalent resistance of several resistors in parallel, i.e. the reciprocal of the equivalent capacitance is equal to the sum of the reciprocals of all the capacitances. As a result, the equivalent capacity is less than the smallest capacity considered in the calculation.

$$\frac{1}{C_{EQ}} = \frac{1}{C_1} + \frac{1}{C_2} + [...]$$

In case there are only two capacitors we will find the final formula:

$$\frac{1}{C_{EQ}} = \frac{1}{C_1} + \frac{1}{C_2} \rightarrow C_{EQ} = \frac{C_1 \cdot C_2}{C_1 + C_2}$$

- **Parallel connection:** it's like having a capacitor with a larger surface area → the capacity increases.

The equivalent capacitance of multiple capacitors in parallel is calculated as the equivalent resistance of multiple resistors in series.

$$C_{EQ} = C_1 + C_2 + [...]$$

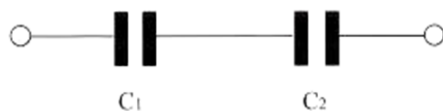
These formulas are valid for all types of capacitors seen. In the case of polarized (electrolytic) capacitors it is important to respect the polarity of the connections. In series connection, the positive pole of the first capacitor must be connected to the negative of the second, and vice versa. In parallel connection, the positive poles are connected to the positive poles and the negative poles to the negative poles.



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Some examples follow:

Example 1 – calculate the equivalent capacitance of two capacitors in series. $C_1 = 10 \text{ mF}$ and $C_2 = 5 \text{ }\mu\text{F}$.



We perform the calculation in microfarads, so we first convert the unit of measurement of C_1 .

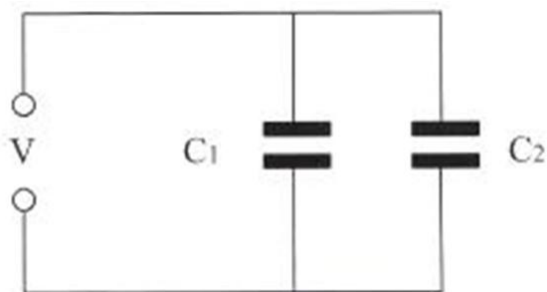
$$C_1 = 10 \text{ mF} \rightarrow C_1 = 10000 \text{ }\mu\text{F} = 10^4 \text{ }\mu\text{F}$$

Let's apply the formula for calculating the equivalent capacity of two capacitors in series:

$$\frac{1}{C_{EQ}} = \frac{1}{C_1} + \frac{1}{C_2} \rightarrow C_{EQ} = \frac{C_1 \cdot C_2}{C_1 + C_2} \rightarrow C_{EQ} = \frac{10^4 \cdot 5}{10^4 + 5} = 4.99 \text{ }\mu\text{F}$$

We note that, although capacitor C_1 has a very high capacity, the equivalent capacity is still less than the smallest capacity (C_2).

Example 2 – calculate the equivalent capacitance of two capacitors in parallel. $C_1 = 10^4 \text{ }\mu\text{F}$ and $C_2 = 5 \text{ }\mu\text{F}$.



Let's apply the formula for calculating the equivalent capacitance of two capacitors in parallel:

$$C_{EQ} = C_1 + C_2 \rightarrow C_{EQ} = 10^4 + 5 = 10005 \text{ }\mu\text{F}$$



Exponential charge and discharge of a capacitor, time constants

Consider a discharged and isolated capacitor. Under these conditions, the plates are electrically neutral because there are the same number of protons and electrons on them.

Charging phase

Now, connect the capacitor to a voltage generator: the current will begin to flow towards the plates, clockwise (remember that the direction of the electrons is opposite to the current, so the flow of electrons is anti-clockwise).

The positive pole of the generator attracts electrons from the armature to which it is connected (the upper one), which becomes positively charged due to the lack of negative charges. The movement of the electrons is the **charging current**. The other plate of the capacitor (the lower one) in the meantime attracts electrons from the negative pole of the generator. A potential difference was therefore created between the plates.

The number of electrons circulating is initially maximum and decreases as the capacitor charges.

When the capacitor reaches its maximum charge, i.e., its voltage is equal to that of the generator, there is an excess of positive charge on the upper plate and an excess of negative charge on the lower plate. Even if the capacitor is disconnected from the generator, the charge on the plates is maintained.

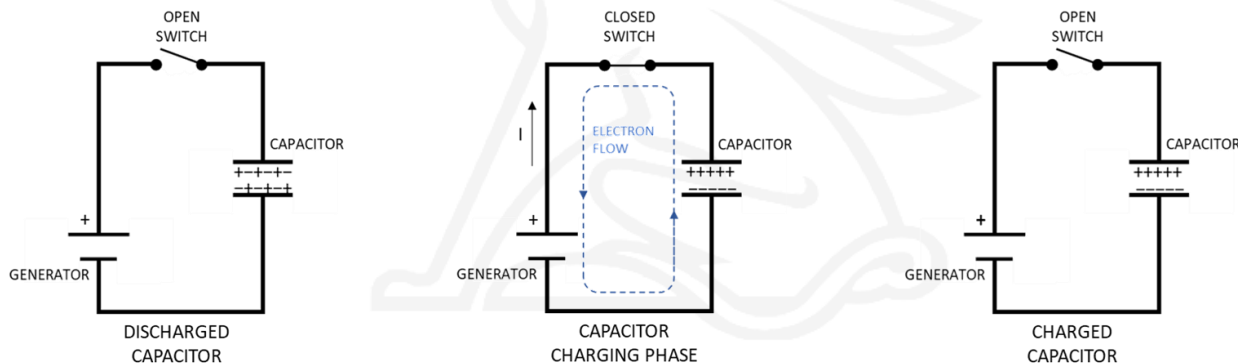


Figure 50 - Charging phase of a capacitor

Discharging phase

At this point the energy stored in the capacitor can be used. If we put the plates in contact by means of a resistor, the capacitor discharges, because the charges can use the electrical circuit to rebalance themselves.

There will therefore be a movement of electrons (*discharge current*) opposite to that present during the charging phase and, at the end of the discharge process, the plates will be as neutral as they were initially. The potential difference goes to zero.

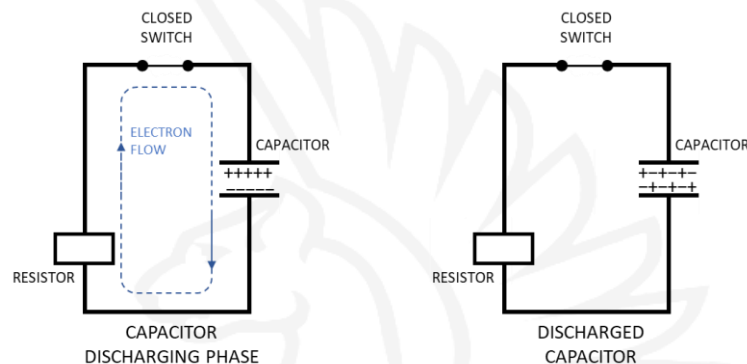


Figure 51 - Discharging phase of a capacitor

The charging and discharging of a capacitor occur in a specific time interval, which depends on the capacitance of the capacitor and the resistance in series with it.

The definition of this time derives from data obtained experimentally. It can be shown that the time required for any capacitor to charge to 63% of its maximum capacity is equal to a time constant, denoted by the Greek letter τ ('tau'), given by:

$$\tau = R \cdot C$$

Where the capacitance C of the capacitor is given in Farads and the resistance R of the load is in Ohms. After a time interval equal to τ , the charging continues more slowly. Experimentally, it can be shown that the capacitor will reach its maximum charge after a time interval 5τ long.

The same thing happens for the discharge, i.e. the capacitor will be 63% discharged after one τ and will be completely discharged after 5τ .

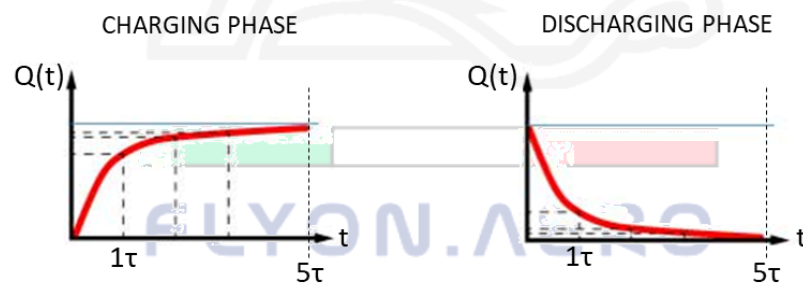


Figure 52 - Charging and discharging rate of a capacitor

Testing of capacitors

The most frequent problem for capacitors is related to the loss of the insulating properties of the dielectric, and occurs due to the time of use, wear and the environmental conditions in which the element is placed (humidity, temperature, etc.).

When the dielectric deteriorates, a certain portion of the charges manages to constitute a current between the plates, and therefore the capacitor does not function correctly. Statistically, the capacitors most subject to this type of problem are those with paper dielectrics and electrolytics.

To check the state of a capacitor you can use a capacitor or a tester. Before this, however, it is always better to do a visual inspection. If an electrolytic capacitor shows swelling or electrolyte leaks, it must be replaced. Checking the parameters is done with the capacitor removed from the circuit and completely discharged, because even a small current could damage the instruments or be dangerous.

By connecting the capacitor to a *capacitance-meter* you can read the actual capacitance value and, by comparing it with the nominal one, you can understand the state of the capacitor.

The capacitance of a worn capacitor may increase or decrease from its nominal value. Usually, the capacitance tends to decrease in low-capacity capacitors, while it increases in high-capacity capacitors. Within certain limits of variation, the capacitor is still usable (remember that the tolerance value exists).

If you use a *tester*, i.e. a multimeter, to check the capacitor, you will not obtain precise capacitance values, but you can still understand whether the capacitor is working or not. The tester is suitable for checking capacitors with medium-high capacities, such as electrolytic ones.

You choose the ohmmeter mode and set the maximum resistance value according to the nominal capacitance value of the capacitor (the tester manufacturer provides tables of reference values). The test is performed by connecting the terminals of the capacitor to the ohmmeter: the resistance value is initially maximum and then decreases towards a certain value and then increases again to the maximum value. The resistance has this trend because the capacitor is charging, so there is a flow of current. When the resistance returns to the maximum value it means that the capacitor is charged and therefore the current does not pass.

By reversing the polarity (i.e. connecting the terminals of the capacitor in opposite ways), we notice a decrease in the resistance value again and a new reaching of the maximum value. In this case the capacitor discharge occurred.

The resistance values obtained can be compared with those contained in the tables provided by the multimeter manufacturer, which define the acceptable limits for each type of capacitor.

If the resistance value decreases without returning to the maximum, it means that the capacitor is short-circuited, i.e. the dielectric is not doing its job.

If the value remains at the maximum and does not vary, it means that the capacitor is an open circuit and is unable to accumulate charges.

If the resistance value remains halfway or returns too slowly from the minimum to the maximum value (remember the time constant τ), it means that the capacitor has losses.

3.10 Magnetism

SECTION A

Theory of magnetism

Magnetism is a physical phenomenon whereby some materials are able to attract ferrous objects (or objects containing cobalt or nickel). These materials are called *magnets*.

A magnet is therefore a body capable of exerting forces that act in the surrounding space, i.e. a *magnetic field*. The magnetic field is indicated with the capital letter **H** and is measured in Amperes per meters (*A/m*).

The sources of this field are the *magnetic poles*: the North (N) pole and the South (S) pole.

Magnetic fields can be generated by permanent (or temporary) magnets or by electromagnets (which fall under the definition of temporary magnets). In the former the magnetic properties are intrinsic and can be traced back to the subatomic structure. Electromagnets, on the other hand, are able to generate a magnetic field thanks to a current flowing along a conductor.



Properties of a magnet

When we talk about *natural magnets*, we mean materials that by their nature can generate a magnetic field. These magnets are composed of iron oxides (such as magnetite and hematite) and their properties derive from the movement of the electrons external to the atoms.

More precisely, the magnetic properties are due to the spin of the unpaired electrons in the orbitals, i.e. the direction with which they rotate around their axis. These electrons have a constant orientation in large regions of matter, called **magnetic domains**, making them polarized and therefore capable of generating a magnetic field.

There are also *artificial magnets*, generated thanks to the interaction of a magnetizable body with an external magnetic field. The magnets obtained can be temporary (electromagnets) or permanent.

Two poles can be identified in magnets. The behavior of the two poles is similar to that of electric charges. Two equal poles (North-North or South-South) repel each other, and two different poles (North-South) attract each other.

By spreading iron filings on a surface and placing the magnet on it, you notice that it assumes particular positions. Its trace highlights the presence of the magnetic field because the filings align with the field lines (or lines of flux or force).

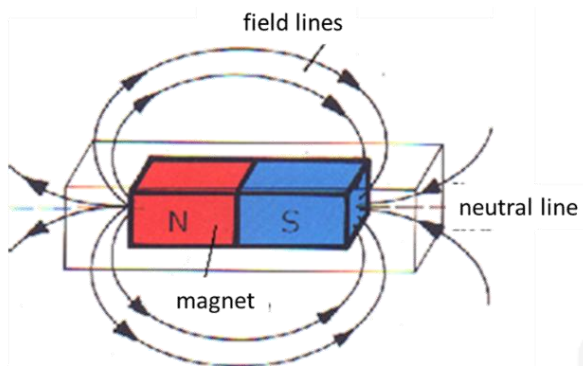


Figure 53 - Magnetic field lines

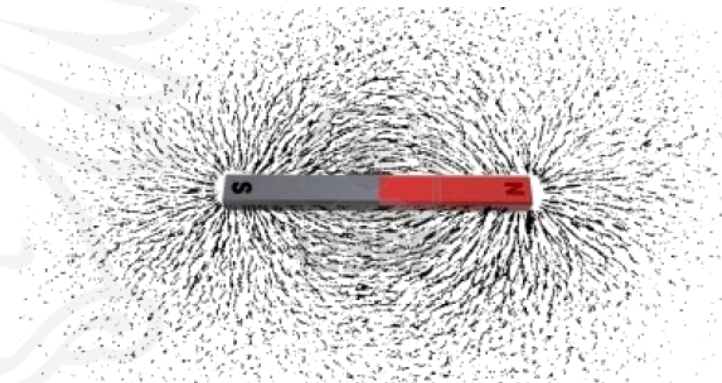


Figure 54 – Magnet on iron filings

Magnetic field lines behave similarly to the electric field lines seen previously. By convention, in fact, they start from the North Pole and arrive at the South Pole, they never intersect each other and are always perpendicular to the pole. The tangent to the field lines, at each point, coincides with the direction that a magnetic needle would take if placed at that point.

The lines are more concentrated in the area of the poles: it means that the magnetic field in those points has the maximum intensity.

When two magnets come together, the field lines of one combine with those of the other, without ever intersecting. By bringing two opposite poles closer, the flow lines strengthen each other, while they weaken by bringing two equal poles closer.

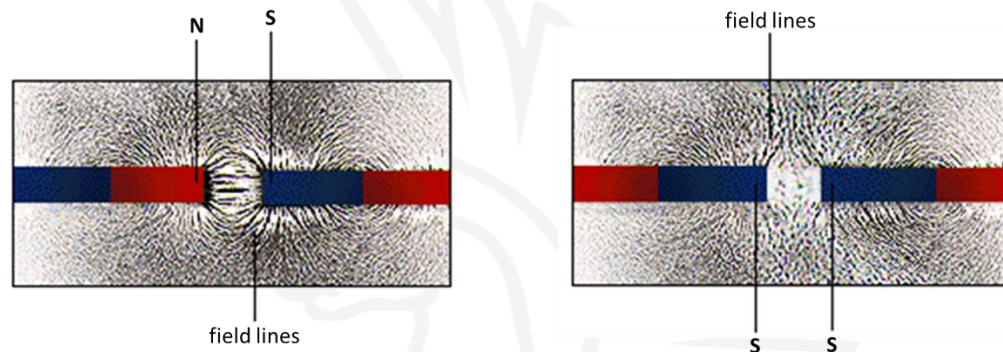


Figure 55 - Interaction between two magnets

The magnetic field lines are always closed, because in every magnet there is always a North and a South pole. The free *magnetic monopole*, although hypothesized theoretically, has never been observed experimentally. In fact, if you take a magnet and cut it into several parts, you will not divide North from South, but you will obtain many smaller magnets, each equipped with two poles. If you divide these magnets further you will obtain further smaller magnets, equipped with North and South poles.

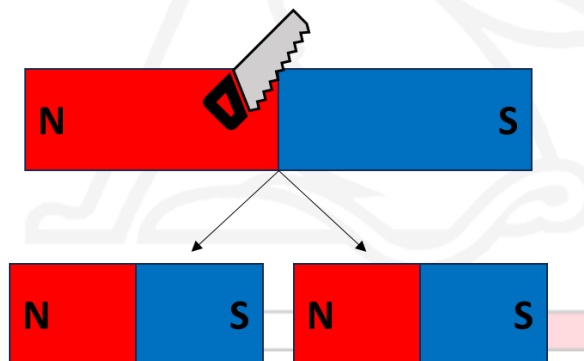


Figure 56 - Cutting of a magnet

Action of a magnet suspended in the Earth's magnetic field

Planet Earth can be considered as a large magnet, as it has a weak magnetic field.

By convention, the Earth's North and South magnetic poles are located respectively in the Northern and Southern hemispheres, even though they do not coincide with the relevant geographic poles. In fact, the latter (through which the Earth's rotation axis passes) are translated by a few hundred kilometers with respect to the Earth's magnetic poles.

We talk about convention because, carrying out a simple experiment, we notice that the magnetic North and South of the Earth, actually, are reversed.

Consider a suspended magnet that is free to move, like a compass needle. It will initially rotate a little but will then align with the Earth's magnetic field. We have seen that equal poles repel each other; therefore, the North pole of the magnet will certainly face the South of the Earth's magnetic field and vice versa. What we notice from the experiment, contrary to what we would expect, is that the north of the magnet is facing the geographical northern hemisphere of the planet. What does it mean?

This can be explained because, in reality, the lines of the Earth's magnetic field exit from the Southern geographic hemisphere and enter the Northern hemisphere, defining a "North pole of the Earth magnet" in the Southern hemisphere and a "South pole of the Earth magnet" in the Northern Hemisphere, as indicated by the small blue and red magnet in the figure. This information is important because it allows us to understand the functioning of compasses, which point towards a North defined as "Earth's magnetic North", but which physically speaking is a magnetic South Pole. In conclusion:

Earth's magnetic North Pole: it is what we consider by convention to be in the Northern Hemisphere, shifted with respect to the corresponding Geographical Pole.

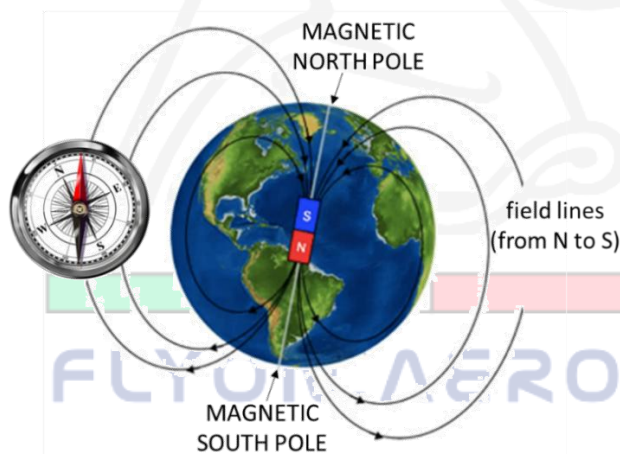


Figure 57 – Magnetic field of Earth

Magnetisation and demagnetisation

Magnets can also influence the behavior of other ferrous objects, making them magnets themselves. In this case we talk about magnetization, and we can define a quantity called *magnetic induction*, indicated with the letter **B** and measured in *Tesla* (**T**) in the SI, or in Gauss ($1 \text{ G} = 10^{-4} \text{ T}$).

We can therefore distinguish natural magnets from artificial ones, which in turn can be temporary or permanent. In magnetizable materials, the magnetic domains are not naturally ordered, but it is possible to establish a constant orientation by making the material interact with a magnetic field applied from outside, generated by a magnet or an electromagnet (current-carrying cable).

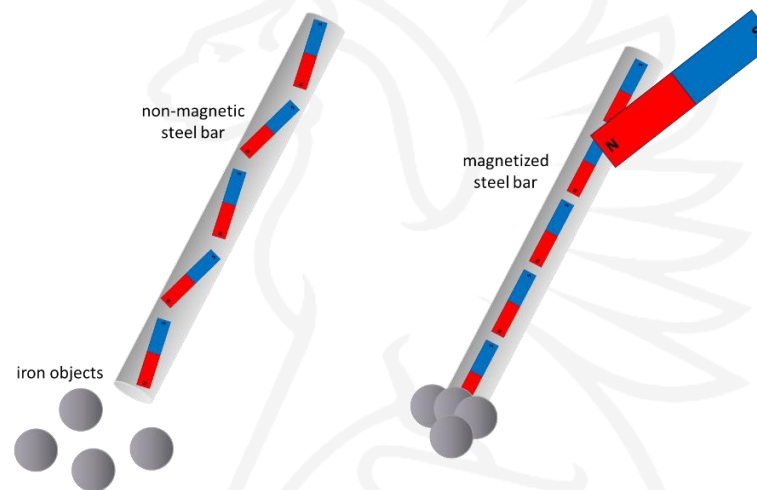


Figure 58 – Magnetisation

Some materials maintain their magnetization even when the magnetic field is no longer active (*permanent magnets*), while others lose it immediately (*temporary magnets*). In fact, among the magnetizable materials, soft iron and hard iron can be distinguished. Soft iron, once the magnetizing field is eliminated, demagnetizes. Hard iron (enriched with carbon = steel) maintains magnetization, becoming a magnet in turn. Artificial magnets generate magnetic fields that are much more intense than natural ones.

Electromagnets are temporary magnets, because once the flow of current that causes the magnetic field has been eliminated, they immediately lose their magnetization.

The magnetization of a body can be eliminated by applying a magnetic field opposite to the one that generated it. Other methods of demagnetization include the administration of heat or the application of intense force. In both cases the magnetization is canceled out because the energy supplied causes the disorder that was originally there to return within the magnetic domains.

Magnetic shielding

Magnetic shielding is used to exclude a magnetic field from a specific area by deflecting its field lines. To do this, permanent magnets can be used to prevent the passage of field lines, or magnetic fields are generated that are opposite to those to be deflected.

Shielding is used to protect equipment and/or people from external magnetic fields.

We mentioned that the passage of electric current generates magnetic fields, therefore, any current-carrying conductor can be associated not only with an electric field but also with a magnetic one. If the current is alternating, the fields generated fluctuate, that is, they expand and contract with each change in polarity of the current. Possible sources on board aircraft include power distribution lines, processors, electric actuators, or maintenance equipment.

To detect these magnetic fields, one or three-axis magnetic field meters, also called gauss-meters, can be used.

There are several techniques to prevent these magnetic fields from interfering with other equipment (such as monitors, microphones, amplifiers, etc.):

- **Shielding with magnetic materials (passive)**
the magnetic field is made to pass through these shields, deviating it from its original direction. The materials most used for this purpose are iron and nickel alloys ("MU" metal), iron and cobalt alloys, soft iron, iron and silicon alloys and low carbon steel.
- **Active shielding**
in this case the magnetic field beyond a certain space is suppressed by emitting a magnetic field opposite to it.
- **Identification, detection, control**
the areas where the magnetic fields interfere are identified. It is ensured that sensitive equipment is far enough away so as not to be affected. Furthermore, anomalous sources of interference are checked, identifying dangerous short circuits and leakage currents.



Various types of magnetic material

We have seen that there are "soft" and "hard" materials based on the ease with which they can be magnetized.

A broader and more precise subdivision can be made based on a quantity called *magnetic permeability*, represented with the Greek letter 'mi' (μ_m). This indicates, in fact, how much a material is inclined to magnetize in the presence of an external magnetic field. It is calculated as the ratio between the magnetic induction field B (T) and the magnetic field H (A/m):

$$\mu_m = \frac{B}{H}$$

To understand this formula well, it is worth making a clarification:

- **Magnetic field H** is the field generated by a magnetic body regardless of the effects it may have on the surrounding matter.
- The **magnetic induction field B** has a definition based on the effects that the magnetic body is able to have on the matter that surrounds it. We will see that B can also be called *magnetic field flux density*, which gives a clearer reference to the fact that "a certain amount" of the field interacts with matter.

Magnetic permeability is measured in Tesla-meters per Ampere (T·m/A). In SI the quantity T·m²/A is called Henry (H) → [μ_m] = H/m.

As in the case of the dielectric constant, the *magnetic permeability* can be *absolute* or *relative*, when compared to that of a vacuum, indicated with μ_0 :

$$\mu_0 = 4\pi \cdot 10^{-7} \text{ H/m} = 1.256 \cdot 10^{-6} \text{ H/m} \quad \mu_r = \frac{\mu_m}{\mu_0}$$

Each material has its own characteristic magnetic permeability, which defines the way in which the material is influenced by surrounding magnetic fields.

We define three classes of materials, based on the relative permeability value:

- **Ferromagnetic materials** – They have extremely high relative permeability. These materials become magnetized very intensely under the action of an external field. At the atomic level this means that the magnetic domains align in the same direction as the external magnetic field. Based on the duration of the magnetization we distinguish hard materials from soft (or soft) ones.
Examples: iron ($\mu_r = 5'000 \div 200'000$), steel ($\mu_r = 100 \div 4'000$), nickel ($\mu_r = 100 \div 600$), MU-metal ($\mu_r = 20'000 \div 50'000$).
- **Paramagnetic materials** – They have a relative permeability slightly greater than unity. These materials magnetize weakly. At the atomic level the magnetic domains align slightly with the external field (although at room temperature thermal agitation counteracts this effect).
Examples: air ($\mu_r = 1.00000037$), aluminum ($\mu_r = 1.000022$), platinum ($\mu_r = 1.000265$).

- **Diamagnetic materials** – They have a relative permeability less than unity.

These materials are weakly magnetized in the opposite direction to the magnetic field acting from the outside, generating a slight repulsive force. At the atomic level the magnetic domains align slightly opposite to the external field.

Examples: water ($\mu_r = 0.999992$), copper ($\mu_r = 0.999994$), silver ($\mu_r = 0.999981$).

Ferromagnetic materials lose their magnetization when they reach a certain temperature called the **Curie temperature**, that is, when the thermal agitation is so high that it eliminates the order in the magnetic domains.

Curie temperature of iron, for example, is 1043°C.



Electromagnets construction and principles of operation

The use of permanent magnets is limited to applications on low-power motors and other small instruments. Today, in most cases, electromagnets are used instead.

The operating principle of electromagnets is as follows. Let's consider a coil of conducting wire (*solenoid*), constructed in a linear or toroidal way (if closed on itself), with a length much greater than the section. If a current is made to flow along the wire that makes up the coil, a magnetic field is generated inside it, directed along the axis of the solenoid, which has an intensity depending on the current and the size of the coil. The solenoid, in effect, becomes a magnet.

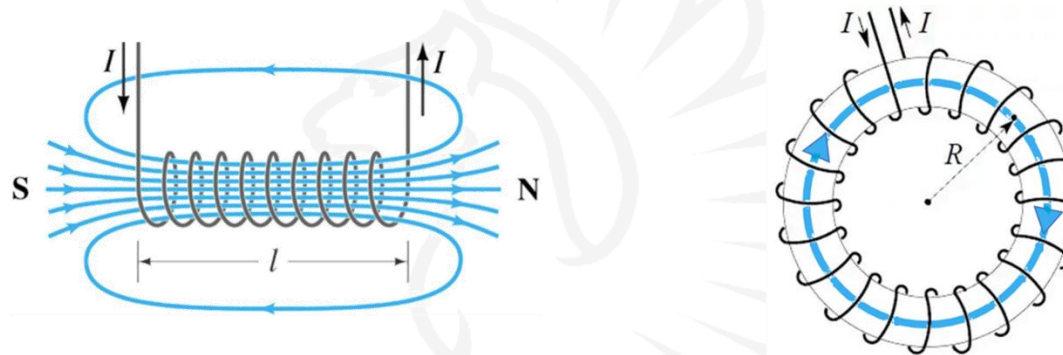


Figure 59 - Current flow through a linear and a toroidal solenoid and consequent magnetic fields (blue)

We note that in the case of the linear solenoid the field lines exit the coil, while in the case of the toroidal solenoid the field lines remain inside.

Unlike the field generated by permanent magnets, the field of a solenoid is adjustable, thanks to the possibility of varying the number of its turns and the intensity of the current that flows through it. In fact, the magnetic field H generated by the solenoid is calculated as the product between the number of turns N (non-dimensional) and the current i (Amperes), divided by the length l (metres) of the solenoid.

What interests us, however, is the way in which the solenoid interacts with the matter inside it; then, we calculate the magnetic induction field B by adding the direct dependence on the magnetic permeability:

$$B = \mu \cdot \frac{N}{l} \cdot i$$

If we are considering a solenoid placed in a vacuum, we will use the value μ_0 . If we are considering the solenoid in air (or other materials) then we will use the specific value of μ . In the case of the toroidal solenoid, the length l will be nothing more than the length of the circumference ($2\pi R$).

NOTE: the magnetic field generated by a solenoid is uniform, that is, it does not depend on the position in which it is measured inside the solenoid.

If we wind the solenoid on a core of soft ferromagnetic material, we get an *electromagnet*.

When current flows through the solenoid, the core inside becomes magnetized, becoming a temporary magnet with a North Pole and a South Pole.

The intensity of the induction field B of the solenoid can be calculated taking into account the value of the magnetic permeability μ of which the core is composed.

As soon as the current is interrupted, the core is almost completely demagnetized.



Figure 60 - Electromagnet

The **advantages** of electromagnets compared to permanent magnets are:

- The ability to activate and deactivate the magnetic field as needed;
- The possibility of regulating the intensity of the magnetic field by acting on the current;
- The possibility of generating very intense magnetic fields even by applying reduced currents (using solenoids with a very high number of windings);
- The possibility of operating at higher temperatures which, if applied to permanent magnets, would risk demagnetizing them.

Electromagnets are used in generators/motors, electronic components, sensors and actuators or in the industrial field.

An example of their use concerns the separation of magnetic materials from other types of materials. The electromagnet is activated and generates a magnetic field so intense that it can lift heavy objects such as car wrecks.



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Hand clasp rules to determine magnetic field around current-carrying conductor

Let us consider the simple case of a straight current-carrying wire. The magnetic field that is generated is cylindrical, with an axis corresponding to the wire, and the field lines are concentric and lie on planes perpendicular to the wire.

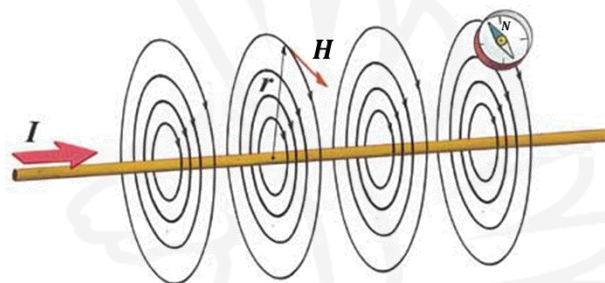


Figure 61 - Straight current-carrying wire

We can easily calculate the **intensity** of the magnetic field H using the **Biot-Savart law** (same as the formula seen before for the rectilinear solenoid):

$$H = \frac{i}{2\pi R}$$

R is the distance (radius) we are considering starting from the wire. As this distance increases, the field will be less intense. To understand in which **direction** the magnetic field moves (clockwise or counterclockwise?) generated by the conductor we can use the **right-hand rule** (or screwdriver rule).

We position the thumb of the right hand in the direction of the current and close the other fingers to form a fist: the direction in which we close the fingers determines the direction of rotation of the magnetic field.

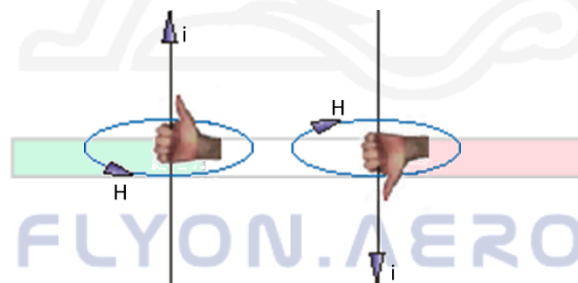


Figure 62 - Right-hand rule

SECTION B

Magnetomotive force, field strength, magnetic flux density, permeability, hysteresis loop, retentivity, coercive force, reluctance, saturation point, eddy currents

Consider a coil of length l made up of N turns, wound on a core of ferromagnetic material with magnetic permeability μ . When current starts flowing in the coil, the magnetic field is generated. The induction field B is equal to the generated field multiplied by the permeability:

$$B = \mu \cdot \frac{N \cdot i}{l}$$

We distinguished the H and B fields based on the fact that the B field depends on the interaction with the matter surrounding the body that generated the H field.

We can quantify this interaction by introducing a quantity called magnetic field flux, represented by the capital Greek letter 'phi' (Φ). Considering the induction field B (T) and a given surface S (m²) of the body with which the field interacts, the **magnetic field flux** (or magnetic flux density) through the surface is defined as:

$$\Phi = B \cdot S$$

Flux is therefore a quantity directly related to the number of lines entering and exiting through a surface. The surface considered is oriented perpendicular to the field lines that cross it. If the surface is inclined with respect to the field lines, we consider a vector decomposition, therefore the flux decreases. We can therefore distinguish two limiting cases: if a surface is parallel to the field lines, then the flux is zero, if it is perpendicular the flux is maximum.

The unit of measurement of flow is the Tesla – square meter (or Ampere-Henry), which in SI takes the name of **Weber (Wb)**.

Returning to the solenoid wound on the ferromagnetic core, we can define the surface S as the section area of the core.

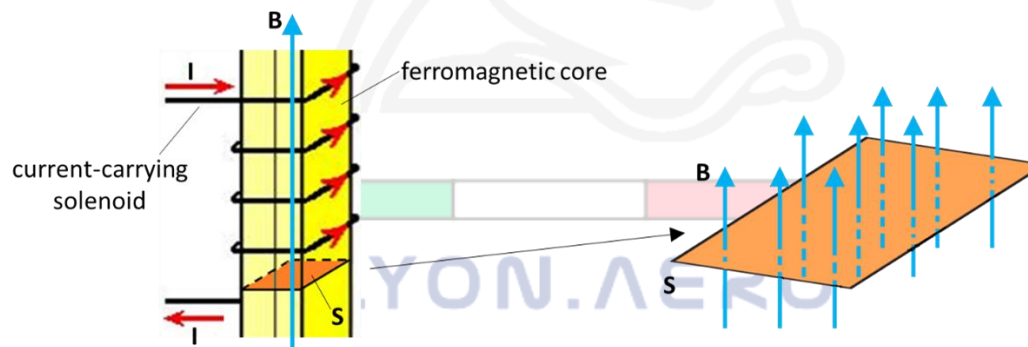


Figure 63 – Magnetic flux

Considering that the field B is the one produced by a solenoid, for this case we can write that the magnetic field flux through the surface S of the nucleus is:

$$\Phi = \mu \frac{N \cdot i}{l} \cdot S$$

Within this equation we can identify a quantity called *magnetomotive force* (F_{MM}) given by:

$$F_{MM} = N \cdot i$$

The magnetomotive force is measured in *Ampere-turns* (At) and is a quantity similar to the electromotive force. We know that the latter is the force that causes the circulation of charges in an electrical circuit. Similarly, the magnetomotive force is the force thanks to which a magnetic circuit manages to magnetize itself, thanks to the passage of the magnetizing field caused by the current.

We can continue with the analogy by defining *reluctance*, that is, a quantity that represents the ability of a material to oppose the passage of a magnetic flux (just as resistance opposes the passage of current). The reluctance is represented with the symbol R_m and is calculated as:

$$R_m = \frac{l}{\mu \cdot S}$$

As in the case of resistance, reluctance depends exclusively on the physical and geometric characteristics of the body. In fact, it is directly proportional to its length and inversely proportional to its section and magnetic permeability. The unit of measurement for reluctance is the Henry a -1 (H^{-1}).

Substituting the expressions of magnetomotive force and reluctance to the flux formula we obtain:

$$F_{MM} = R_m \cdot \Phi$$

The equation above is called *Hopkinson's law* and clearly shows us the analogy with electrical circuits; in fact, it is also called “magnetic Ohm's law”.

It defines the magnetomotive force (necessary to magnetize the circuit) in a manner directly proportional to the reluctance (the opposition of the circuit to the magnetization) and to the flux (the cause of the magnetization). The magnetomotive force could also be defined as a magnetic voltage.



Hysteresis loop, retentivity, saturation point, coercive force

The process of magnetization and demagnetization of a ferromagnetic material as a function of the external magnetic field is called a *magnetic hysteresis cycle*. Hysteresis is the property of the ferromagnetic material to react to instantaneous conditions but also to previously applied conditions. This concept allows us to explain why ferromagnetic materials can become permanent magnets.

NOTE: so far, for simplicity, we have considered linear ferromagnetic materials, i.e. materials in which the magnetic induction B varies linearly with the applied magnetic field H . The constant of proportionality is the absolute magnetic permeability of the medium: $B = \mu \cdot H$.

To define the hysteresis loop, the linear relationship is not enough because, as we will see, the graph will not be a straight line.

In fact, we consider non-linear materials, so we can define, in addition to the H and B fields, also another field called **magnetic polarization** or magnetization (to talk about magnetization we had only considered the B field). The magnetic polarization is indicated by the letter M and gives a measure of how the magnetic domains are polarized as a function of the external field H applied. In fact, the formula for magnetic induction for a nonlinear field is: $B = \mu_0 \cdot H + M$.

If we applied this formula to a linear material, we would return exactly to the formula already seen previously.

Let's consider an object made of ferromagnetic material and an external magnetic field H (we are not interested in the source) and see how the magnetic induction B varies as a function of it. We represent the closed line of the magnetic hysteresis loop on an H - B graph:



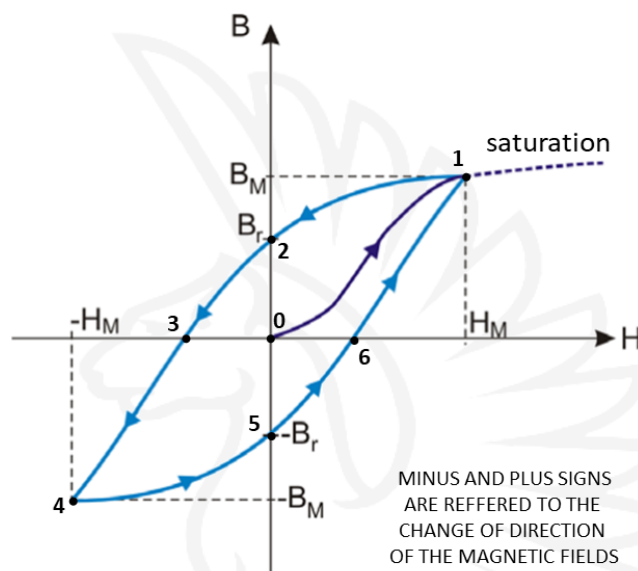


Figure 64 - Magnetic Hysteresis Loop

Point 0: the external magnetic field H is 0, therefore the magnetic induction B is also 0.

Section 0 – 1: the H field grows in intensity and with it the B field.

This section is called the **first magnetization curve** and stops at **point 1**, where **magnetic saturation** (B_{MAX}) is reached. It is the maximum value of magnetic induction that can be reached by the ferromagnetic material considered: even if the H field continues to grow, the B field has reached its limit.

Section 1 – 2: the H field is reduced until it is canceled at **point 2** (the vertical axis is reached).

We can note that field B , however, is reduced but not canceled out. It means that there is **residual magnetization** in the ferromagnetic material. We denote the corresponding magnetic induction by B_r .

Section 2 – 3: to completely cancel the remaining magnetic induction, it is necessary to apply an H field inverse to the one applied initially, for this we go to the negative part of the graph. This H field is called the **coercive field**. We note that **point 3** is on the horizontal axis, demonstrating that the application of a field opposite to the initial one brings B to zero.

Section 3 – 4: we can further increase the magnetic field H (always considering the direction of the previous section) until we reach the saturation of the ferromagnetic material again, at **point 4**, this time indicated with $-B_{MAX}$ due to the direction considered.

Note that the coordinates of the two saturations are symmetrical with respect to the origin.

Section 4 – 5: we cancel the H field again, finding a residual magnetization $-B_R$ at **point 5**. Also in this case, there is symmetry with respect to the origin.

Section 5 – 6: to eliminate the magnetization we apply the H field by inverting it again. We get to **point 6**, where field B returns to zero.

Section 6 – 1: at this point we can re-start the cycle. If we increase the H field further, we reach saturation of the ferromagnetic material and so on. We note that the first magnetization curve can no longer be followed, just as it is impossible to return to point 0.

At points 2 and 4, i.e. where we see the residual magnetic induction, the external magnetic field is zero. This is the condition that allows us to create artificial magnets.

A measure of “how much magnetism” remains inside a material when the external field is removed is given by *magnetic retentivity*, which depends on the characteristics of each material. Iron certainly has a higher retentivity than carbon-rich steel.

As for the graph, soft magnetic materials have a tight hysteresis loop, because it is easy to magnetize and demagnetize them. In contrast, hard materials have a large hysteresis loop.

We have seen that at a microscopic level magnetization occurs thanks to the alignment of magnetic domains. Continuous cycles of magnetization and demagnetization cause continuous movement of these domains, causing the material to heat up due to internal friction.

Eddy Currents

Parasitic induced currents, also called Foucault currents or eddy currents ("vortex currents"), are electric currents that are generated in bodies of conducting materials when they are hit by a magnetic flux that varies over time.

As will be studied in greater detail in the next chapter, the variation in the magnetic field causes the flow of current. The moving electrons, in turn, generate a magnetic field, and so on.

The currents circulate along the sections of the conductors, and the larger the section the greater the intensity of these currents. For this reason, to limit its effect, the conducting body can be divided into many thin plates, parallel to the flux lines of the magnetic field. In this way the section is reduced (see Figure 65) and consequently the resistance is increased. Another way to increase resistance is to increase the resistivity, therefore changing the type of material used.

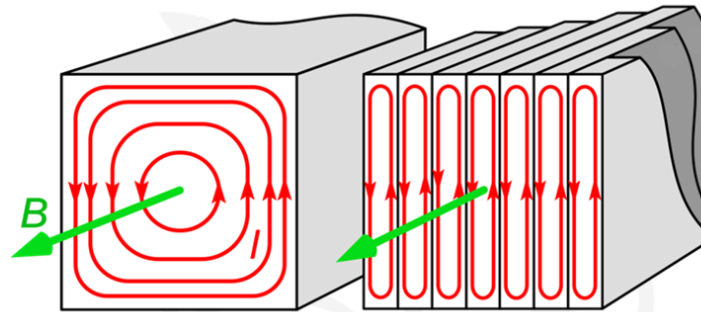


Figure 65 - Eddy Currents: solid body vs laminated body

Eddy currents can be exploited in non-destructive testing (*Eddy Current Testing – EDT*).

Let's consider a magnetic component to inspect and an EDT instrumentation. The instrumentation consists of a probe (a coil of conducting wire) and a screen.

The coil is crossed by alternating current (i.e. with a direction that varies over time) and, consequently, generates an oscillating induced magnetic field which influences the nearby metallic component, on which eddy currents begin to flow (the intensity of which also depends on the material). These currents, in turn, generate a variable “secondary” magnetic field. This last magnetic field interacts with the probe winding. Based on the values detected, it is possible to understand whether discontinuities are present in the metal component.

This method is used for surface and slightly sub-surface inspection of components.

Precautions for care and storage of magnets

To prevent permanent magnets from becoming damaged or interfering with other equipment, they must be transported and stored properly.

It is essential that magnets are stored away from metal shelving and ferromagnetic objects that are free to move. Furthermore, they must be distant from electrical equipment, which could cause problems such as interference, saturation of the transformer core, cancellation of magnetic memories, etc.

We have seen that magnets can undergo demagnetization if subjected to excessive heat or mechanical stress. To prevent this from happening, or the magnets from weakening each other if held close together or, in the case of horseshoe magnets, by their own opposing poles, the magnets are stored in special armatures called *magnet keepers* or *magnet shunts*.

A magnet keeper is a bar of soft iron or steel, which placed between the poles of a permanent magnet completes the magnetic circuit and helps maintain the strength of the field.

Linear magnets are stored in pairs, positioned opposite each other, sometimes separated from each other by an insulating support such as wood. At their ends there are two keeper bars. In the case of horseshoe magnets, a single bar placed between the two poles is sufficient.

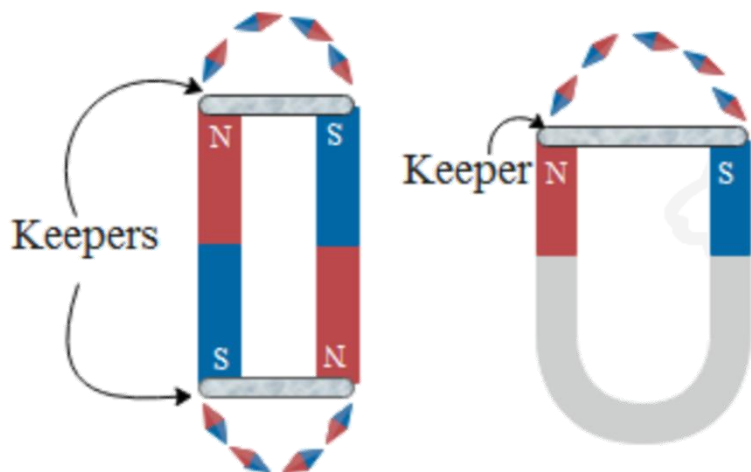


Figure 66 - Schematic of magnet keepers



Figure 67 - Magnet keepers



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3.11 Inductance/Inductor

Faraday's Law

Electromagnetic induction is a phenomenon that occurs when a variable magnetic field interacts with a conductor, causing the production of an electromotive force (voltage), and therefore a current, within it.

The Faraday-Neumann-Lenz law describes this phenomenon and states that:

The electromotive force induced in the conductor is equal to the opposite of the variation of the magnetic field flux over time:

$$EMF = -\frac{\Delta\Phi}{\Delta t}$$

By combining the above formula with Ohm's law, we can calculate the value of the induced current:

$$I = -\frac{1}{R} \cdot \frac{\Delta\Phi}{\Delta t}$$

Action of inducing a voltage in a conductor moving in a magnetic field

Consider a magnetic field and a conductor moving within it. The lines of force of the magnetic field that intersect with the conductor ("concatenated flux") vary over time, because the conductor is in motion. Therefore, with respect to the conductor, the magnetic field can be considered variable and consequently a current is generated inside the conductor.

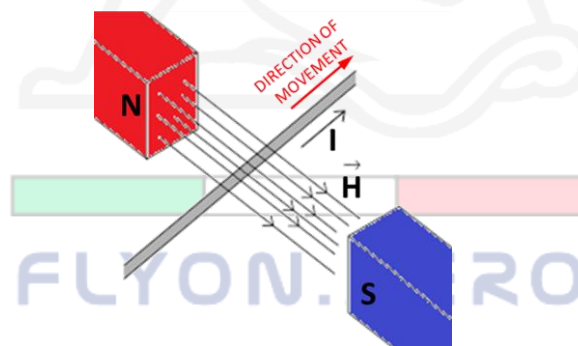


Figure 68 – Induction effect on a conductor moving in a magnetic field

The minus sign within the Faraday's Law formula indicates that the current inside the conductor opposes the cause that generated it. In practice, the forces of the magnetic field disturb the initial equilibrium of the conductor, which is recovered thanks to the onset of the electromotive force.

The magnetic field flux is greater the faster the conductor moves within the field. Consequently, the current will also be more intense under the same conditions. If we consider that the same conductor carrying a current generates a magnetic field, it is easy to understand that the direction of the current "is always opposite to the cause that generated it" because, if this were not the case, the magnetic field generated by the conductor would strengthen the one in which it is immersed, moving further and further away from its condition of equilibrium.

Induction principles

To obtain the effects of induction it is not always necessary for the conductor to move within the magnetic field: it is enough to have a variation in flux. In general, electromagnetic induction is achieved:

- Moving a conductor inside a magnetic field
- Moving a magnet inside a conductor
- By varying the intensity or direction of magnetic field over time
- By varying the conductor surface over time, for example using a deformable coil or one that rotates with respect to the magnetic field lines.

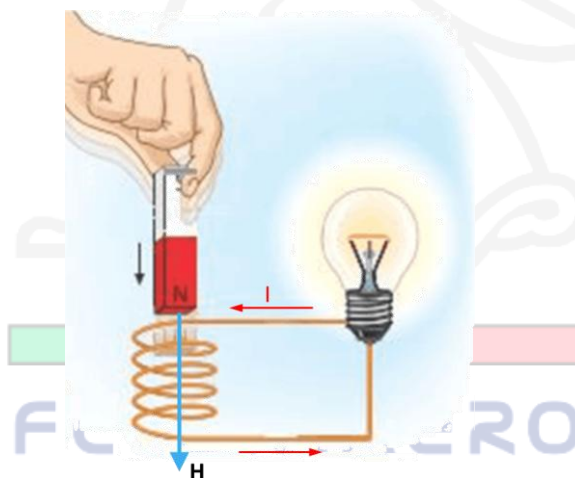


Figure 69 - Electromagnetic induction

Effects of the following on the magnitude of an induced voltage: magnetic field strength, rate of change of flux, number of conductor turns

From the magnetic field flux formula, we can see that it depends on the dimensions of the conductor considered (number of turns, section area) and on the intensity of the magnetic field. The value of the induced voltage, therefore, depends on:

- The intensity of the magnetic field: as the magnetic field increases, the value of the flux increases, and consequently the induced voltage.
- The rate of change of the flux: the faster the magnetic field varies over time, the greater the electromotive force produced to oppose the magnetic forces.
- The number of windings of the conductor (therefore its dimensions): if the surface crossed by the field lines increases, it means that the flux increases and consequently the induced voltage.

The correct formula of the electromotive force induced in the case of a coil also considers the number (N) of its windings:

$$EMF = -N \cdot \frac{\Delta\Phi}{\Delta t}$$

Mutual induction

Let's now consider two windings of conductors, placed one close to the other. We call one conductor "**primary**" and the other "**secondary**".

If a current flows in the primary winding, we know that a magnetic field is generated. If this current is variable over time, then the magnetic field will also be variable. The magnetic field lines intersect with the secondary winding, inducing an electromotive force in it and therefore variable currents (because the chained field flux varies over time).

These induced currents, in turn, will produce another variable magnetic field, which will intersect with the primary winding and cause the onset of other induced currents, and so on. So:

Mutual induction (or inductance) is that phenomenon that occurs between two electrically separate circuits, when the magnetic field generated by one generates an electromotive force on the other, and vice versa.



The effect the rate of change of primary current and mutual inductance has on induced voltage

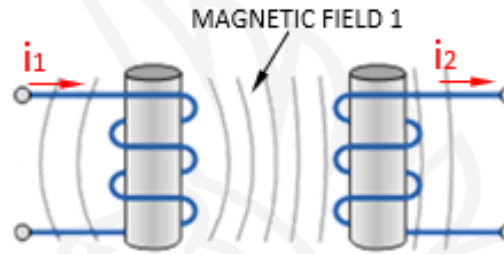


Figure 70 – double winding circuit

Taking the figure as a reference, we hypothesize that the first winding is initially crossed by a variable current I_0 , producing a magnetic field whose flow is linked with the turns of the second winding, generating the current I_2 in it. At this point:

- current I_2 flowing in the second winding is variable and produces a magnetic field
- the flux of the magnetic field produced by the secondary winding, which is concatenated with the primary winding, can be indicated with Φ_{21}
- the variable current that is consequently generated in the primary is I_1
- the magnetic field flux (produced by I_1) which concatenates with the secondary winding is Φ_{12} .

We can define a **mutual induction coefficient**, indicated with the capital letter M , which is:

$$M = \frac{\Phi_{12}}{I_1} = \frac{\Phi_{21}}{I_2}$$

So, M can be calculated as the ratio of the concatenated flux and the current that generated that magnetic field, regardless of the winding we consider. The mutual induction coefficient has Henry (H) as its unit of measurement and depends on the material of the windings and their dimensions (section and number of turns).

We distinguish the current I_0 from the others because this can be defined as **primary current**, i.e. the one which, circulating in the first winding (thanks to a generator) starts the entire mutual induction cycle, from which the currents I_1 and I_2 derive instead. The two induced currents, and therefore the relative electromotive forces, are directly proportional to the primary current because, as the latter increases, the intensity of the magnetic field increases and consequently the concatenated flux.

Furthermore, the flux is directly proportional to the mutual induction coefficient therefore the induced electromotive force also increases as the value of the coefficient M increases.

Factors affecting mutual inductance: number of turns in coil, physical size of coil, permeability of coil, position of coils with respect to each other

The effects of mutual induction, as well as those of generic induction, vary depending on various factors.

- As the number of turns in the windings and their size increases, the magnetic field flux increases (and consequently the induction).
- If the magnetic permeability of the material of which the winding is made is greater, the intensity of the magnetic field and induction increase.
- If the windings are closer together, the effects of mutual induction are more intense because there are fewer losses.

Another way to intensify the effects of mutual induction is to place the windings around soft iron cores, so as to form a magnetic circuit that allows for lower flux losses, as in the figure:

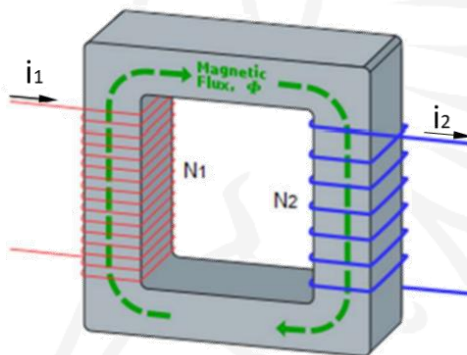


Figure 71 - Mutual induction between two coils wrapped on an iron core



Lenz's Law and polarity determining rules

Lenz's law establishes that the electromotive forces induced inside a conductor are such as to oppose the cause that generated them. From this law derives the minus sign of the Faraday-Neumann-Lenz law seen at the beginning of the chapter.

To determine the direction of the induced electromotive force (and therefore of the current), you can use the right-hand or right-handed screw rule.

First, it is necessary to establish the direction of the induced magnetic field vector, which is always opposite to the magnetic flux variation vector. Let's consider the example of the loop in figure 72, using a positive upward reference system.

B1 is an initial value assumed by the variable magnetic field and B2 is another of its values, assumed later in time. The vector ΔB is the variation of the magnetic field, obtained from the difference (remembering the rules of operations between vectors) between B2 and B1.

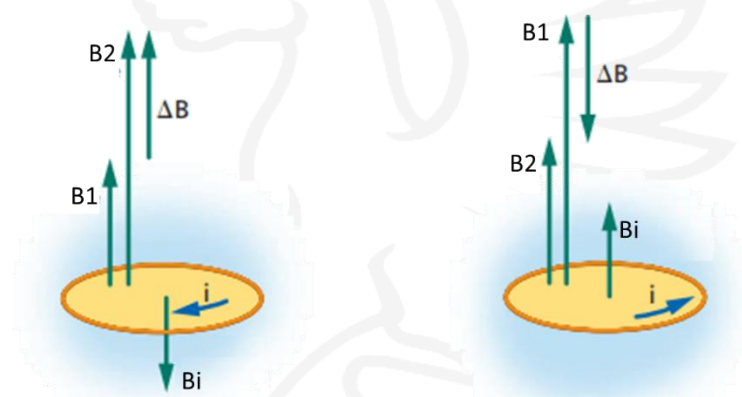


Figure 72 – Rules for determining the direction of the induced current

In the first case (on the left), the vector B2 is larger than the vector B1: it means that the magnetic field has increased, therefore, the vector ΔB is positive and pointing upwards. The induced magnetic field B_i (i.e. the one generated by the induced current) in the loop must oppose the cause that caused the current to arise and will consequently be directed downwards. At this point, to determine the direction that the induced current has assumed, simply position the thumb of the right hand in the direction of the induced magnetic field B_i and the direction in which the other fingers rotate will indicate the direction of the current, which in this case is clockwise.

In the second case (right), the magnetic field is decreasing (ΔB downwards) because B1 was larger than the successive value B2. The magnetic field B_i will therefore be directed upwards, and the induced current will be counterclockwise.

In these examples the magnetic field is positive, but the rule also applies in the case of vectors pointing downwards: it's sufficient to consider a vector B_i always opposite to the variation vector ΔB .

Back EMF, self-induction

When we talked about mutual induction, we neglected the inductive effects that a current-carrying circuit has on itself. In fact, the principles of induction are also applicable to a single circuit, and in this case, we are talking about *self-induction*.

Let's consider a winding of conducting wire carrying a variable current. This current generates a magnetic field which, intersecting with the coils of the same circuit, generates another current in it, the self-induced one. The electromotive force linked to this current is called *counter-electromotive force* (abbreviated CEMF), or back-EMF, because, as in the other cases, it opposes the cause that generated it.

We define a self-induction coefficient (also called *inductance*) indicated with the capital letter L and measured in Henry (H), as the ratio between the self-flow and the primary current that generated it:

$$L = \frac{\Phi}{I}$$

The coefficient L depends on the material and size of the conductor.

The back electromotive force can be calculated as:

$$CEMF = -L \cdot \frac{\Delta I}{\Delta t}$$

NOTE: the last formula can also be used to calculate the induced electromotive force in the case of mutual induction (M will be used instead of L).

Self-induction is the basis of the operation of *inductors* used in circuits. These components are formed by a winding of conductor (often around a ferromagnetic core) and generate a magnetic field when the electric current passes.

Basically, inductors can store electrical energy in the form of a magnetic field and slowly release it into the circuit when needed. They are effective in absorbing current peaks.

Saturation point

The *saturation point of an inductor* is the limit beyond which the inductor can no longer store energy. When the current passing through it exceeds a certain value, the magnetic flux saturates the core, therefore, even if the current increases there is no further increase in the magnetic flux.

This phenomenon should be avoided because it could cause an unexpected increase in current that the inductor would not be able to absorb. This can lead to damage to the circuit.

Principle uses of inductors

Inductors are windings of conducting wire (usually copper) of linear or toroidal or other shapes. There are different types of inductors; a generic distinction can be made between those with an air core, in which the winding is empty, and those with an iron core, in which the conductor is instead wound on a block of ferromagnetic material. The inductance values (L) of inductors on the market are of the order of microhenries (μH).



Figure 73 – Inductor

Inductors can be used as:

- **Filters.** They are used in combination with capacitors and resistors, to filter certain signal frequencies and in general in radio frequency communications.
- **Energy stores.** They can be used to absorb energy in the form of a magnetic field and release it when needed, but also to smooth out voltage variations (absorbing current peaks). The energy stored in an inductor can be calculated as:

$$E = \frac{1}{2} \cdot L \cdot I^2$$

- **Trasformers.** Inductors are the basis of transformers, used to transfer electrical energy.
- **Wireless chargers.** Thanks to contactless energy transmission, inductors are used for wireless charging of electronic devices.

Like other elements seen previously, inductors can be fixed or variable. In the latter case, within certain limits, it is possible to vary the inductance value through mechanical control.

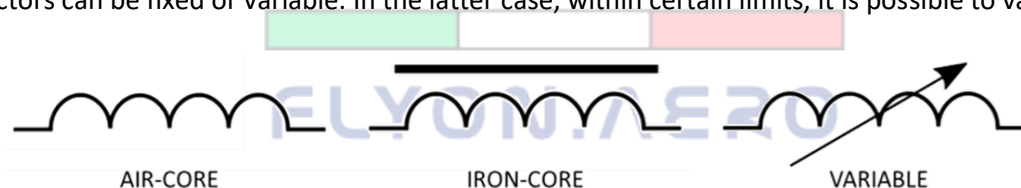


Figure 74 - Symbols for inductors in circuits

Series and parallel connections

As in the case of resistors and capacitors, inductors can also be connected in series and parallel.

By connecting multiple **inductors in series** (same current), a higher inductance value is obtained. This allows to improve protection against over-voltages. The equivalent inductance of multiple inductors in series is calculated as the equivalent resistance of the resistors in series:

$$L_{EQ} = L_1 + L_2 + \dots$$

A **parallel** connection (same voltage drop) of multiple inductors allows to increase the maximum current that the circuit can handle, without the risk of saturating a single inductor. The equivalent inductance is calculated as:

$$\frac{1}{L_{EQ}} = \frac{1}{L_1} + \frac{1}{L_2} + \dots$$

When we only have two inductors in parallel, we can apply the formula already used for resistors:

$$L_{EQ} = \frac{L_1 \cdot L_2}{L_1 + L_2}$$



3.12 DC Motor/Generator Theory

Basic motor and generator theory

Direct current machines (DC) can be used reversibly as motors or as generators (dynamos), based on the relationship between the direction of rotation and the direction of the induced magnetic fields/inductors. These machines base their operation on the physical principles of induction. The motor converts electrical energy into mechanical energy, while the generator does the opposite, allowing electrical energy to be obtained from a movement.

The simplest form of motor/generator consists of a single coil of conducting wire rotating within a magnetic field.

DC Motor

Let's consider a rectangular loop traversed by a direct current I in an anti-clockwise direction and immersed in a uniform magnetic field B , with the field lines coplanar to the loop, as in the figure:

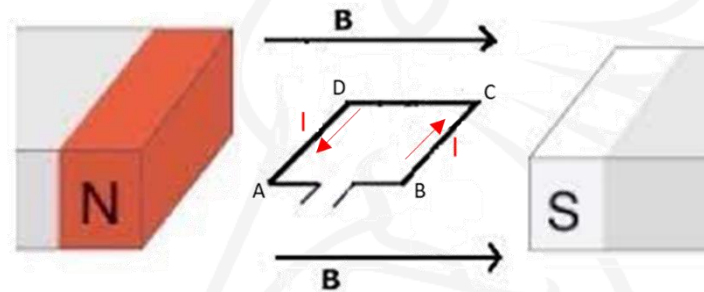


Figure 75 - Current-carrying rectangular turn immersed in magnetic field – 1/4

A magnetic force (**Lorentz force**) will act on each side of the loop:

$$F = I \cdot L \cdot B \cdot \sin \vartheta$$

Where L is the length of the side of the loop considered and ϑ is the angle formed by that side and the magnetic field lines.

If we consider the sides AB and CD, parallel to the magnetic field, the force is zero, because $\vartheta=0^\circ$ and $\sin 0^\circ=0$; while if we consider the sides BC and DA, perpendicular to the field, the force takes on the maximum value, because $\vartheta=90^\circ$ and $\sin 90^\circ=1$.

The direction of the magnetic force on BC and DA can be found using the right-hand rule, placing the thumb in the direction of the current I and the index finger in the direction of the magnetic field B : the direction of the middle finger will be that of the force F .

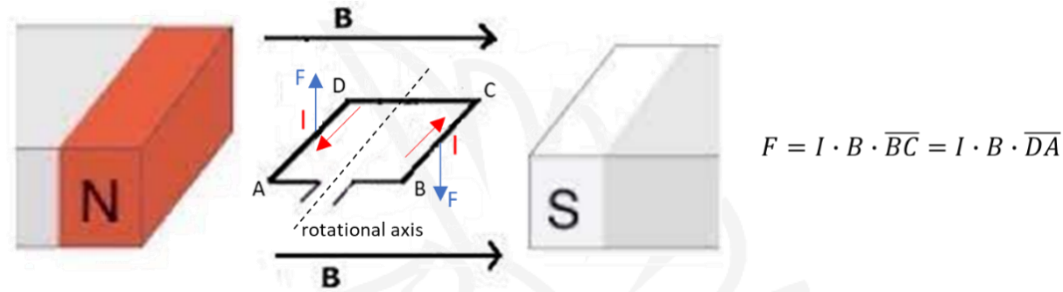


Figure 76 - Current-carrying rectangular turn immersed in magnetic field – 2/4

The sides BC and DA have the same length and are crossed by the same current, therefore the forces acting on them are equal and opposite to each other, constituting a couple. This means that the forces due to the interaction between the current and the magnetic field are able to rotate the coil around its axis.

The torque due to a couple of forces can be calculated as the sum of the products of the forces for the respective arms ($AB/2$ is equal to $CD/2$).

We can replace F with its equation $F = I \cdot L \cdot B$ (we still consider $\vartheta = 90^\circ$, therefore $\sin 90^\circ = 1$), where L is indifferently the length of BC or DA:

$$M = F \cdot \frac{\overline{AB}}{2} + F \cdot \frac{\overline{AB}}{2} = F \cdot \overline{AB} \quad \rightarrow \quad M = I \cdot \overline{BC} \cdot B \cdot \overline{AB}$$

The loop is rectangular, so the product of its sides BC and AB is nothing other than its surface S .

In general, we can state that on any current-carrying loop immersed in a magnetic field, with any relative angle, a mechanical moment (torque) will act. It is equal to the product of the current, the magnetic field, the surface of the loop and the sine of the angle ϑ between the current and the magnetic field lines:

$$M = I \cdot B \cdot S \cdot \sin \vartheta$$

In this formula, the $I \cdot S$ contribution constitutes a quantity called **magnetic moment (m)**, measured in Amperes for square meters and oriented according to the right-hand rule (with the thumb positioned in the direction of the current, the direction of the index finger indicates that of m , exiting or entering the loop).

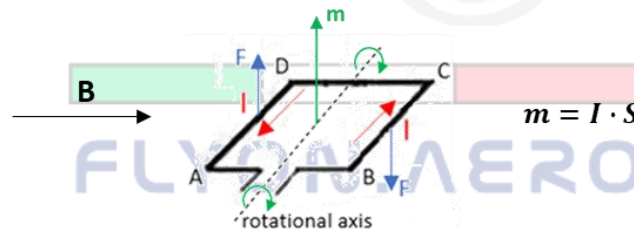


Figure 77 - Current-carrying rectangular turn immersed in magnetic field – 3/4

NOTE: the magnetic moment can be associated with the magnetic field produced by the current-carrying loop; therefore, the loop rotates under the effect of a counter-electromotive force.

The coil, under the action of the torque, rotates clockwise until it is positioned perpendicular to the magnetic field lines.

In this position, the two forces F no longer constitute a couple capable of rotating the coil, because they lie on the same line. In fact, they cancel each other out, placing the coil in equilibrium. The loop then stops. The magnetic moment m , at equilibrium, is parallel to the magnetic field B .

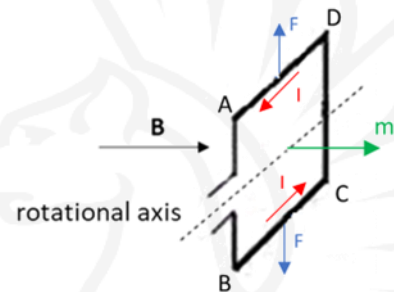


Figure 78 - Current-carrying rectangular turn immersed in magnetic field – 4/4

In practice, equilibrium is reached when the magnetic moment of the coil manages to align itself with the magnetic field acting from the outside. Obviously this is not enough to constitute an electric motor, because the coil stops after a maximum of 180° of rotation, while a motor needs continuous rotation.

The construction solution of the motor is to use multiple coils oriented differently, so that the inertia of the rotation of the other coils always keeps in rotation the one which reaches equilibrium at a given moment. In addition to this provision, it is also necessary to apply a reversal of the polarity of the current to the coils. When a given loop slightly exceeds its equilibrium position, it is ensured that it is unable to return to it by reversing the direction of the current using commutators.

If before the current was counterclockwise, now it is made to circulate clockwise. In this way, once the vertical position has been overcome, the forces do not tend to bring the coil back to equilibrium but generate a twisting moment again in a clockwise direction, until it reaches a new equilibrium position 180° later, where the polarity will be inverted once again.

In this way, the rotation will always occur in the same direction and the motor can function as such.



DC Generator

Also in the case of the simple generator, we can consider a coil immersed in a magnetic field.

The coil can be made to rotate through various sources of mechanical energy. When the coil begins to rotate within the magnetic field, the magnetic field flux through it varies and, based on the laws of electromagnetic induction, an induced electromotive force and therefore a current is generated in it.

This current, however, is produced in an alternating manner, because as in the case of the motor, the direction of the current is reversed at every half turn. The solution consists in using collectors/commutators that reverse the polarity of the coil at every half turn, ensuring that the current is always produced in the same direction.

Furthermore, the current is produced with lesser or greater intensity depending on the angle that the coil forms with the magnetic field. This problem is solved by collecting the current only when it reaches its maximum peak, i.e. when the loop is perpendicular to the field lines.

By applying these techniques, the output current assumes an almost constant value.



Construction and purpose of components in a DC generator

DC generators and motors are made up of three elements:

- An inductor (stator) – is the source of magnetic field.
- An inducted or armature (rotor) – is the system of conductors in which electromotive forces manifest themselves.
- A commutator or collector – allows the polarity to be reversed.

The **inductor** can be made up of permanent magnets (especially in small machines) or windings used as electromagnets. In any case it produces a magnetic field with constant direction. The field intensity is fixed in the case of permanent magnets, while it can be varied in the case of the electromagnet.

We have seen that it is indifferent which part moves: it can be the magnetic field or the conductor, because to obtain the effects of induction it is sufficient to have a relative motion between them. Usually, however, in direct current machines the inductor is always the fixed part (**stator**).

The rotating part (**rotor**) is the **armature**, which is made up of windings of conductive material around a lamellar ferromagnetic core (to avoid excessive eddy currents).

The **commutator** or **collector** is an element made up of copper or bronze blades (semi-collectors) which form the two halves of a cylinder integral with the rotor shaft. At each 180° rotation of the coil, the two halves of the cylinder change polarity, because they alternately touch the two ends of an electrical circuit. The collector has the dual function of collecting the generated current and converting it into direct current. The current is collected via graphite contacts, called **brushes**, which are integral with the electrical circuit and slide over the two semi-collectors.

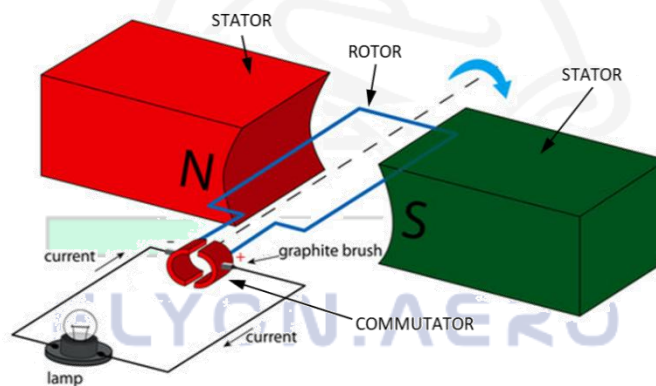


Figure 79 – Simple schematic of a DC generator

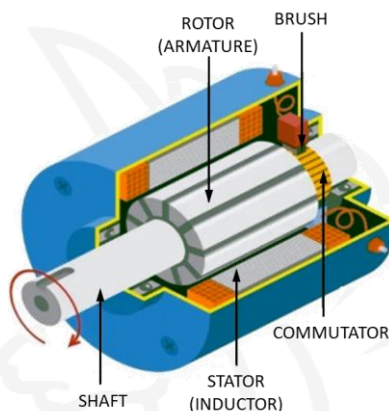


Figure 80 – DC motor/generator

The stator can be composed of two or more alternating North and South magnetic poles, arranged around the rotor and connected to each other by a **crown**. If they are electromagnets then each pole is made up of a **polar core** (solid or laminated ferromagnetic material), around which the coil on which direct current flows is wound. Each pole has a **polar expansion**, built in laminar ferromagnetic material, which has the task of distributing the magnetic field over a larger portion of the surface. The polar flow which will generate the current is established through two adjacent magnetic poles and the portion of the winding close to them.

The rotor is made up of a laminated cylinder of ferromagnetic material which has grooves (**slots**) into which the conductors are inserted longitudinally. The sheets that make up the cylinder may have holes that aid heat dissipation. The conductors connect two by two at the front and rear of the rotor, forming the turns.

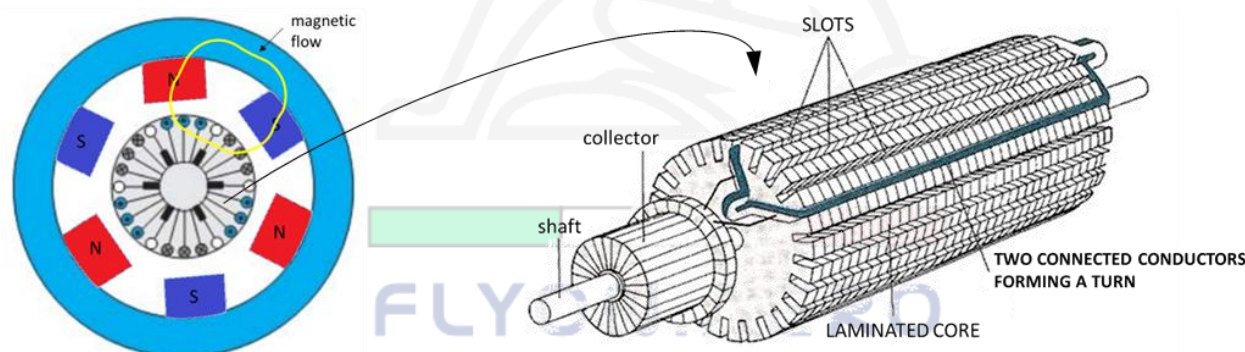


Figure 81 – Details of stator and rotor

Operation of, and factors affecting the output and direction of current flow in DC generators

The intensity of the electromotive force induced in the armature can be calculated using the formula:

$$E = k \cdot \Phi \cdot N$$

Essentially, therefore, the intensity of the induced electromotive force and current depends in a directly proportional way on three factors:

- The construction parameters of the machine, expressed by the constant k. The constant includes aspects relating to the number of stator poles and the number of rotor turns. As the latter increases, the induced electromotive force increases.
- The magnetic flux Φ of each pair of poles.
- The rotation speed N (in rpm) of the rotor.

The direction of the output current can be determined using the left-hand rule, placing the thumb in the direction of rotation of the conductor and the index finger in the direction of the magnetic field. The middle finger will indicate the direction of the current.

The efficiency of a DC generator can be reduced by several types of **losses**:

- Losses due to the Joule effect ('Copper loss').
The current flowing through the conductors causes them to heat up, dissipating a certain amount of energy in the form of heat.
- Losses due to eddy currents.
Due to induction in the iron core of the rotor, currents are established. To reduce them, the core is built using many plates rather than using a single block.
- Losses due to magnetic hysteresis.
The magnetic domains of the armature constantly change orientation due to rotation. This causes the material to heat up. To reduce the effects of this problem, armatures are constructed from materials that have very low hysteresis losses.
- Mechanical losses.
They can be caused by the friction of parts in contact such as bearings and brushes but also by the friction of the air with the rotor during rotation. It can be reduced with good maintenance (lubrication, cleaning, etc.) and by ensuring the correct positions of the elements in relative motion to each other.

Operation of and factors affecting the output power, torque, speed and direction of rotation of DC motors

Among the advantages of DC motors, we find the possibility of easily regulating the speed (and therefore starting and stopping them) by varying the input voltage.

It should be noted that there are two forces in the motor: one is the Lorentz force, generated by the interaction between the current circulating in the rotor and the external magnetic field; the other is the counter electromotive force that is generated in the rotor, due to the variation in magnetic flux through the coils once the armature has started to rotate.

The force that makes the motor work is the Lorentz force, which therefore must be intense enough to oppose the counter-electromotive force.

We therefore understand that the rotation speed of the motor decreases as the externally induced magnetic field increases, because the counter-electromotive force increases. The back EMF can also be reduced by using armature coils with a higher resistance.

Practical information is that at the moment of the starting, the rotor is stationary and consequently does not produce any counter-EMF force. The armature behaves like a simple resistance and initially absorbs the starting current, which is of very high intensity. To avoid damage, you can start the motor using a reduced voltage until the motor produces a high enough Lorentz force by increasing revolutions per minute.

Once the motor has started to rotate, by varying the direction of the voltage on the power supply circuit, the direction of rotation can also be varied. Furthermore, by regulating the intensity of the voltage and with the help of a sensor, you can easily adjust the speed. The rotation speed also depends on the current absorbed by the rotor (based on its resistance) and, obviously, on the applied load.

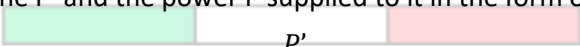
The torque delivered by the motor is directly proportional to the magnetic field and armature current.

The power delivered by the motor can be calculated as the product of the torque T and the rotation speed ω of the motor:

$$P = T \cdot \omega$$

The work developed is practically the same as that of two forces (couple or torque) applied to the ends of a diameter of the rotating part. We have seen that torque can be calculated as the sum of the products of each force for the respective arm. Remember also that power P (W) is nothing other than work W (J) divided by time t (s): $P=W/t$. Therefore, for the same amount of work delivered, the machine that takes less time to deliver it has greater power.

By calculating the ratio between the power supplied by the engine P' and the power P supplied to it in the form of electrical energy we obtain the efficiency h of the engine:


$$h = \frac{P'}{P}$$

As always, efficiency has a value less than 1, due to friction, hysteresis, eddy currents, etc.

Series wound, shunt wound and compound motors

Direct current motors (therefore also the generators) can be subdivided as follows:

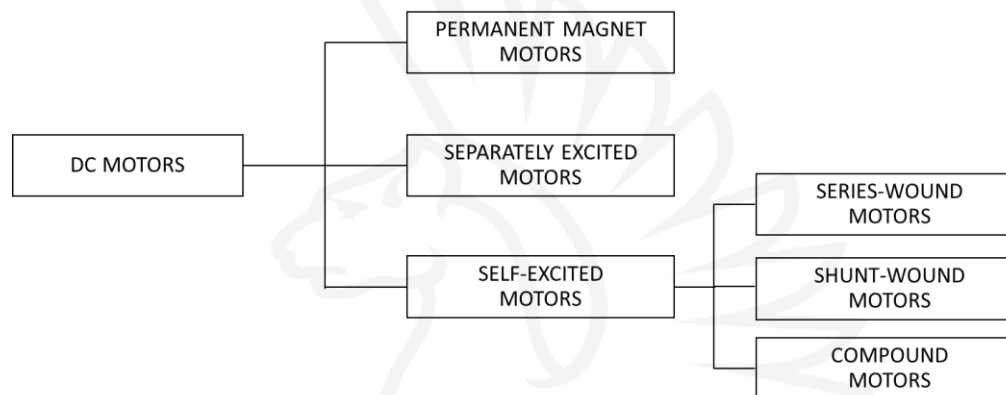


Figure 82 - Classification of DC machines

Permanent magnet machines differ from the other two types because they are the only ones in which the inductor is not made up of electromagnets. The power supply is therefore connected only to the armature.

Separately excited machines have a stator winding powered separately from that of the rotor. This way the supply voltage can be adjusted more freely. Another advantage is that the armature current does not flow through the inductor windings.

In self-excited machines, instead, the stator winding can be connected in series ("series-wound") or in parallel ("shunt-wound") to that of the rotor, or there can be a compound solution. Shunt motors are the most used.



Excitation in series: series-wound machines

In the **series-wound motor**, the field stator winding (inductor) and the armature (rotor) are connected in series. The current flowing on the armature also flows equally in the field winding, therefore, the intensity of the magnetic field depends on the intensity of the current also circulating on the rotor.

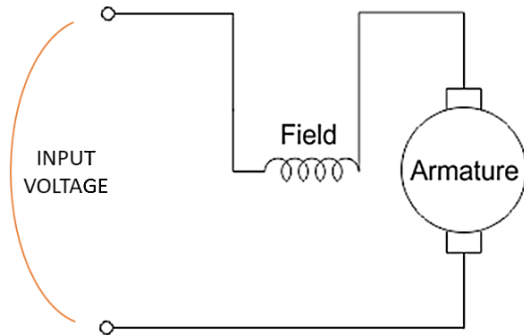


Figure 83 - Equivalent circuit of a series-wound machine

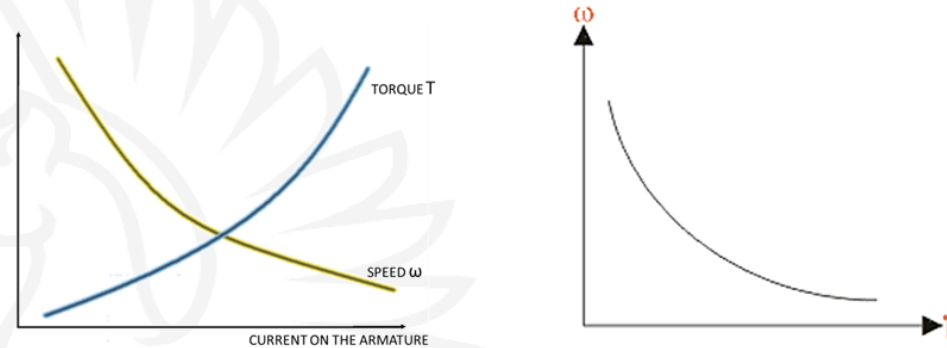


Figure 84 - Variation of speed and torque in a series-wound machine

With this type of connection, the speed is significantly influenced by the load applied to the motor.

If the load is heavy, it slows down the rotor. Therefore, there is an increase in current due to the decrease in the counter-EMF, which causes an increase in the Lorentz force and therefore in the torque produced. However, the current is the same as that passing through the inductor (armature and field winding are connected in series), so an increase in it causes also an increase in the intensity of the magnetic field. This leads to a further decrease in speed (we have seen that the rotation speed is inversely proportional to the magnetic field).

On the contrary, when the applied load is light, the rotor accelerates. There is therefore an increase in counter-EMF and a consequent decrease in current in the armature. This means that the torque generated is less, but also that the magnetic field of the inductor has a lower intensity. Following the reasoning made in the previous case, the rotation speed increases further.

The winding resistance of these types of motors is rather low, therefore, upon starting (when the rotor is still stationary), the rotor absorbs a very high current, immediately generating a high torque.

In fact, series-wound motors are used in starters. Furthermore, they can be found in lifting mechanisms and DC actuators. Some series-wound motors have two separate windings, so that the direction of rotation can be reversed more easily.

Series-wound DC motors are the most widely used on board aircraft.

Excitation in parallel: shunt-wound machines

In the *shunt-wound motor*, the field stator winding (inductor) and the armature (rotor) are connected in parallel. The field windings (stator) are exposed to the full voltage supplied to the rotor. The field current remains constant if the supply voltage is constant.

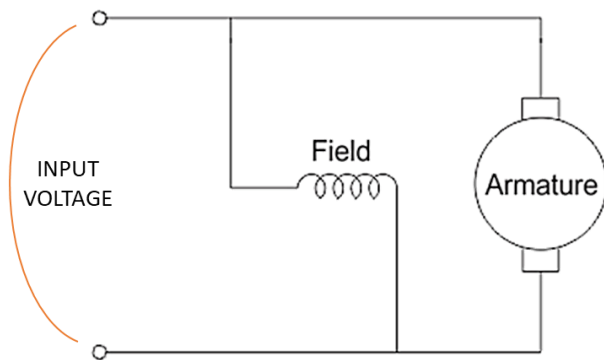


Figure 85 - Equivalent circuit of a shunt-wound machine

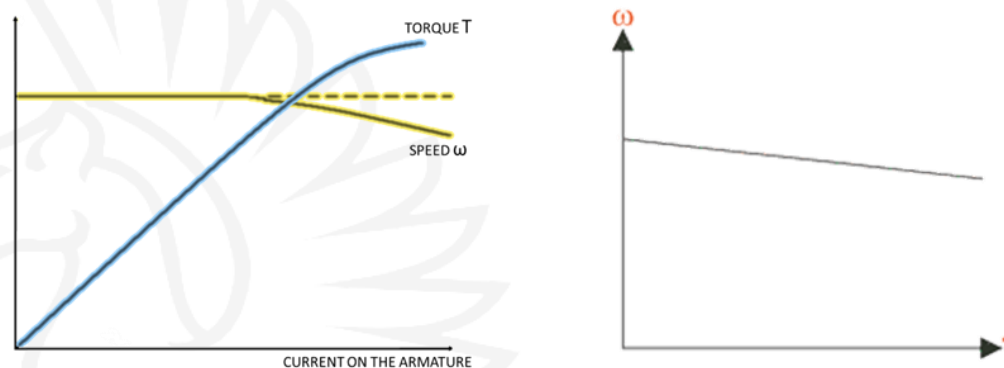


Figure 86 - Variation of speed and torque in a shunt-wound machine

In this type of motor, the speed can be controlled more effectively than in series motors and is influenced more lightly by the load. In this configuration, the current on the field windings is constant, because the connection with the power supply is in parallel and is not influenced by the resistance of the armature.

When the load increases, as in the previous case, the speed decreases, but by a very small amount. Also in this case, there is a decrease in the counter-EMF, but now the current only increases in the armature, while the inductor current maintains its fixed value. This means that the rotor speed is not affected by the magnetic field as much as it was in the series motor. However, the increase in armature current leads to an increase in torque.

As the load decreases, the speed increases (also in this case, by a very small amount). Therefore, the counter-EMF increases and the current decreases and, consequently, the torque.

In general, in these types of motors, small reductions in speed cause large increases in current. If a comparison is made between the no-load speed and the full load speed, the speed can be considered almost constant, so large variations are not necessary to obtain different torque values.

Shunt-wound motors are used in fans and centrifugal pumps.

Excitation in series plus parallel: compound machines

In the *compound motor* there are two field windings, one in series and one in parallel to the armature. The one in parallel usually has more turns.

It is the most adopted solution for a wide variety of applications as it combines both characteristics of the two types of excitation. In fact, thanks to the series field it is able to deliver greater torque at starting and, thanks to the parallel field, it allows better speed control.

The compound motor can be of the differential type, if the two magnetic fields are opposite to each other, or cumulative, if they are in the same direction.

This type of DC motor is used for driving machines, thanks to its high stability even in the event of load variations.



Starter Generator construction

Many starting systems for aircraft engines (both piston and turbine) use starter generator systems.

This type of construction has two functions:

- Engine start
- Current generator.

The fact of having two functions in a single element allows to save space and weight, but also to have a lower cost.

The starter generator is built as a shunt machine, with an additional winding in series. This series winding allows the production of a very intense magnetic field and therefore a large starting torque. The starter generator is always connected to the engine via a gear box.

Starter

The connection with a battery allows the passage of current on the field windings and on the armature: the starter generator works as a motor, until the aircraft engine connected to it reaches the minimum speed necessary to self-sustain. During the operation as a starter, only the series field is used, while the parallel field is used for the generator function.

Generator

Once the aircraft engine has reached minimum speed, the starter generator disconnects from the battery power supply (there is a relay) and begins to work as a generator, thanks to the rotation imposed by the connection of its shaft with the aircraft engine. The generator powers the electrical systems and recharges the battery itself.

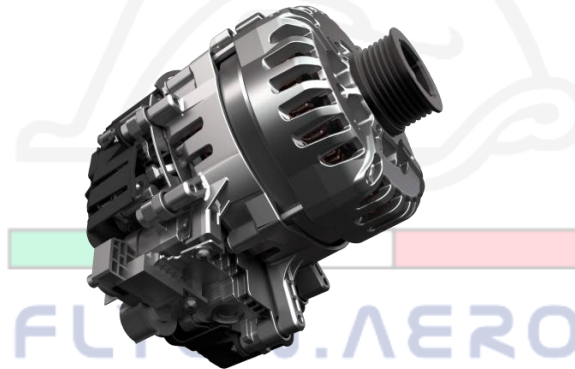


Figure 87 - A starter generator

3.13 AC Theory

Alternate Current (AC) is a current characterized by the variation of its direction over time.

Sinusoidal waveform: phase, period, frequency, cycle

In general, if a quantity takes on different values over time it is defined as variable. If these values repeat at constant time intervals, then the quantity is defined as periodic. Many periodic quantities, such as alternating current, can be defined via sinusoidal functions:

$$e(t) = E_{MAX} \cdot \sin \alpha \quad \rightarrow \quad e(t) = E_{MAX} \cdot \sin (\omega \cdot t)$$

In the function, $e(t)$ represents the value of the quantity considered at a given instant (t), E_{MAX} represents the maximum value that the quantity can assume (amplitude). α is the **phase angle**, i.e. a specific angular position relative to the instant t . In fact, it is calculated as the product of the angular velocity (or pulsation) by the time.

A sine wave is practically the trajectory of a point moving along a circumference with a constant angular velocity ω , projected onto a diameter of the circumference. The projection is a function of the angle α traveled along the circumference, therefore, it is also a function of time and angular velocity.

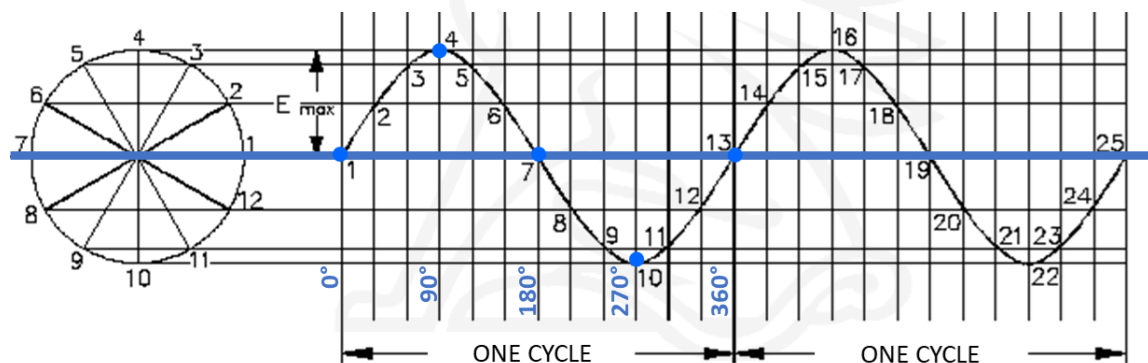


Figure 88 – Sinusoidal wave

The numbers (1→13, and so on), indicated in the figure above, identify the various **phases**, that is, each of the positions that the moving point occupies during its trajectory. We therefore note that the phase angle is determined by the ray passing through the point at that moment. The maximum amplitude value of the wave corresponds to the angles of 90° ($\sin 90^\circ = 1$) and 270° ($\sin 270^\circ = -1$), which are respectively the positive peak and the negative peak.

Every time the point travels a complete circle (360°) it has completed a **cycle**. The projection of this distance is the **wavelength**. The time it takes to make one complete revolution is the **period** (T), measured in seconds. The **frequency** (f) represents the number of cycles performed in one second and is measured in Hertz ($1 \text{ Hz} = 1/\text{s}$).

Instantaneous, average, root mean square, peak, peak-to-peak current values and calculations of these values in relation to voltage, current and power

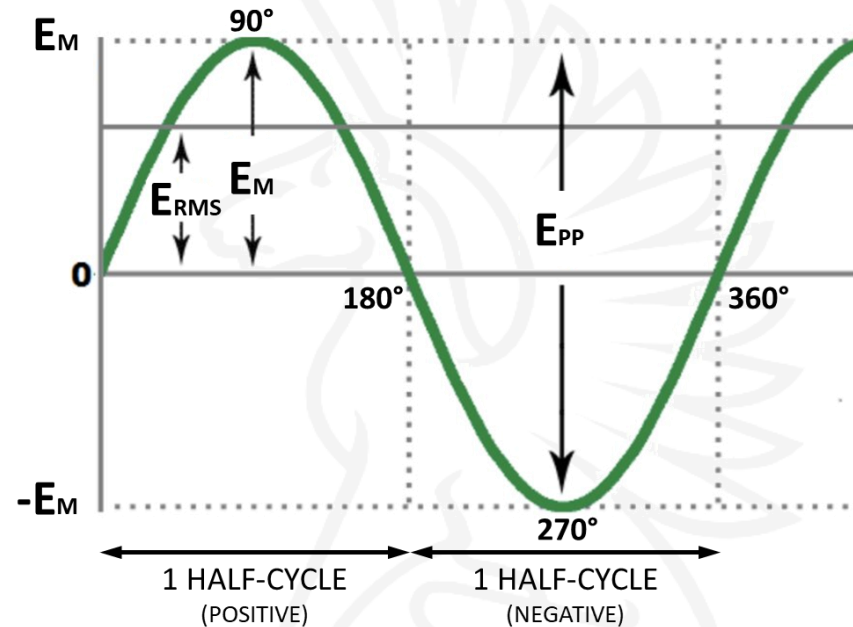


Figure 89 - Alternate current wave

The alternating current (and therefore also the voltage) moves along a sinusoidal wave. Considering any instant during a cycle, the value assumed by the current/voltage at that instant is called the *instantaneous value*. Obviously, this value can vary in a range between the negative and positive peaks.

The *peak value* E_M (or E_P) is the largest instantaneous value that the wave can reach. As already mentioned, the peak value can be positive (90°) or negative (270°).

The *average value* E_m is the average of all values that can be assumed by current/voltage during a half cycle (half-cycle). It is for sure a value between zero and the maximum value. It can be calculated as:

$$E_m = \frac{2 \cdot E_M}{\pi} \cong 0.637 \cdot E_M$$

The peak-to-peak value is the difference between the positive peak value and the negative peak value of the wave. Practically it is the maximum vertical distance that can be measured within the wave graph, in fact it is calculated as:

$$E_{PP} = 2 \cdot E_M$$

To carry out calculations using alternating quantities it is not possible to use instantaneous values, because they vary too quickly compared to the detection speeds of the measuring instruments. One might think of using average values, but the average value of a purely sinusoidal wave calculated over an entire cycle is always zero.

We therefore use current/voltage values called *effective values* E_{eff} , which are equal to the value that a continuous signal of equal average power would have compared to the alternating one considered. Practically, an effective value is an alternating current value that produces the same thermal effects as an equivalent direct value. This not only allows to carry out calculations, but also to make comparisons between various quantities. The effective value is also called the "Root Mean Square" (E_{RMS}) value because it is calculated as the square mean of all the instantaneous values of the wave.

$$E_{eff} = \frac{E_M}{\sqrt{2}} \cong 0.707 \cdot E_M$$

Average and effective values are stable over time and are independent of the amplitude of the wave at a given instant.



Triangular/Square waves

There are also other types of periodic waves besides sine waves: rectangular/square waves and triangular waves. In this type of waves, it is useful to calculate the peak-to-peak values, because they could be symmetrical or asymmetrical with respect to the horizontal axis.

Triangular waves have triangle-shaped peaks: the wave proceeds linearly with abrupt changes in slope.

A *rectangular wave* has only two voltage levels. There is a very short transition time between one level and another, which can be a "rising edge" (rise time) or a "falling edge" (fall time). Let's consider the rectangular wave in the figure.

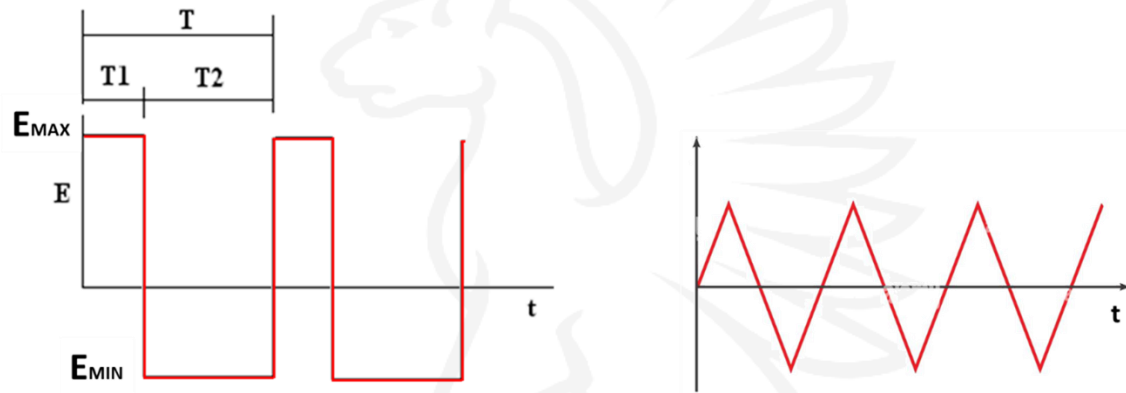


Figure 90 – Rectangular and triangular waves

The wave takes on two values: the positive peak E_{MAX} in the T_1 period interval and the negative peak E_{MIN} in the T_2 interval. The wave repeats every period $T = T_1 + T_2$.

If the two peak values are equal, the rectangular wave is said to be symmetrical.

If in a symmetric wave the two period fractions are equal, then we talk about a *square wave*.

The average value of the wave (considering the different distribution of amplitude over time) and the effective value can be calculated. In the calculation of the effective value the concept of "**duty cycle**" appears, which is a measurement of the percentage of wave above the horizontal axis:

$$\text{duty cycle} = \frac{T_1}{T}$$

The duty cycle can be calculated for all types of waves. In the case of the rectangular wave, if the duty cycle is equal to 0.5 (50%), it is a square wave.

However, the perfectly triangular/rectangular/square wave is actually a theoretical concept, because in the transmission of a signal there is always some noise that contributes to making the wave irregular. The fluctuations we are talking about, in any case, are limited, because otherwise they would lead to the loss of information.

Single-phase/Three-phase principles

An alternating current system can consist of one or more phases, depending on the usage needs.

Normally, the most common energy systems produce three-phase current, but homes receive energy from only one of the three phases (single-phase power).

A single-phase circuit is made up of two conductors: one is the phase, and the other is the neutral (it has zero potential and is used to give an output path to the current). The single-phase power supply gives lower power, in fact it is used for lighting, heating and small motors.

The three-phase system has three conductor wires plus a possible neutral conductor. In both single-phase and three-phase systems the earth conductor is always present.

In single-phase systems the voltages always vary following the same wave, while in three-phase systems the voltage waves have three different phases. When the system is symmetrical and balanced, the waves have the same intensity and frequency. What varies is the initial phase, that is, "the starting point" of one wave compared to the other. The three waves are distributed over 360° , so each of them is 120° out of phase with respect to the previous one and the next. For this reason, the amplitudes assumed at any moment by the three waves are different from each other (but the maximum amplitude is always the same).

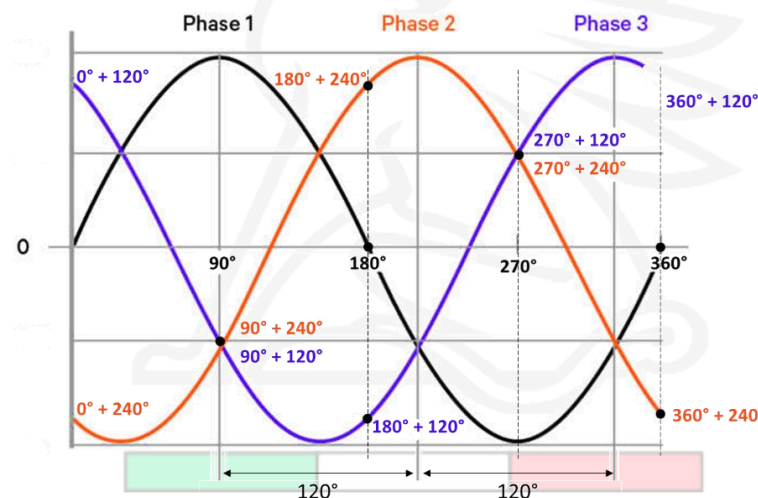


Figure 91 - Three-Phase quantity

From the figure above you can see that each of the waves passes through the point of zero voltage (horizontal axis) when it is out of phase with respect to the previous one by 120° . Using phase 1 (in black) as a reference, phase 2 (orange) arrives at the same point after 120° and phase 3 (blue) after 240° .

The sum of the three phases considered at any instant is 0° (or a multiple of 360°).

3.14 Resistive (R), Capacitive (C) and Inductive (L) Circuits

Phase relationship of voltage and current in L, C and R circuits, parallel, series and series parallel

In alternating current circuits, depending on the elements present, the relationship between the voltage wave and the current wave varies. We can distinguish three limit cases:

- Purely resistive circuits (or “purely Ohmic”), in which only resistances (R) appear.
- Purely capacitive circuits, in which only capacitors (C) appear.
- Purely inductive circuits, in which only inductors (L) appear.

NOTE: Remember that the current and voltage values considered from now on will always be the effective ones (the effective voltage is indicated with E).

Purely resistive AC circuits

In purely Ohmic circuits, the only elements that connect the voltage and current values are the resistors. In fact, Ohm's law is valid:

$$E = R \cdot I$$

This means that the voltage and current are directly proportional to each other, in function of a constant (the resistance value R). Consequently, the two quantities are defined as **in phase**, because they increase and decrease together. In fact, the maximums of the voltage wave correspond to the maximums of the current wave and the same is true for the zeros and minimums.

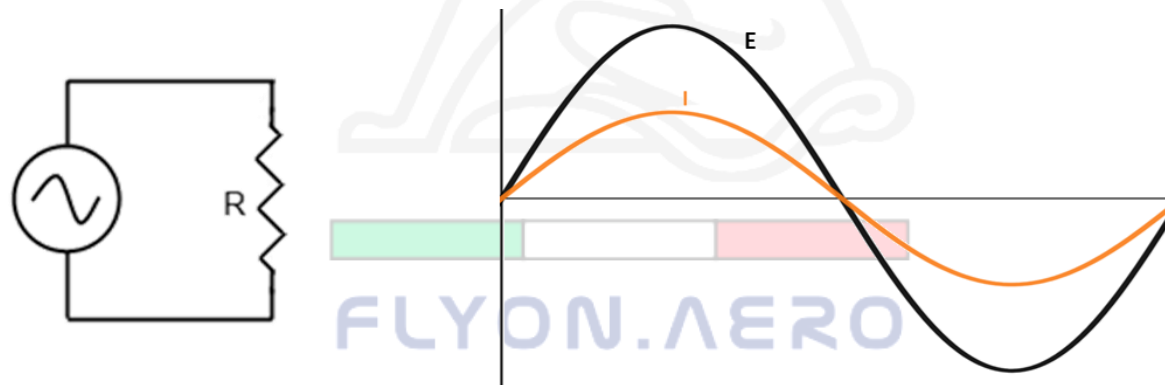


Figure 92 – R-circuit: voltage and current in phase

Purely capacitive AC circuits

NOTE: fully capacitive or inductive circuits are theoretical concepts, because there is always a minimum of resistance.

In purely capacitive circuits, the current and voltage waves always have the same frequency, but the current wave is 90° ahead of the voltage wave. The current is defined as in **quadrature of advance** to the voltage, because it reaches its maximum earlier.

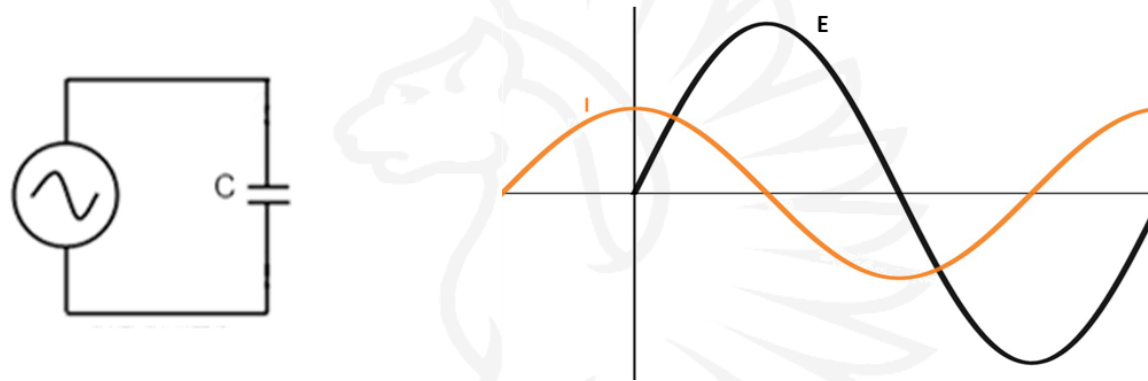


Figure 93 – C-circuit: current in quadrature of advance compared to the voltage

In this case, the proportionality between voltage and current is defined by a quantity called **capacitive reactance**, indicated by X_C and measured in Ohm. Reactance expresses the fact that the capacitor opposes the flow of current. The value of X_C is not constant, but depends on the frequency f (ω is the pulsation or angular velocity):

$$X_C = \frac{1}{\omega \cdot C} = \frac{1}{2\pi f \cdot C}$$

As the frequency of variation of the current increases, the reactance decreases, opposing its passage less (the passage of current is not through the dielectric of the capacitor, but through the circuit with the influence of the electric field). If the frequency decreases the reactance increases.

NOTE: In direct current electrical circuits, the frequency is zero; therefore, the calculation of the capacitive reactance would give a value tending to infinity. It is for this reason that direct current stops flowing once the capacitor is charged.

Also in this case, therefore, we can apply Ohm's law and define the relationship between voltage and current:

$$E = X_C \cdot I$$

Purely inductive AC circuits

In purely inductive circuits, the current wave lags the voltage wave by 90°. The current is defined as in **quadrature of lag** to the voltage.



Figure 94 – L-circuit: current in quadrature of lag compared to the voltage

The self-induced electromotive force in the circuit limits the value of the current.

In this case we define the **inductive reactance**, also measured in Ohms and indicated by X_L :

$$X_L = \omega \cdot L = 2\pi f \cdot L$$

The inductive reactance opposes the variations in current (the magnetic field produced by the passage of the current generates another self-induced current), in fact as the frequency increases the reactance also increases and therefore the opposition to the passage of the primary current.

Applying Ohm's law again we obtain the proportionality between voltage and current:

$$E = X_L \cdot I$$

RLC circuits

In real circuits resistors, capacitors and inductors can appear in various combinations (series, parallel, series-parallel).

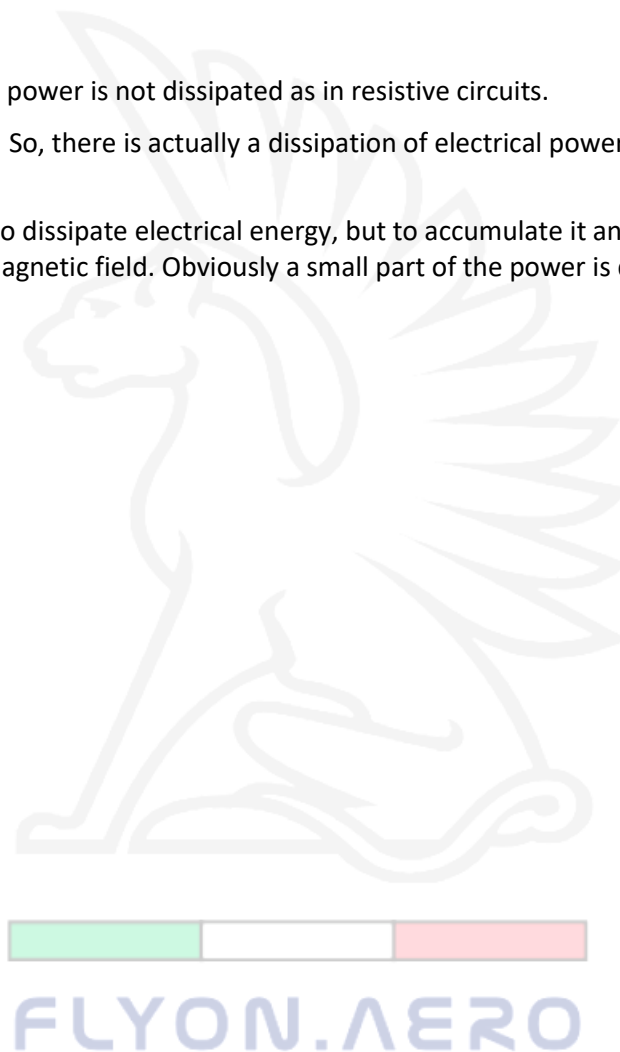
The calculation of the total opposition to the passage of current considers the contributions of the three elements. The quantity that expresses this opposition is called impedance. The equivalent impedance of multiple elements in series and/or multiple elements in parallel is calculated in the same way as the equivalent resistance.

Power dissipation in L, C and R circuits

In purely inductive and purely capacitive alternating circuits the power is not dissipated as in resistive circuits.

In resistive circuits, resistors have the task of absorbing current. So, there is actually a dissipation of electrical power, which is dispersed into the environment in the form of heat.

Capacitors and inductors, on the other hand, are not designed to dissipate electrical energy, but to accumulate it and use it at a later time. Capacitors accumulate it in the form of an electrostatic field, while inductors in the form of a magnetic field. Obviously a small part of the power is dissipated in the form of heat, but it is an inevitable "side effect" due to the flow of current.



Impedance, phase angle, power factor and current calculations

We have seen that in the case of purely resistive or capacitive or inductive circuits the phase angle between current and voltage was 0° , $+90^\circ$ and -90° respectively.

In circuits that combine the three elements, the phase angle takes on different values. We can define a quantity called **power factor** (indicated with PF) which is the cosine of the **offset angle** φ ('phi') formed by current and voltage:

$$\text{PF} = \cos \varphi$$

Power factor gives a measure of how effectively electrical energy is converted into other forms of energy. In the next paragraph we will see that it depends on two types of power: the active one and the reactive one. The first is the one that is actually used to do work, while the second is the one that is used to make the components of the circuit such as capacitors and/or inductors work.

The maximum value that the PF can assume is one, which therefore corresponds to an angle φ equal to zero, i.e. when the AC circuit is purely resistive.

Knowing the value of the PF we can obtain the value of the angle by calculating the arc-cosine function.

Power factor is also considered when we talk about **impedances**. Impedance, like resistance, is the measure of how much the circuit opposes the passage of current, however in RLC circuits it considers both the contribution of the resistors and that given by the reactance of capacitors and inductors.

Impedance, indicated by the letter Z and measured in Ohms, is represented by a complex number. In short, complex numbers are made up of a real part (in this case the resistance) and an "imaginary" part (the reactance). The second part is called imaginary because it is a contribution given by the product of a number by a constant j, called "imaginary unit", which is equal to $\sqrt{-1}$ and, as we know, the square root of a negative number does not exist (in the set of real numbers).

The impedance formula, therefore, is:

$$Z = R + j (X_L - X_C) \quad \text{con} \quad j = \sqrt{-1}$$

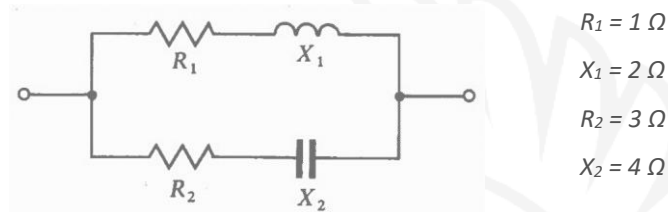
Each resistance and reactance can be associated with the corresponding impedance value.

If an impedance has only a resistive component, then it will only have the real part. If, however, it only has the reactive component (capacitive and/or inductive) then it will only have the imaginary part. If R-L-C are present in the circuit, then the impedance has both components.

Having made this consideration, the calculations for the series and parallel impedances already seen for the resistances remain valid. So, if we have to calculate an equivalent impedance of multiple elements in series, we will make a sum and if it involves multiple elements in parallel we will calculate the reciprocal of the sum of the reciprocals.

The calculation is rather complicated due to the nature of the numbers considered.

NOTE: The following **example** is just to give us an idea of how we obtain the impedance value that we will actually use for solving AC circuits.



Each element can be associated with an impedance. Impedance associated with R_1 and R_2 will only have the real component, while that associated with X_1 and X_2 will only have the imaginary component (furthermore, the component due to the capacitor must have a minus sign):

$$Z(R_1) = 1 \Omega \quad ; \quad Z(X_1) = j 2 \Omega \quad ; \quad Z(R_2) = 3 \Omega \quad ; \quad Z(X_2) = -j 4 \Omega$$

We can calculate the equivalent impedance of the entire circuit by first calculating the equivalent impedance of the series in branch 1 and of the series in branch 2:

$$\begin{aligned} \text{ramo 1:} \quad Z_{EQ,1} &= Z(R_1) + Z(X_1) \quad \rightarrow \quad Z_{EQ,1} = (1 + j 2) \Omega \\ \text{ramo 2:} \quad Z_{EQ,2} &= Z(R_2) + Z(X_2) \quad \rightarrow \quad Z_{EQ,2} = (3 - j 4) \Omega \end{aligned}$$

We can note that, in both cases, the calculated impedance has a real and an imaginary part, due to the fact that one component is resistive, and one is reactive. At this point we can calculate the equivalent impedance of the entire circuit ($Z_{EQ,1}$ and $Z_{EQ,2}$ are in parallel with each other):

$$Z_{EQ} = \frac{Z_{EQ,1} \cdot Z_{EQ,2}}{Z_{EQ,1} + Z_{EQ,2}} \quad \rightarrow \quad Z_{EQ} = \frac{(1 + j 2) \cdot (3 - j 4)}{(1 + j 2) + (3 - j 4)}$$

Let's carry out the calculations, remembering that j^2 is equal to -1 :

$$= \frac{3 - j 4 + j 6 - j^2 8}{4 - j 2} = \frac{3 + j 2 - (-1) \cdot 8}{4 - j 2} = \frac{11 + j 2}{4 - j 2}$$

Let's eliminate the denominator:

$$= \frac{11 + j 2}{4 - j 2} \cdot \frac{4 + j 2}{4 + j 2} = \frac{44 + j 22 + j 8 + (-1) \cdot 4}{16 - (-1) \cdot 4} = \frac{40 + j 30}{20} = 2 + j 1.5$$

The total impedance value of the circuit is:

$$Z_{EQ} = (2 + j 1.5) \Omega$$

Solving a circuit in alternating regime means knowing the effective values of the voltage and current, and therefore of the equivalent impedance Z of the circuit.

If we know the value of the equivalent impedance in the form of a complex number, to compare it with the voltage and current values we must calculate its modulus, a bit as if it were a vector. In this regard, we refer to the "**impedance triangle**", which relates the resistance and reactance values with the correspondent impedance value: one cathetus is the real component of the number, the other one is the imaginary component.

The hypotenuse will be equal to the modulus of the total impedance, which can be used for calculations. We find the value of Z simply by applying the Pythagorean Theorem.

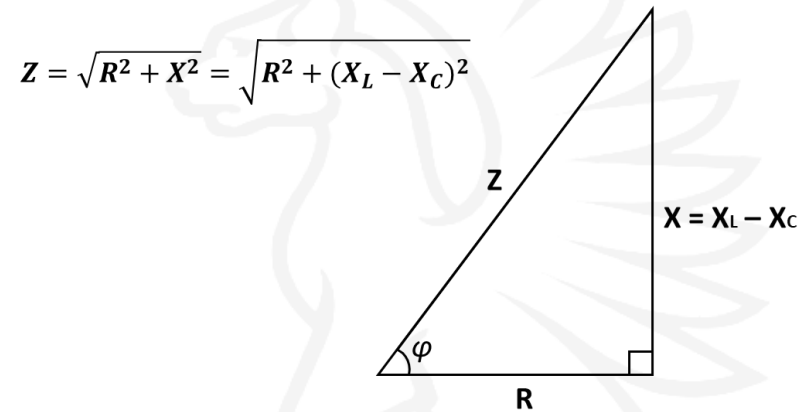


Figure 95 - Impedances triangle

Example:

$$Z = (2 + j 1.5) \Omega \quad \rightarrow \quad R = 2 \Omega ; X = 1.5 \Omega \quad \rightarrow \quad Z = \sqrt{2^2 + 1.5^2} = \sqrt{4 + 2.25} = \sqrt{6.25} = 2.5 \Omega$$

The value of Z thus found can be used in Ohm's law to find the effective voltage or current values:

$$E = Z \cdot I$$

We can note that in the figure above, the slope of the hypotenuse of the triangle is nothing other than the angle φ of the power factor. We can calculate it as:

$$\text{PF} = \cos \varphi = \frac{R}{Z}$$

True power, apparent power and reactive power calculations

In AC circuits, it must be considered that the voltage and current do not reach their maximum at the same time, because they are waves out of phase with each other.

For this reason, by analyzing the instantaneous power equation (given by the product between voltage and current) we can identify a value that remains constant all the time and one that fluctuates. The first value is the active power, i.e. the power transformed into heat by the Joule effect, therefore actually used and dissipated. The second, which is an oscillating value, is the reactive power, which is alternately stored and released by the circuit.

True power

True power is the average value of instantaneous power. It is indicated with the letter **P** and is measured in **Watts**. We calculate it as the scalar product between voltage and current (RMS values):

$$P = E \cdot I \cdot \cos \varphi \quad \text{or} \quad P = I^2 \cdot R$$

Reactive power

Reactive power cannot power loads but is used by the magnetic fields of the inductors and the electrostatic fields of the capacitors. When the power is greater than 0 (wave above horizontal axis) the behavior is inductive, if it is less than 0 the behavior is capacitive.

Reactive power is indicated with the letter **Q** and is measured in **Reactive Volt-Amperes (VAR)**. It is calculated as:

$$Q = E \cdot I \cdot \sin \varphi \quad \text{or} \quad Q = I^2 \cdot X$$

Reactive power is power that cannot be exploited: it is unwanted, and we try to reduce it to a minimum.

The power factor is therefore a measure of how inefficient the circuit is, because as the phase shift angle ϕ between current and voltage increases, the value of the apparent power increases (if ϕ increases, the value of the sine increases).



Apparent power

Active power and reactive power are respectively the real and imaginary component of another quantity: the *apparent power*, which in practice is the total power supplied to the circuit by the source. It therefore results in $S = P + j Q$.

What interests us about apparent power is its module, which, as in the case of impedance, can be calculated starting from the geometric relationship between P, Q and S, defined by the **power triangle**:

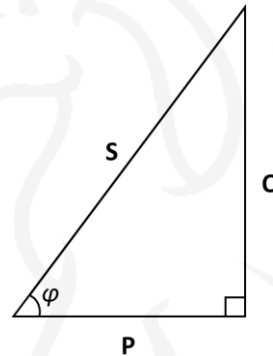


Figure 96 - Power triangle

Applying the Pythagorean Theorem and the property of trigonometry ($\cos^2 \varphi + \sin^2 \varphi = 1$) we obtain the formula for calculating the apparent power, which is indicated with the letter **S** and is measured in **Volt-Amperes**:

$$S = \sqrt{P^2 + Q^2} \quad \rightarrow \quad S = E \cdot I \quad \text{or} \quad S = I^2 \cdot Z$$

Power Factor

The most used definition of *power factor* also derives from the geometric construction:

$$PF = \cos \varphi = \frac{P}{S}$$

The formula above shows us more explicitly that the PF is a measure of the efficiency of the circuit, i.e. how much of the total power produced (apparent) is actually used to power the loads (true power). Let's look at some specific cases.

PURELY RESISTIVE CIRCUIT

In a circuit in which only resistances appear ($Z = R$), there is no phase shift between voltage and current. Therefore, angle φ is null and the power factor is equal to 1, i.e., its maximum value:

$$\varphi = 0^\circ \quad \rightarrow \quad \text{PF} = \cos \varphi = 1$$

Let's calculate powers:

$$P = E \cdot I \cdot \cos 0^\circ = E \cdot I$$

$$Q = E \cdot I \cdot \sin 0^\circ = 0$$

$$S = E \cdot I$$

$$\rightarrow S = P$$

We understand that the purely resistive circuit is the most efficient, because the true power P used to power the load corresponds to all the apparent power S produced.

PURELY CAPACITIVE CIRCUIT

In a circuit in which only capacitors appear ($Z = X_C$), the current leads by 90° with respect to the voltage, so the PF reaches its minimum value:

$$\varphi = 90^\circ \quad \rightarrow \quad \text{PF} = \cos \varphi = 0$$

Let's calculate powers:

$$P = E \cdot I \cdot \cos 90^\circ = 0$$

$$Q = E \cdot I \cdot \sin 90^\circ = E \cdot I$$

$$S = E \cdot I$$

$$\rightarrow S = Q$$

This type of circuit is not efficient, because the power S produced is only absorbed and released cyclically by the capacitor, in the form of electrostatic field.



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PURELY INDUCTIVE CIRCUIT

In a circuit in which only inductors appear ($Z = X_L$), the current lags the voltage by 90° . Also in this case the PF has its minimum value:

$$\varphi = 90^\circ \rightarrow \text{PF} = \cos \varphi = 0$$

Let's calculate powers:

$$P = E \cdot I \cdot \cos 90^\circ = 0$$

$$Q = E \cdot I \cdot \sin 90^\circ = E \cdot I$$

$$S = E \cdot I$$

$$\rightarrow S = Q$$

Like the previous circuit, this type of circuit is not efficient. The power S produced is only absorbed and released cyclically by the inductor in the form of a magnetic field.

RLC CIRCUITS

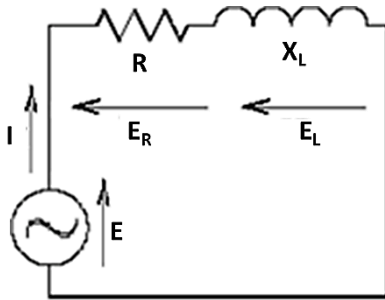
In circuits where resistors, capacitors and inductors appear, the phase angle between voltage and current is any angle between 0° and 90° , so the PF is a value between 0 and 1. Both true power and reactive power have a value other than 0 and the apparent power takes both contributions into account.

A smaller phase shift generates a greater PF and therefore greater true power (more efficiency).



Examples

Example 1 – calculate the voltage drop across the resistor and inductor, the PF, the true, reactive and apparent power. Data: $R = 6 \Omega$; $X_L = 8 \Omega$; $E = 100 \text{ V}$ (RMS value).



We can calculate the total impedance Z using the impedance triangle:

$$Z = \sqrt{R^2 + X_L^2} \rightarrow Z = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = 10 \Omega$$

We need the total impedance to calculate the current flowing in the circuit, with Ohm's law:

$$E = Z \cdot I \rightarrow I = \frac{E}{Z} \rightarrow I = \frac{100}{10} = 10 \text{ A}$$

At this point we can calculate the two voltage drops (remembering that the impedance of the resistor is equal to R and that of the inductor is equal to X_L):

$$E_R = Z_R \cdot I \rightarrow E_R = R \cdot I \rightarrow E_R = 6 \cdot 10 = 60 \text{ V}$$

$$E_L = Z_L \cdot I \rightarrow E_L = X_L \cdot I \rightarrow E_L = 8 \cdot 10 = 80 \text{ V}$$

Let's find the power factor:

$$PF = \frac{R}{Z} \rightarrow PF = \frac{6}{10} = 0.6 = \cos \varphi$$

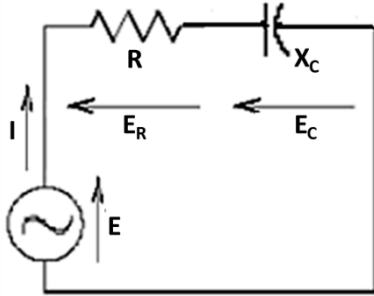
Let's calculate the powers using the formulas

$$P = E \cdot I \cdot \cos \varphi \rightarrow P = 100 \cdot 10 \cdot 0.6 = 600 \text{ W}$$

$$Q = E \cdot I \cdot \sin \varphi = E \cdot I \cdot \sqrt{1 - (\cos \varphi)^2} \rightarrow Q = 100 \cdot 10 \cdot \sqrt{1 - 0.6^2} = 100 \cdot 10 \cdot 0.8 = 800 \text{ VAR}$$

$$S = E \cdot I \rightarrow S = 100 \cdot 10 = 1000 \text{ VA}$$

Esempio 2 – Calculate the voltage drop across the resistor, the reactance of the capacitor, the powers, the PF and the phase angle. Data: $R = 3 \Omega$; $E_C = 4 \text{ V}$; $I = 2 \text{ A (RMS)}$.



The voltage drop across the resistor is:

$$E_R = Z_R \cdot I \rightarrow E_R = R \cdot I \rightarrow E_R = 3 \cdot 2 = 6 \text{ V}$$

The reactance of the capacitor:

$$E_C = Z_C \cdot I \rightarrow X_C = \frac{E_C}{I} \rightarrow X_C = \frac{4}{2} = 2 \Omega$$

To calculate the powers we need the voltage value of the generator, which we can obtain from Ohm's law once the total impedance of the circuit has been calculated

$$Z = \sqrt{R^2 + X_C^2} \rightarrow Z = \sqrt{3^2 + 2^2} = \sqrt{9 + 4} = 3.6 \Omega$$

Therefore:

$$E = Z \cdot I \rightarrow E = 3.6 \cdot 2 = 7.2 \text{ V}$$

We can calculate powers without knowing the value of the power factor using the other formulas given on text:

$$P = I^2 \cdot R \rightarrow P = 4 \cdot 3 = 12 \text{ W}$$

$$Q = I^2 \cdot X \rightarrow Q = 4 \cdot 2 = 8 \text{ VAR}$$

$$S = I^2 \cdot Z \rightarrow S = 4 \cdot 3.6 = 14.4 \text{ VA}$$

At this point we can find the value of the power factor using the impedances or powers, and then the value of the angle by applying the arc-cosine function:

$$PF = \frac{R}{Z} = \frac{P}{S} \rightarrow PF = \frac{3}{3.6} = \frac{12}{14.4} = 0.83 = \cos \varphi$$

$$\varphi = \cos^{-1} 0.83 = 33.9^\circ$$

3.15 Transformers

Transformer construction principles and operation

Transformers are static electric machines (they have no moving parts) that work in alternating regime and exploit the phenomena of electromagnetic induction to transfer electrical energy from one circuit to another.



Figure 97 - A mono-phase transformer

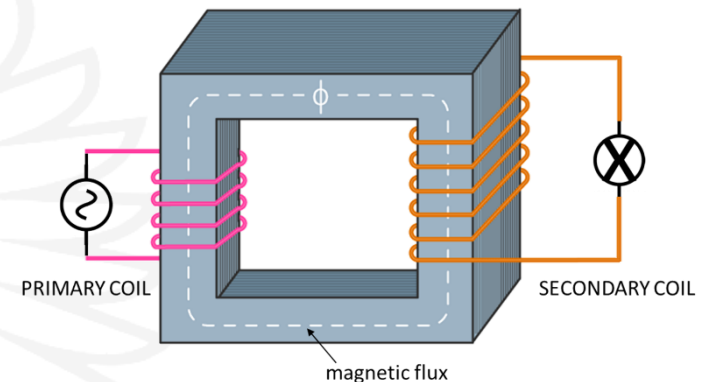


Figure 98 – Schematic of a mono-phase transformer

A **mono-phase transformer** has three main parts:

- Ferromagnetic core
- Primary winding
- Secondary winding with a different number of coils.

The two windings are not connected to each other but are both in contact with the ferromagnetic core. The primary winding is connected to the power supply, while the secondary is connected to the users and receives energy thanks to inductive effects.

In fact, the primary circuit is powered with an alternating current, so a magnetic field is generated around it which contracts and expands with the same frequency as the current. The variable field lines that cross the windings of the secondary generate an induced electromotive force in it and therefore an induced current, with the opposite direction compared to the current variation of the primary winding. The ferromagnetic core on which the two circuits are wound serves to facilitate the passage of the magnetic flux.

The two windings are reversible, that is, each of them can have the function of both primary and secondary.

Transformation is nothing other than the transfer of electricity from one winding to another by exploiting the variation in the magnetic field. The transfer depends on the number of windings of primary and secondary.

We can distinguish:

- **Voltage transformers:** the voltage reaching the load is lower (step-down) or higher (step-up) than the primary supply voltage. In these transformers the primary winding is connected in parallel to the power supply. They are the most used.
- **Current transformers:** the current reaching the load is lower or higher than the primary supply current. The power supply and the primary winding are connected in series.

On an application level we distinguish:

- **Power transformer:** they transmit large amounts of power, working with voltages in the kilovolt range. They are usually found in power plants.
- **Distribution transformer:** they transmit electricity at the level of local networks, working with lower voltages (ex. 230 V).

NOTE: in any case, when we talk about transformers, the power output from the secondary does not increase compared to that input to the primary.

Another distinction of transformers can be made by considering the method of heat dissipation. Dry-type transformers simply use ambient air to cool. Oil-filled transformers, on the other hand, have the coils contained in a shell filled with dielectric mineral oil which exchanges heat with the environment.

Three-phase transformers are basically obtained by combining three single-phase transformers and applying constructional measures. The three ferromagnetic cores are connected to each other and may be coplanar or not. The most common configuration is the one with coplanar cores, which we see in the figure, because it allows it to occupy less space. The primary (A, B, C) and secondary (a, b, c) of each phase are wound on the same column.

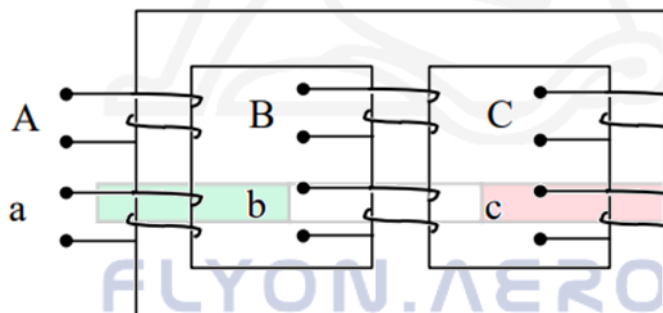


Figure 99 - Three-phase transformer

Transformer losses and methods for overcoming them

Transformers are very efficient machines, with efficiencies of up to 99%.

They therefore have very low losses, which can be due to various phenomena:

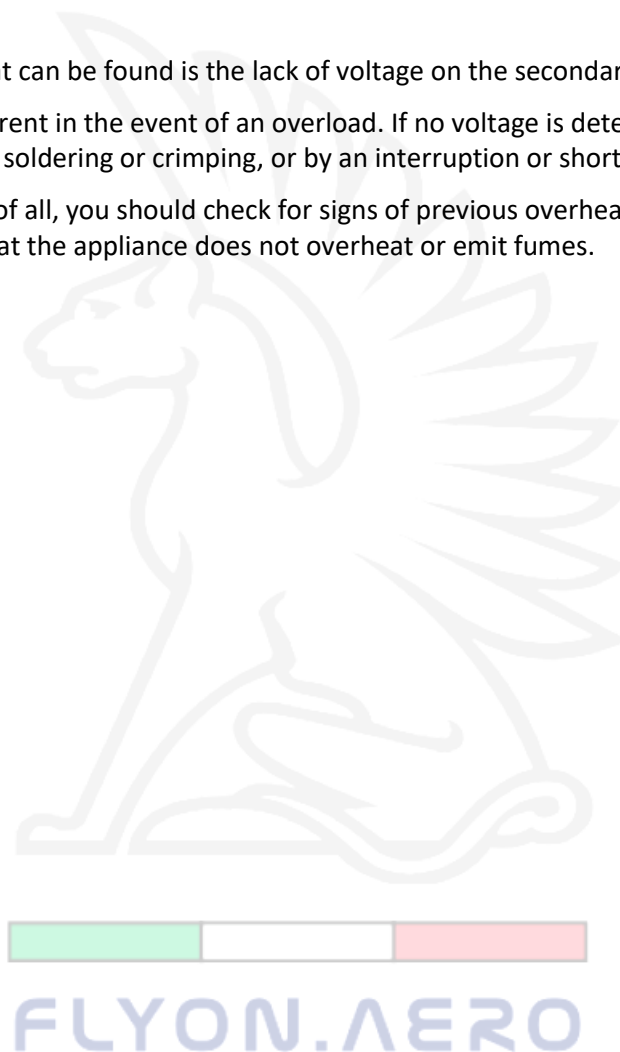
- **Ohmic Losses:** in the conductors, due to **Joule effect** ("copper loss").
The windings have a certain resistance, so some of the energy is lost in the form of heat to overcome this resistance. These losses are independent of the frequency of the current.
→ They can be reduced by using wire windings with a larger section, which have less resistance. Furthermore, the two windings are sized in such a way as to have approximately equal losses due to the Joule effect.
- **Magnetic Losses:** in the core, they include losses due to **hysteresis** and **eddy currents** ("iron loss" or "core loss").
The former are due to the fact that the magnetized and demagnetized material continuously heats up, dissipating part of the energy supplied. This type of loss depends on the frequency of the current, because as the frequency increases the magnetization and demagnetization cycles increase. Eddy currents, on the other hand, are the unwanted currents caused by induction in the ferromagnetic core.
→ The hysteresis problem is solved by working at constant frequencies and building the core using materials with tight hysteresis loops such as silicon iron (stalloy), and mu-metal.
→ As regards eddy currents, the magnetic core is constructed with a series of laminations rather than with a single block, in order to reduce their intensity. In high power transformers the core is made of ceramic material with ferromagnetic particles.
- Losses due to **magnetic flux dispersion**.
Due to the fact that not all the flow generated by the primary is coupled with the secondary, therefore the voltage does not reach the theoretical maximum value. In modern transformers this loss is negligible.
→ The flow can be redirected using magnetic shields and taking other construction measures such as increasing the section of the ferromagnetic core.
- **Magnetostriction** and **mechanical** losses.
A minimal part of the losses is due to the continuous (slight) **contraction and expansion** of the **ferromagnetic** core under the action of variable magnetic fields (dissipation of energy in friction). Additionally, electromagnetic forces cause nearby metallic objects to vibrate, further dissipating small amounts of energy.
→ Magnetostriction losses can be reduced by keeping the core at a lower temperature.
→ Energy dissipation due to vibrations can be reduced by using isolation devices that avoid transmitting forces to nearby elements.

Defects

As regards defects on transformers, the most common fault that can be found is the lack of voltage on the secondary.

Some transformers have a fuse which interrupts the flow of current in the event of an overload. If no voltage is detected for this reason, simply replacing it will fix the problem. The lack of voltage can also be caused by poorly done soldering or crimping, or by an interruption or short circuit in the windings.

To check the transformers, you can do some simple tests. First of all, you should check for signs of previous overheating or defects. Then just power the transformer and check that the output has an acceptable value, also checking that the appliance does not overheat or emit fumes.



Transformer action under load and no-load conditions

No-load condition

No-load operation of a transformer refers to the condition in which there is no load connected to the secondary winding. In practice, the secondary is an open circuit.

The open circuit test allows the performance of the transformer to be evaluated. It is carried out by powering the primary winding at the nominal voltage (for example 230 V). The current on the primary (**no-load current**) has a very low value, which is sufficient to maintain the magnetic field flux. On the secondary, being an open circuit, no current flows.

The consequence is that the ohmic losses in these conditions are very low, practically negligible.

The magnetic field generated by the primary current, however, generates self-inductive effects on the primary itself. The **no-load voltage** is simply the self-induced electromotive force on the winding. We understand that, since ohmic losses are negligible, all losses evaluated in these conditions are connected to magnetic induction phenomena (eddy currents and hysteresis). In fact, the true power during an open circuit test is very low and the supplied power is transformed almost exclusively into reactive power.

You can also do a short circuit test (always without load). In this test, the primary is powered with a reduced voltage (the primary current is close to zero), such that the nominal current circulates in the secondary. The secondary current cannot be absorbed by a load; therefore, it is completely used to generate the magnetic field which will have a high value. Consequently, the current induced in the primary will also be very high. In these conditions, the magnetic losses are negligible compared to the ohmic ones and the power is mainly true power.

The reactive and true power values measured at no load in the two tests mentioned are used to calculate the efficiency of the transformer.

Loaded condition

When a load is connected to the secondary winding, an induced current I_2 flows through it. This current generates a variable magnetic field which induces another current on the primary (I_1), adding to the supply current (which we call I_0).

The magnetic field generated by the primary winding has self-inductive effects on itself. This means that there is a self-induced electromotive force ("back EMF") that reduces the current I_1 . The magnetic field generated by the secondary, however, is opposite to that generated by the primary and consequently allows the current to resist this reduction.

The current I_1 increases, increasing the current I_2 and therefore the secondary field which then further increases I_1 . These variations continue until the magnetic field generated by the primary reaches its original intensity, i.e. the one not reduced by the interaction with the secondary magnetic field. In this condition, the primary and secondary are balanced.

Power transfer, efficiency, polarity markings

Ideal Transformer

In an ideal transformer, all losses are considered negligible. This means that there is no resistance in the windings, that the ferromagnetic material is not influenced by hysteresis and eddy currents, and that all the magnetic flux links perfectly with the windings.

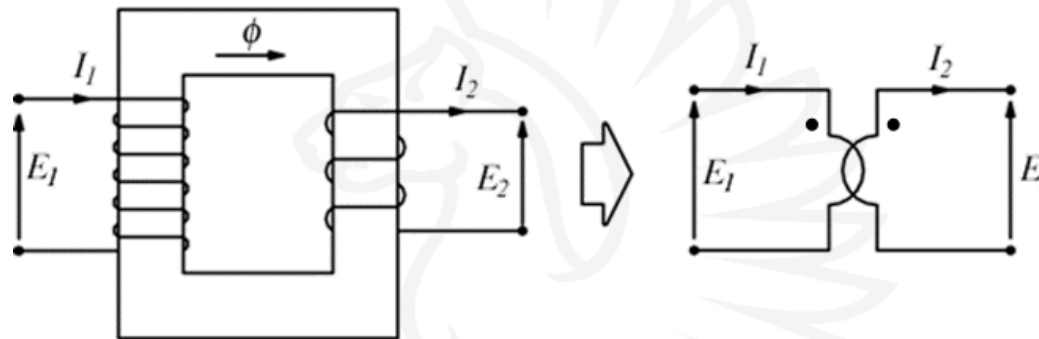


Figure 100 - Ideal transformer equivalent circuit

In an ideal transformer, all the **power** supplied (therefore the apparent power) at the input is transferred to the output:

$$S_1 = S_2 \quad \rightarrow \quad E_1 \cdot I_1 = E_2 \cdot I_2$$

This means that there is no power dissipated or absorbed, in fact $R=0$ and $X=0$, and consequently the true and reactive power are zero. The voltage and current can vary (if one increases the other decreases and vice versa), but their product remains constant.

The **efficiency** of a transformer is given by the ratio between the power absorbed by the secondary and that supplied to the primary. Logically, for an ideal transformer, the efficiency is 100%.



Practical transformer

In practical transformers, resistances and reactance are not negligible.

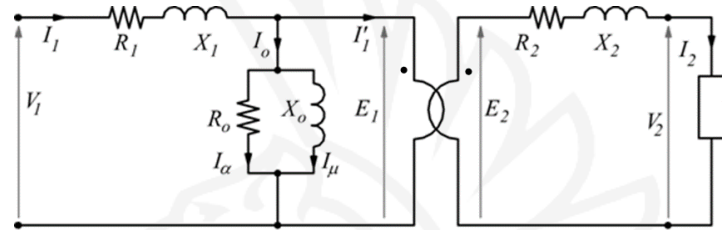


Figure 101 – Practical transformer equivalent circuit

In the equivalent circuit we can notice several more elements compared to the ideal equivalent circuit. X_1 and X_2 represent the leakage inductances, due to the magnetic fluxes dispersed to the primary and secondary. Resistors R_1 and R_2 take ohmic losses into account. The resistance R_0 considers the magnetic losses and the reactance X_0 of the current necessary to maintain the flow.

The practical transformer absorbs both true power P and reactive power Q .

The efficiency or performance can be calculated as the ratio between the true output power (delivered, P_2) and the true input power (absorbed, P_1). Remember that part of the absorbed power is dissipated (P_D) in the form of heat, therefore $P_1 = P_2 + P_D$. This formula could lead to an incorrect calculation because, since the transformer is a very efficient machine ($\eta \approx 99\%$), even small errors in the measurement of the power would cause the actual value of the efficiency to vary too much.

For this reason, the conventional efficiency of the transformer is normally calculated, given by:

$$\eta = \frac{S_n}{S_n + P_{Fe} + P_{Cu}}$$

In the numerator there is the nominal apparent power S_n that can be obtained from the transformer, i.e. the output power. Its value is provided by the manufacturer as plate data. In the denominator there is the apparent input power. The output power is nothing more than the input power without the copper and iron losses. For this reason, the input power is the sum of the nominal output power S_n and of the losses P_{Fe} and P_{Cu} .

- P_{Fe} is the power that is measured in the open circuit no-load test. We have seen that in those conditions all the losses can be traced back to magnetism ("Fe" indicates the "iron losses"), while the ohmic ones are negligible.
- P_{Cu} is the power measured in the short circuit test. The magnetic losses are negligible compared to the ohmic ones ("Cu" indicates the "copper losses").

Polarity

The polarity of a transformer refers to the relative direction of the voltages induced between the windings. Depending on how the coils are wound around the core, the phase difference between voltage/current at the primary and voltage/current at the secondary can be 0° (waves in phase) or 180° (waves in phase opposition).

In the representation of circuits with transformers, the polarity is indicated with a dot “•” or with an “X” or with the “+” sign. The dot is also physically indicated on the transformers.

- When both windings have the dot at the top it means that the polarity is the same, i.e. that the current flows in the same direction. This means that the two currents and the two voltages are in phase.
- When the two dots are positioned oppositely (one at the top and the other at the bottom or vice versa) it means that the two currents are in phase opposition (out-of-phase).

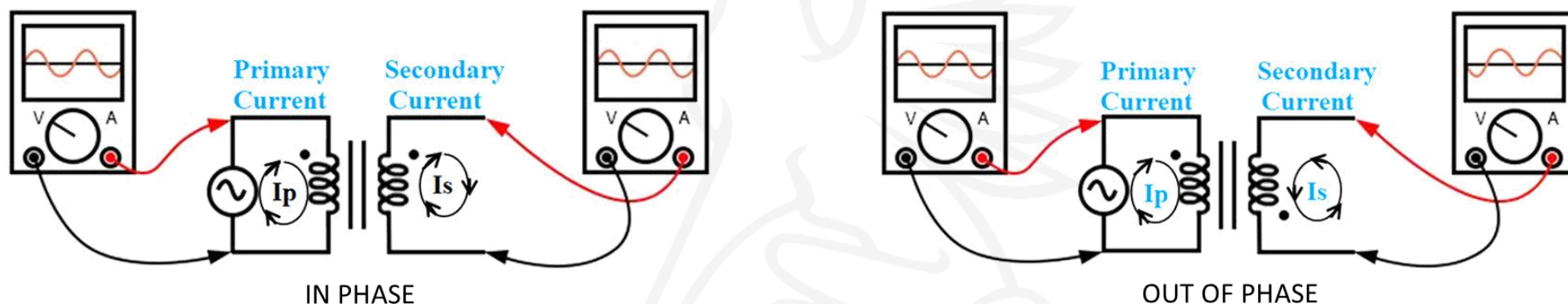


Figure 102 – Polarity of transformers

However, the polarity of each winding can be reversed by reversing the power supply terminals.

This discussion is important because a transformer works correctly only if the quantities on the primary are in phase with the quantities on the secondary.

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Line and phase voltages and currents

In general, in a three-phase system, the three conductors plus the possible neutral of the three-phase system can be represented as follows:

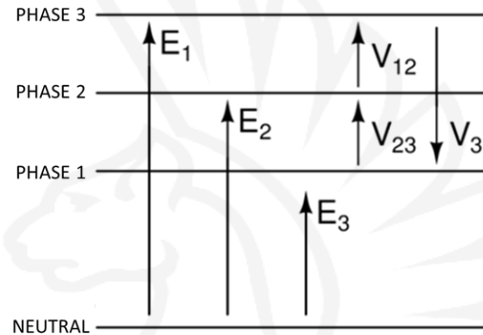


Figure 103 - Line and phase voltages

The load connected to a three-phase system is made up of three impedances (resistance, capacitance, inductance). If the three impedances are equal to each other, the load is said to be **balanced**. If the load is balanced, the presence of the neutral conductor is not necessary.

In the generic circuit of the figure above we can identify:

- **Line voltages (V)** – the voltage drops measured between one phase and another;
- **Phase voltages (E)** – the voltage drops measured between the single phases and the neutral conductor;

Moreover, there will be:

- **Line currents** – the currents that flow through the conductors of the individual phases (there can also be the neutral current);
- **Phase currents** – they are the currents that flow through the impedances of the loads present.

To better understand these concepts, we distinguish the two main methods by which conductors can be connected to each other:

- Star (or “wye”) connection (Y)
- Triangle (or “delta”) connection (Δ – delta).

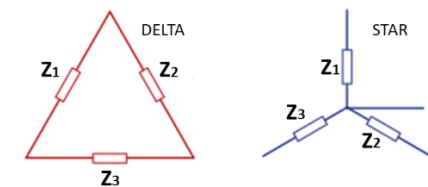


Figure 104 - Delta and star connections

The following two figures can be interpreted as a depiction of the primary windings.

In the *star connection*, the three windings of the primary (the impedances Z) are connected to the three phase conductors and to a common terminal (if the load is not balanced the common terminal is connected to the neutral), i.e. we find them along the single conductor.

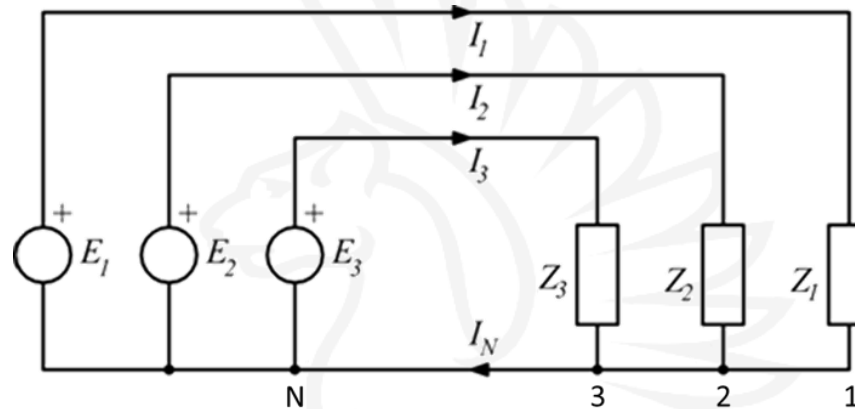


Figure 105 - Star connection

Assuming that the three impedances are equal to each other ($Z_1 = Z_2 = Z_3 = Z$), we can say that the current on the neutral conductor is zero ($I_N = 0$). We have:

Phase voltages: $E_1 - E_2 - E_3$

Line voltages: $V_{12} - V_{23} - V_{13}$

Let's consider a symmetrical and balanced system, i.e. all the generators have the same voltage "E" and are out of phase with each other by 120° :

$$E_1 = E_2 = E_3 = E = V_{PHASE}$$

It can be demonstrated that:

$$V_{12} = V_{23} = V_{13} = V_{LINE} = \sqrt{3} \cdot V_{PHASE}$$

(The line voltage is equal to the vector sum of the phase voltages).

As regards the currents, instead, given that those that flow through the individual conductors are the same as those that flow through the individual impedances, it is:

$$I_{LINE} = I_{PHASE}$$

In the *delta connection* the windings (the impedances Z) are connected two by two to the phase conductors, i.e. we find them between one conductor and the other.

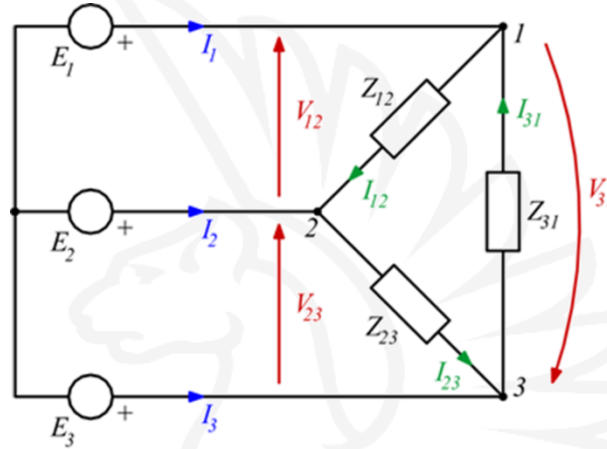


Figure 106 - Delta connection

We assume also in this case that the three impedances are equal to each other.

The three phase voltages and the three line voltages, as before, are respectively $E_1 - E_2 - E_3$ and $V_{12} - V_{23} - V_{13}$. Also in this case we consider the phase ones (E) to be equal to each other and consequently also the line ones (V).

Line currents: $I_1 - I_2 - I_3$

Phase currents: $I_{12} - I_{23} - I_{13}$

Regarding the voltages, we can see that each impedance is in a mesh with two generators. It means that each impedance is always subjected to chain voltages. Consequentially:

$$V_{LINE} = V_{PHASE}$$

The line currents are all equal to each other (because E is always the same value), and consequently also the phase ones (because Z is always the same). It can be demonstrated that:

$$I_{LINE} = \sqrt{3} \cdot I_{PHASE}$$

(Line current is equal to the vector sum of the phase currents).

Therefore, applying the discussion to transformers: in the star connection there is a terminal common to all three windings; in the case of delta connection, the end of one winding is connected to the beginning of the other.

Three-phase transformers can be built with different combinations:

- Star – star (Y-Y)
- Delta – delta (Δ - Δ)
- Star – delta (Y- Δ)
- Delta – star (Δ -Y).

Regardless of the type of connection, each primary phase is wound in the same column as the relevant secondary phase. Hence, the concatenated magnetic flux is the same. This means that the electromotive force induced in the primary of a given phase is in phase or in phase opposition (depending on the dot) with the EMF induced in the relevant secondary.

Furthermore, the induced electromotive force has a value that is very close to the value of the phase voltage, both at the primary and at the secondary. Consequently, the phase voltages at the secondary are always approximately in phase (or in phase opposition) with the phase voltages of the primary.

Instead, for line voltages the type of connection considered is important. If the connection is delta, the line voltages are equal to the phase voltages. If the connection is star, the line voltages are different from the phase ones.

Therefore, the line voltages at the secondary are out of phase with respect to the corresponding line voltages at the primary. The phase shift is always a multiple of 30° .

NOTE: in three-phase transformers, the nominal voltage/currents are always the line values.



Power in a three phase system

To calculate the powers of a three-phase system, we simply calculate three times the powers of a single-phase system.

In three-phase systems it is necessary to distinguish between phase and line quantities, but in the single-phase system it makes no sense.

The quantities indicated in the power formulas of the single-phase transformer will be those of the phase of the three-phase system:

$$P = E \cdot I \cdot \cos \varphi \quad ; \quad Q = E \cdot I \cdot \sin \varphi \quad ; \quad S = E \cdot I$$

For better understanding, we denote the phase voltage as $E = V_{PH}$ and the phase current as $I = I_{PH}$

We obtain the powers of the three-phase system:

$$P = 3 \cdot V_{PH} \cdot I_{PH} \cdot \cos \varphi \quad ; \quad Q = 3 \cdot V_{PH} \cdot I_{PH} \cdot \sin \varphi \quad ; \quad S = 3 \cdot V_{PH} \cdot I_{PH}$$

As regards transformers, however, the powers are usually indicated according to the **nominal values**, which by definition are the line values of the secondary. Then we distinguish two cases:

- Star connected secondary: $V_L = \sqrt{3} \cdot V_{PH} \rightarrow V_{PH} = \frac{1}{\sqrt{3}} \cdot V_L \quad ; \quad I_{PH} = I_L$
- Delta connected secondary: $I_L = \sqrt{3} \cdot I_{PH} \rightarrow I_{PH} = \frac{1}{\sqrt{3}} \cdot I_L \quad ; \quad V_{PH} = V_L$

Writing the line values with nominal value notation at the secondary (I_L becomes I_{2n} and V_L becomes V_{2n}) we obtain the formulas for the calculation of powers in three-phase transformers:

$$P_{2n} = \sqrt{3} \cdot V_{2n} \cdot I_{2n} \cdot \cos \varphi \quad [W]$$

$$Q_{2n} = \sqrt{3} \cdot V_{2n} \cdot I_{2n} \cdot \sin \varphi \quad [VAR]$$

$$S_{2n} = \sqrt{3} \cdot V_{2n} \cdot I_{2n} \quad [VA]$$

In all cases the factor $\sqrt{3}$ (≈ 1.73) appears. It is valid both in the case of star connection and in the case of delta because it derives either from the calculation of the voltage or from that of the current.

(mathematically we obtain the root of 3 because we rationalize the product $3 \cdot (1/\sqrt{3})$).

The power factor is always obtained in the same way: it is related to the phase angle between line voltage and line current at the secondary.

Primary and secondary current, voltage, turn ratio, power, efficiency

Each primary-secondary coupling can be defined by a value called the **turn ratio**, which is simply the ratio between the number of turns in the primary winding and the number of turns in the secondary winding. It is indicated with the letter n :

$$n = \frac{N_1}{N_2}$$

The characteristic of the transformer is often referred to as $N_1:N_2$. In the case of the single-phase transformer, we can also state that:

$$n = \frac{N_1}{N_2} = \frac{V_1}{V_2} = k$$

Where k is defined as the **transformation ratio**. We can distinguish two cases:

- **$k < 1$** : the transformer is a **voltage booster**: if the number of turns in the secondary is greater, then the output voltage is higher than the input voltage.
- **$k > 1$** : the transformer is a **voltage reducer**: the number of turns in the secondary is smaller, therefore, the output voltage is lower than the input voltage.

The above two formulas are valid for both ideal and practical single-phase transformers.

Only in the case of the ideal single-phase transformer, also the following equality is valid, where the relationship between voltages and currents derives from $S_1=S_2$ (apparent input power equal to output power).

$$k = n = \frac{N_1}{N_2} = \frac{V_1}{V_2} = \frac{I_2}{I_1}$$

For currents the opposite considerations apply to voltages. If $k < 1$, the transformer is a current reducer; while if $k > 1$, it is a current booster.

Remember that we always consider the RMS values.

Example: an ideal transformer with 4800 windings on the primary and 800 on the secondary is powered by 115 V and 1 A. Let's calculate the voltage and current on the secondary: V_2 and I_2 .

1) the characteristic of the transformer (the turn ratio) is: $n = 4800/800 \rightarrow 6:1$

2) the following equations are valid:

$$\frac{N_1}{N_2} = \frac{V_1}{V_2} \rightarrow V_2 = \frac{V_1 \cdot N_2}{N_1} \rightarrow V_2 = \frac{115 \text{ V} \cdot 1}{6} = 19 \text{ V} ; \quad \frac{N_1}{N_2} = \frac{I_2}{I_1} \rightarrow I_2 = \frac{I_1 \cdot N_1}{N_2} \rightarrow I_2 = \frac{1 \text{ A} \cdot 6}{1} = 6 \text{ A}$$

In the case of three-phase transformers, for each phase we can calculate the turns ratio as the ratio between the phase voltages (E), while we can calculate the transformation ratio as the ratio between the line ones (V). We understand that, depending on the connection type of primary and secondary, the relationship between n and k varies.

$$n_n = \frac{N_{1n}}{N_{2n}} = \frac{E_{1n}}{E_{2n}} \quad ; \quad k_n = \frac{V_{1n}}{V_{2n}}$$

(the n at the subscript refers to the phase number).

We therefore distinguish the four cases and obtain the relationship between n and k starting from that between line and phase voltages:

Y – Y	1-2: LINE = $\sqrt{3}$ PHASE	k = n
$\Delta - \Delta$	1-2: LINE = PHASE	k = n
Y – Δ	1: LINE = $\sqrt{3}$ PHASE ; 2: LINE = PHASE	k = $\sqrt{3} \cdot n$
$\Delta - Y$	1: LINE = PHASE ; 2: LINE = $\sqrt{3}$ PHASE	k = n / $\sqrt{3}$

In high-voltage transformers, a star connection is preferred, because if we consider the phase voltages, those in the delta connection are higher than those in the star connection (by a factor of $\sqrt{3}$). The star connection is therefore more economical, because it allows you to use a smaller amount of insulation. Furthermore, this connection is preferred in case the addition of a neutral connection is necessary (if the loads are unbalanced). If the secondary currents are unbalanced it is better to have a Δ -Y connection.

The nameplate data of a transformer provided by the manufacturer includes the nominal apparent power, the transformation ratio, the short circuit voltage, the copper loss and the iron loss (the last three data can be expressed in terms of percentage).

The nominal voltages refer to the guaranteed insulation level of the dielectrics used (if these voltages are exceeded the dielectric can be perforated); the nominal currents refer to the thermal effects on the conductors.

Regarding the calculation of power and efficiency, refer to the previous paragraphs of this chapter.



Auto-transformers

The autotransformer (single- or three-phase) has a single winding. Practically, the primary winding ($N_1 = N_2 + N$) has a certain number of coils ($N_2 = N_2$) in common with the secondary and therefore it is not isolated from it.

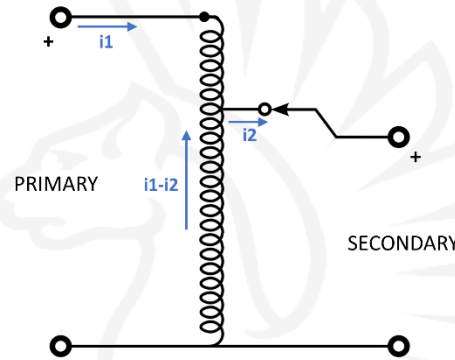


Figure 107 – (Step-down) auto-transformer scheme

The turns ratio then results:

$$n = \frac{N_1}{N_2} = \frac{N + N_2}{N_2} = \frac{V_1}{V_2} = k$$

The transformation ratio is always given by the voltage on the primary divided by the voltage on the secondary.

The current I_1 flows both in the part of the turns of the primary only and in the turns in common. Current I_2 only flows in the common part, where it adds vectorially with the contribute given by I_1 .

The advantages of an autotransformer compared to an equivalent conventional transformer (same apparent power) are the reduced weight and dimensions, the lower cost, the higher efficiency.

The disadvantages are that, since there is no insulation between primary and secondary, it is necessary to take more safety precautions. If an autotransformer were to fail, the load connected to the secondary could be subjected to the full (high) voltage of the primary. For this reason, its use is limited to transformation ratios not exceeding 1:3.

Autotransformers are used in the ignition of combustion engines as well as in electrical energy distribution systems.

3.16 Filters

Operation, application, and uses of the following filters: low pass, high pass, band pass, band stop

A filter is a circuit operating in an alternating regime, designed to receive input signals of different frequencies and transfer only the signals of the desired frequencies without introducing distortions. The frequencies that pass are part of the "bandwidth" (passing band), while those that do not pass constitute the "blocked band".

In a passive filter the output voltage is always lower than the input voltage. Active filters contain elements such as amplifiers.

The filters we consider in this chapter are passive ones, made up of passive components such as resistors, capacitors, and inductors.



Figure 108 - Example of a filter circuit

In alternating circuits, the current takes on different values depending on the instant considered. Remember that each time interval corresponds to a frequency interval. There are intervals (bands) of frequencies in which the current has higher values and intervals in which the current has lower values.

Different types of filters exist:

- Low-pass (LPF)
- High-pass (HPF)
- Band-pass (BPF)
- Band-stop (notch).

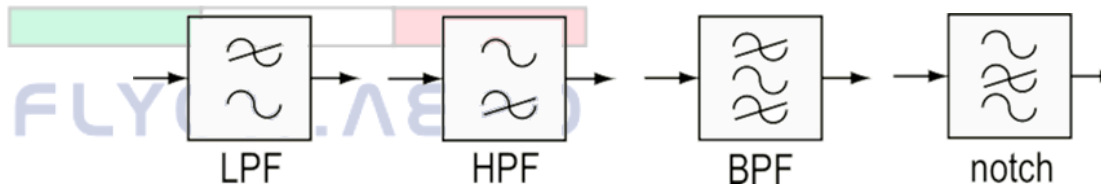


Figure 109 - Circuit symbols for passive filters

Low-pass filter

The low-pass filter, powered in alternating mode, only allows the passage of frequencies below a certain threshold, called the *cut-off frequency* and indicated by f_T . Frequencies higher than this value are blocked.

The low-pass filter is the simplest and is basically an RC circuit (resistor in series and capacitor in parallel at the input) or RL (inductor in series and resistor in parallel).

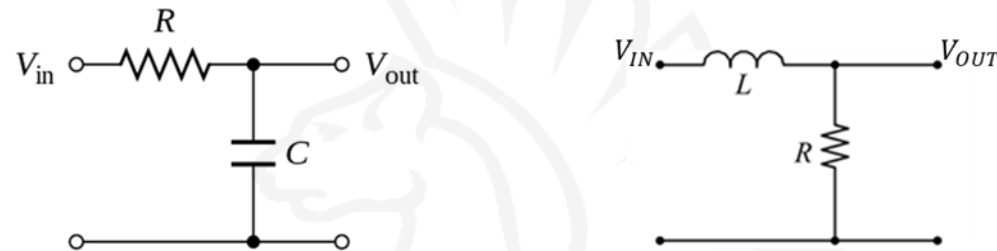


Figure 110 – Low-pass filter circuits

The value of the cut-off frequency is established in the design phase and depends on the resistance and capacitance values. The cut-off frequency is defined as the frequency to which an **attenuation** (=ratio between output voltage and input voltage) of $1/\sqrt{2}$ corresponds:

$$ATT. = \frac{V_{OUT}}{V_{IN}} = \frac{1}{\sqrt{2}} \cong 0.707 \quad \leftrightarrow \quad f_T = \frac{1}{2\pi RC} \quad \text{or} \quad f_T = \frac{R}{2\pi L}$$

In the RC circuit, the impedance of the capacitor is inversely proportional to the frequency of the signal. If the input signal has a low frequency, the impedance of the capacitor is high and therefore also the voltage across it, which corresponds to the output voltage. We therefore understand that, if the frequency is low, the output signal is not attenuated.

If the frequency increases above the cutoff limit, the input signal is attenuated.

In the LR circuit, the inductor impedance is directly proportional to the frequency. If it is low then there is a greater voltage drop across L than across R, and the output voltage (equal to that across R) is reduced.

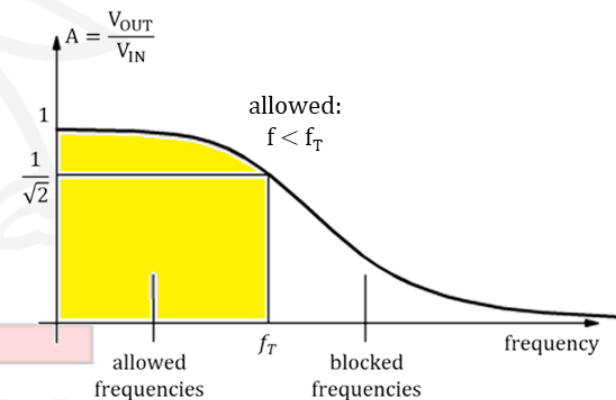


Figure 111 – Diagram of frequencies in a LPF

High-pass filter

The high-pass filter only passes frequencies higher than the cutoff frequency f_T . **Lower** frequencies are blocked, meaning that the corresponding output voltages suffer large attenuations.

The circuit is an RC (consisting of a capacitor in series and a resistor in parallel at the input) or RL (a resistor in series and an inductor in parallel). In the RC circuit, the output voltage is the same as that on the resistor. The capacitor has an impedance inversely proportional to the frequency, therefore, if the frequency increases, there is a lower voltage drop across it, while it is higher across the resistor (it is not attenuated). In the case of low frequencies there is a greater voltage drop on C and therefore the output voltage is attenuated.

In the RL circuit the output voltage is equal to that on the inductor. The opposite applies to that made for the low-pass LR circuit.



Figure 112 - High-pass filter circuits

Also in this case the reference value is the attenuation of $1/\sqrt{2}$. All frequencies **above** the cutoff frequency f_T corresponding to that attenuation value pass. The values are the same as those of the low-pass filter, therefore:

$$V_{OUT} = \frac{V_{IN}}{\sqrt{2}} \quad \leftrightarrow \quad f_T = \frac{1}{2\pi RC} \quad \text{or} \quad f_T = \frac{R}{2\pi L}$$

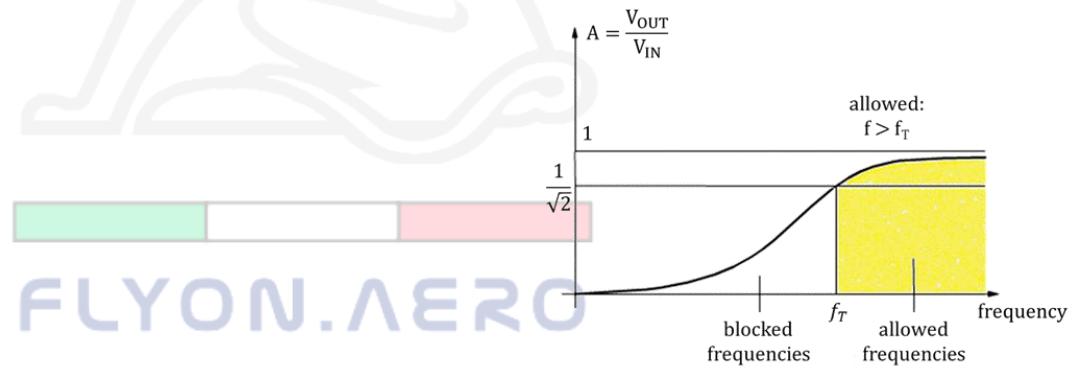


Figure 113 - Diagram of frequencies in a HPF

Band-pass and band-stop filters

The band-pass filter passes signals with frequencies between two extreme values: a higher cutoff frequency, which we indicate with f_H , and a lower one, which we indicate with f_L . The **bandwidth** (B) will then be identified by:

$$B = f_H - f_L$$

This filter can be composed by placing a high-pass filter and a low-pass filter in series, or by using resonant RLC circuits (resonance is a phenomenon that occurs when, at a given frequency, the capacitive and inductive reactance have the same modulus, and the electrical energy oscillates between the magnetic field of the inductor and the electrostatic field of the capacitor).

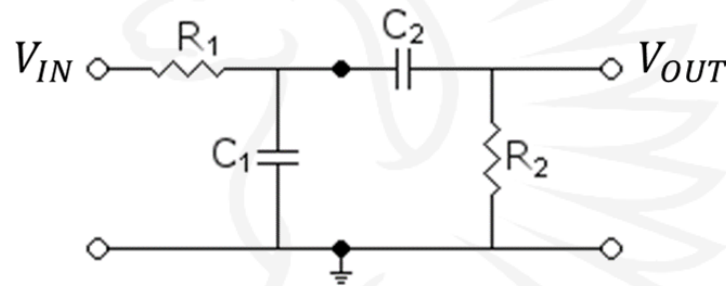


Figure 114 - Band-pass filter circuit

The band-eliminating filter, on the other hand, blocks the frequencies between the two limit frequencies, allowing all the others to pass. It consists of a high-pass and a low-pass circuit in parallel.

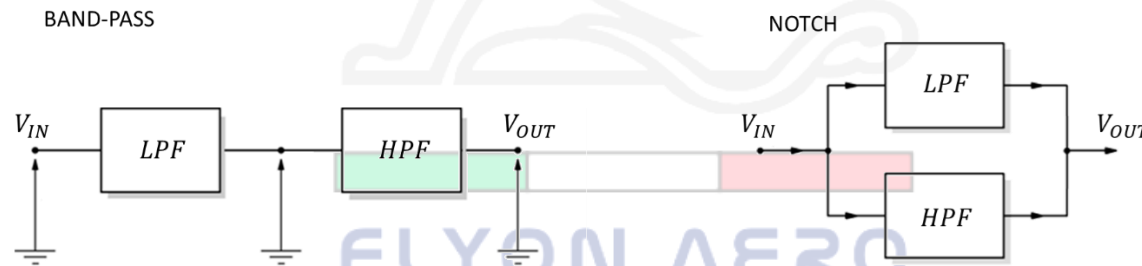


Figure 115 - Schematic of band-pass and notch filters

3.17 AC Generators

Rotation of loop in a magnetic field and waveform produced

We have seen that in a coil which is rotating within a magnetic field, an induced electromotive force and therefore a current is generated.

We consider the magnetic field within which the coil rotates to be constant, and we also consider the rotation speed ω of the coil to be constant. ω is defined as the ratio between the angular position of the coil (ϑ) and the instant of time (t).

The flux of the magnetic field (Φ) through the loop is calculated in a direction perpendicular to its surface, therefore, we apply the vector product considering the cosine of the angle ϑ (which we write as a function of speed and time $\vartheta = \omega t$). The variation in time of the magnetic flux through the loop is defined by the Faraday-Lenz-Neumann law as the electromotive force induced on the coil:

$$\Phi = B \cdot S \cdot \cos(\omega t) \quad ; \quad emf = -\frac{\Delta\Phi}{\Delta t}$$

For a series of mathematical considerations, the induced electromotive force can be written as:

$$emf = E \cdot \sin(\omega t)$$

Therefore, the electromotive force induced in a rotating coil immersed in a magnetic field has a sinusoidal trend. This was also true in the case of the direct current generator, in which the commutator for rectifying the current was present.

In the case of alternating mode generation, the commutator is not necessary.

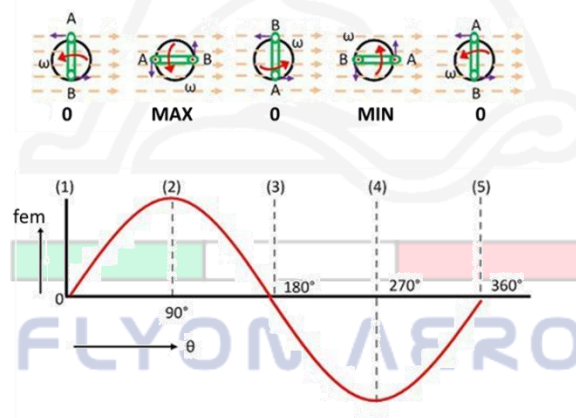


Figure 116 - Waveform produced by an alternator

Operation and construction of revolving armature and revolving field type AC generators

An alternator has two basic parts:

- An inductor (the source of magnetic field)
- An armature (the system of conductors).

Unlike the DC generator, in the alternator the rotor and stator can consist of both the inductor and the armature.

The rotor, in any case, is the rotating part of the alternator, which receives mechanical energy from an external source. It can coincide with the armature if the voltage is generated in it or with the inductor if it is the source of the magnetic field. The stator will correspond to the remaining part. As we have already seen for DC machines, to obtain inductive effects it is sufficient for an element between the magnetic field and the conductor to be in motion to obtain an electromotive force. The current is "collected" via the *slip rings*.

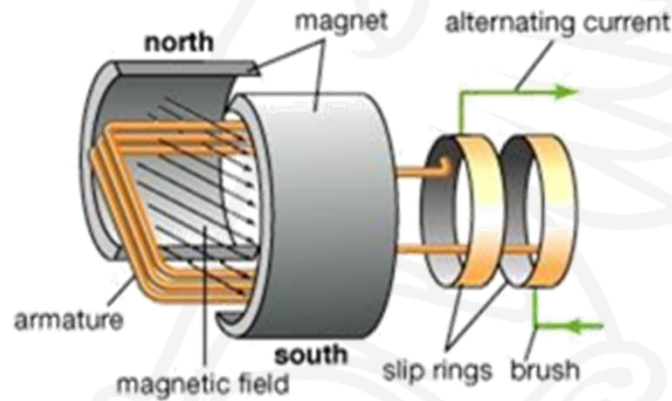


Figure 117 - Alternator's parts

So, we can distinguish alternators based on which part corresponds to the rotor:

- **Fixed inductor** alternator – the rotor is the armature ('revolving armature type').

The revolving armature generator is very similar to the DC generator. The difference, as already mentioned, lies in the absence of the commutators. In this case the voltage is supplied to the load directly by means of the brushes which slide on the slip rings.

This type of construction is not widely used and finds application low-power alternators.

- **Fixed armature** alternator – the rotor is the inductor ('revolving field type').

The rotor, i.e. the magnetic field source, is powered by a DC source. The flux of the magnetic field through the armature varies with the rotation of the inductor. The advantage of having the armature at the stator is that it can be directly connected to the load to be powered, without having to use additional contacts such as brushes. Consequently, there are fewer risks (formation of electric arcs), and less maintenance is required.

Revolving field alternators are the most used, especially in the aviation field, and can reach high voltage values.

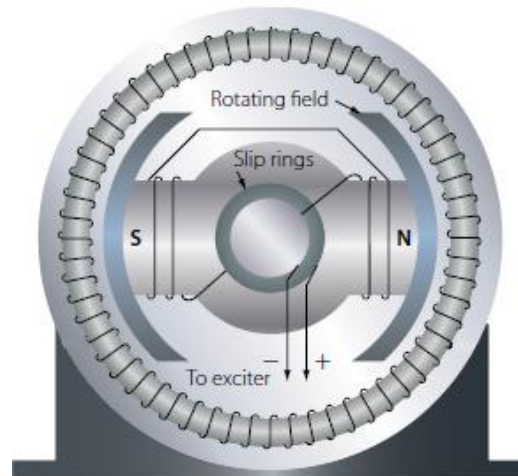


Figure 118 - Rotating field AC generator

In both cases, the magnetic field can be generated by permanent magnets or electromagnets.



Single-phase, two-phase, and three-phase alternators

If an alternator produces a single alternating signal it is called **single-phase**. It generally consists of two stator coils, but it is also possible to find four. In any case, the stator circuit is single and there are only two output terminals, i.e. the positive and negative output of the current/voltage produced by the generator.

The number of rotor poles depends on the number of stator coils. When the pole expansion is closest to the winding, the interaction is maximum and therefore the sinusoid reaches its peak.

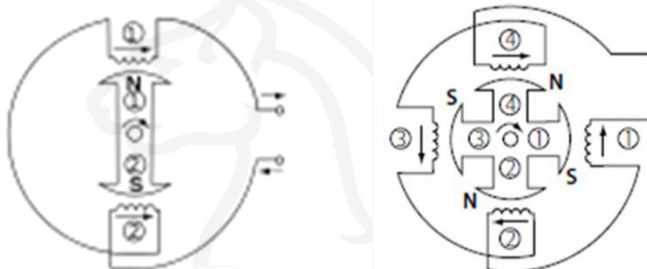


Figure 119 - Examples of single-phase alternator

If the windings are separated from each other and there are two pairs of terminals at the output, then the generator is **two-phase**. The stator coils are positioned at 90° to each other. The pole expansions are not equal in number to the windings, because otherwise they would reach the peaks at the same time and there would not be a phase shift. In fact, there is a 90° phase shift between the two outgoing waves.

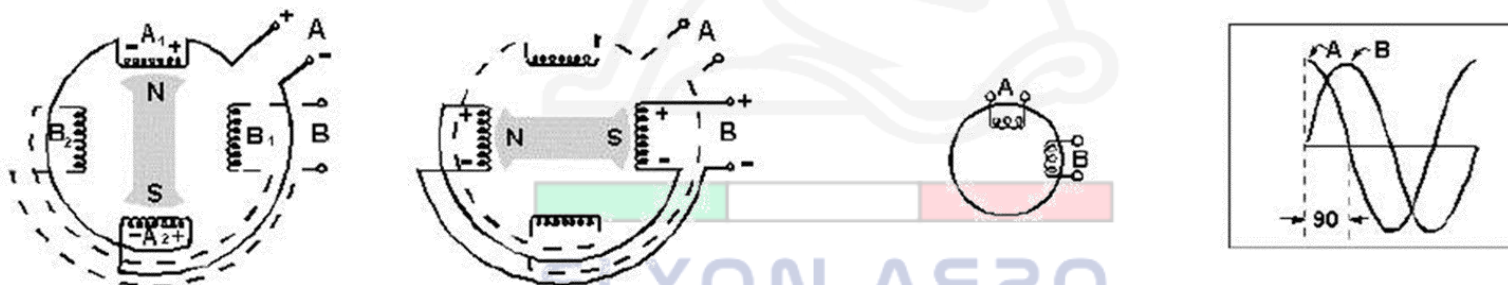


Figure 120 - Two-phase alternator

The three-phase alternator has a stator made up of three single-phase windings, separated from each other by 120°, so as to have three alternating signals out of phase with each other. Each phase has two output terminals. Therefore, a three-phase generator will feature 6 output terminals.

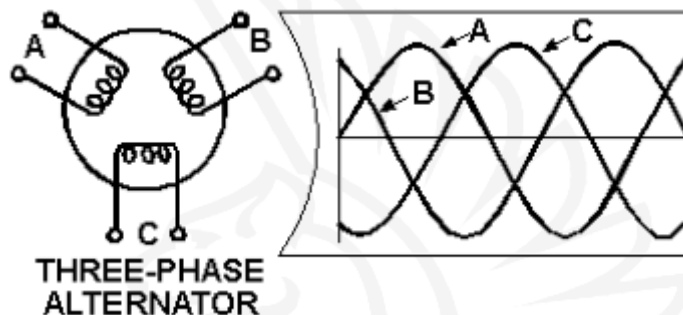


Figure 121- Three-phase alternator

The output of a three-phase generator is symmetrical, because the three voltages/currents have the same maximum amplitude and oscillate with the same frequency.

The frequency (Hz) of the electromotive force induced in a three-phase generator can be calculated as:

$$f = \frac{p \cdot n}{60}$$

Where n is the rotation speed of the machine in rpm and p is the number of pole pairs. As a result, the speed of the machine has a value that depends on the frequency (synchronism speed – detailed later with AC motors).

The frequency used on board aircraft is 400 Hz, so we can use the inverse formula and calculate the product between speed and number of pairs of poles, which is practically a constant:

$$p \cdot n = 60 \cdot f \quad \rightarrow \quad p \cdot n = 60 \cdot 400 = 24000$$

Knowing this value, we can calculate the number of pole pairs needed for a machine to reach a certain speed.

For example, if a machine rotates at 4000 rpm, the number of pole pairs to have a frequency of 400 Hz must be:

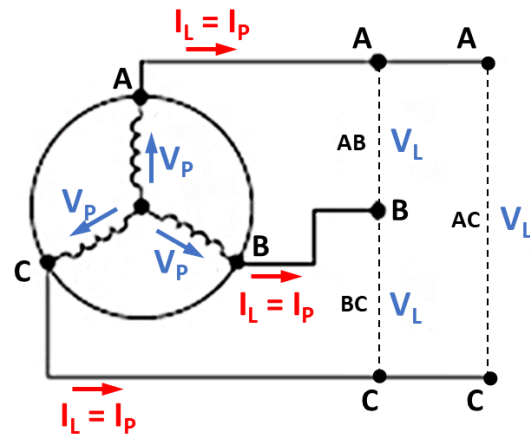
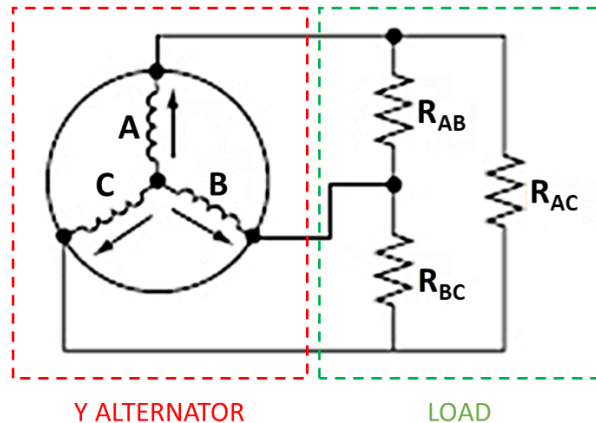
$$p = \frac{24000}{4000} = 6$$

So, 6 pairs of poles (i.e. 12 poles).

Three-phase star and delta connections advantages, and uses

The three phases of the generator can be connected in star or delta.

In the star connection, the three phases all have a common connection (the star center) and there may be a fourth conductor, the neutral.



The following are true:

$$I_{LINE} = I_{PHASE} \quad ; \quad V_{LINE} = \sqrt{3} \cdot V_{PHASE}$$

The phase voltage is practically the induced voltage due to the rotation of the magnetic field.

The line voltage, i.e. the one that reaches the load, is almost double compared to the phase voltage ($\sqrt{3} \approx 1.73$). This means that, for the same line voltage, a star-connected winding can have fewer turns than a delta connection.

So as an advantage there is a smaller quantity of conductor, but also a smaller quantity of insulation required.

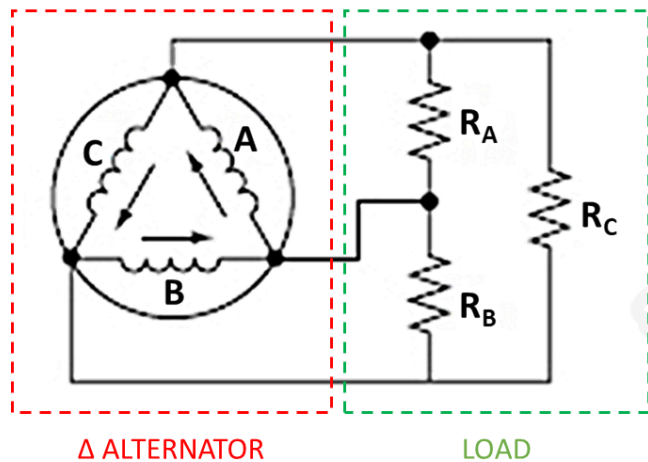
Another advantage linked to having two voltage values is that it is possible to connect single-phase loads to the individual voltages produced, therefore, the two different values can be exploited. In fact, independent single-phase loads can be connected to the phases and three-phase loads to the lines.

Additionally, the neutral wire can be grounded, giving a greater level of safety.

On the neutral, if the load is balanced, the potential difference with the ground wire is 0 V.

Figure 122 – Three-phase alternator – star connection

In delta connection, one of the ends of each phase is connected to one of the ends of the next.



The followings are true:

$$I_{LINE} = \sqrt{3} \cdot I_{PHASE} \quad ; \quad V_{LINE} = V_{PHASE}$$

This type of connection is used when a greater current supply is required.

Delta-connected generators need more windings and more insulation (higher cost) for the same line voltage, compared to star-connected generators.

These generators can exploit only one voltage value.

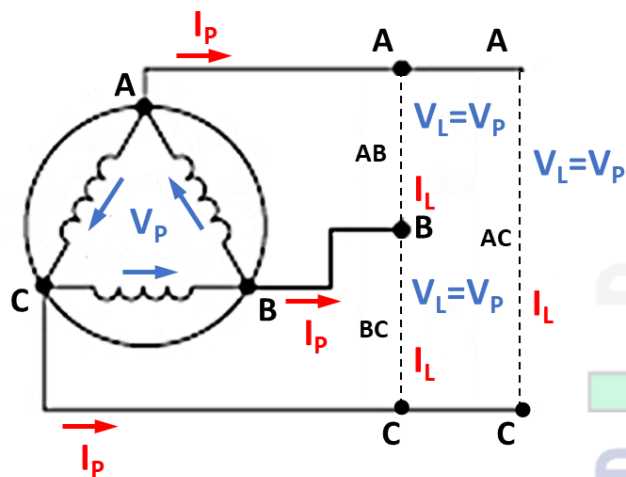


Figure 123 – Three-phase alternator – delta connection

Permanent Magnet Generators

Permanent Magnet Alternators are referred to as PMA or PMG (PM Generator).

In these alternators, the magnetic field is not generated by current-carrying coils, but, in fact, by permanent magnets. These magnets are often made of neodymium (a metal belonging to the rare earth group) or ferrite.

Typically, the magnets are mounted on a rotor placed internally (whereas electromagnet rotors are usually external).



Figure 124 - Permanent Magnet Alternator (PMA/PMG)

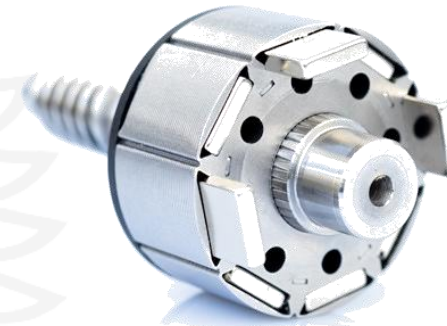


Figure 125 - Rotor made of permanent magnet

A big difference compared to using electromagnets is that the intensity of the magnetic field cannot be adjusted.

Thanks to the absence of electrically powered coils, this type of construction has many advantages, including lower noise production, and reduced dimensions and weight (for the same amount of energy produced). Furthermore, the efficiency of permanent magnet generators is higher, even reaching 90%, because it is not necessary to provide energy to power the electromagnet coils.

The disadvantages are linked solely to the characteristics of the magnets. Neodymium magnets are the most powerful on the market, but they are sensitive to high temperatures. In fact, they can already demagnetize at temperatures around 80°C even if, by applying certain treatments, they can resist up to 200°C. Ferrite magnets are slightly less powerful but have a higher temperature resistance.

As already seen for other components, also in this case the magnets are subject to hysteresis losses. This problem can be mitigated by using a set of laminations rather than a single block.

Furthermore, permanent magnets are more expensive than electromagnets, but still require less maintenance due to the absence of parts in contact (in fact they are brushless machines - without brushes).

3.18 AC Motors

Construction, principles of operation and characteristics of: AC synchronous and induction motors both single-phase and polyphase

AC motors have the same parts as AC generators, i.e. a rotor and a stator. Remember that the electric motor works because the armature produces a magnetic field opposite to the variable one in which it is immersed, thanks to which it rotates. We therefore understand that, in the case of the AC motor, the rotor part is always the armature, and the stator part is always the field.

Starting of AC motors occurs only if the magnetic field produced by the stator is rotating. The *rotating field* is obtained naturally in polyphase motors, while the same is not true for single-phase ones. In fact, in the latter case, an external element is required that allows the rotation of the magnetic field.

In the stator of **three-phase motors** there are three identical windings, spatially set apart from each other by 120° and connected in a star or delta. The windings are powered by a symmetrical trio of voltages (therefore the voltages are equal and 120° out of phase with each other, as are the currents).

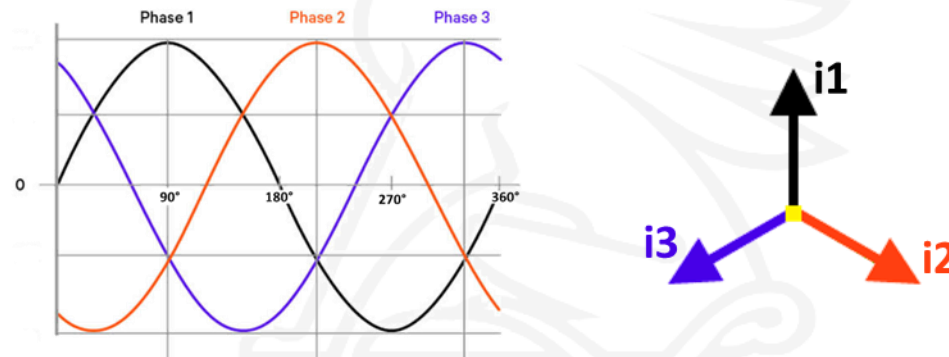


Figure 126 - Three-phase currents

Using the figure above as a reference example, there will be a moment, corresponding to 90° , where the current of the first winding has its maximum value. The second current, 120° out of phase with respect to the first, at that moment has a negative value (equal to half the maximum value) and is growing. The third current, further out of phase by 120° , has the same negative value, but is decreasing. The three currents out of phase with each other each generate a magnetic field with different directions and intensities.

Remember that the quantities considered in the three-phase alternating regime can be represented using vectors (more precisely called "phasors"). At the moment we are considering, the three currents can be represented as in the figure above right.

As the sine wave progresses, the three phasors change direction:

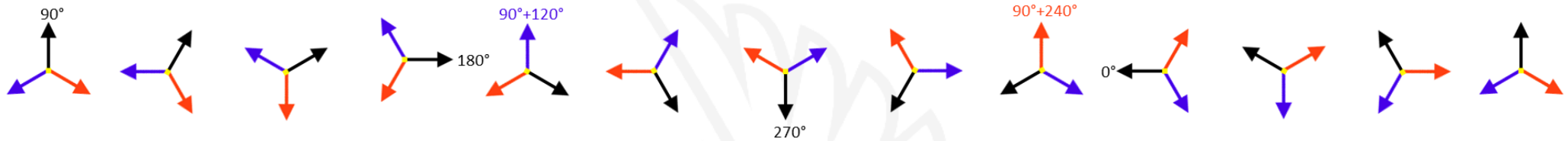


Figure 127 - Current phasors rotation

This means that the overall magnetic field also changes direction, effectively becoming a rotating magnetic field.

The rotating magnetic field can also be obtained starting from two currents of the same frequency and effective value, out of phase with each other by 90° (application in the **two-phase motor**).

In any case, the motor transforms the electrical energy supplied into mechanical energy, i.e. into a torque necessary to overcome the resistive torque relating to the load.

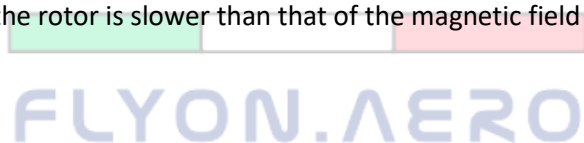
AC motors (both single-phase and three-phase) can be classified based on the relationship between rotor speed n_R and rotation speed of the magnetic field, which we define as "synchronism speed" and indicate with n_S . The latter is measured in rpm and calculated as:

$$n_S = \frac{60 \cdot f}{p}$$

Where f is the frequency (Hz) of the voltage used and p is the number of North-South pole pairs of the stator. The number 60 is the constant that allows you to go from cycles per second of frequency to cycles per minute of speed.

So, we can distinguish:

- Synchronous motors – the rotation speed of the rotor is synchronized with the rotation speed of the magnetic field produced by the stator ($n_R = n_S$).
- Asynchronous or induction motors – the rotation speed of the rotor is slower than that of the magnetic field ($n_R < n_S$).



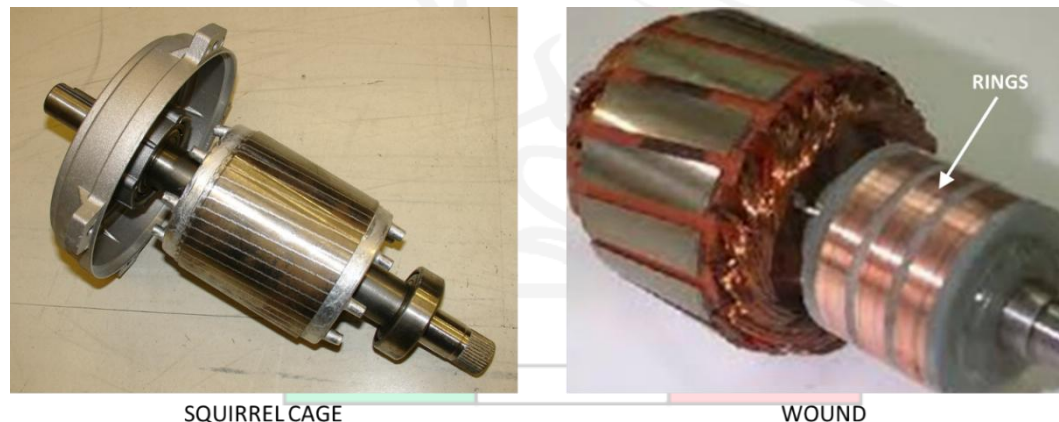
AC asynchronous or induction motors

The asynchronous or induction motor has a stator made up of windings (in the case of three-phase, connected in a star or delta) carrying a current that generates the magnetic field.

The rotor is placed inside the stator, leaving a distance of a few millimeters, and is integral with the motion transmission shaft. In the external slots of the rotor there are the windings (conductors), which are traversed only by the current induced by the stator (hence the name of this type of motor).

We can distinguish two types of rotors:

- “Squirrel cage” or “short-circuited” rotor.
It is the simplest type and can be applied to both single-phase and multi-phase motors. Copper or aluminum bars are placed in each channel of the rotor, which completely fill the channel. The ends of all the bars, on both sides of the rotor, are all connected to each other by means of a copper ring.
- “Wound rotor” or “slip ring rotor”.
Generally on three-phase motors. It is made up of many circular crowns of grooved magnetic sheets, in which the windings are inserted, in a number equal to the number of polarities of the stator. In the rotor we find rings (to which the free terminals of the phases are connected) on which the brushes slide. The minimum number of rings must correspond to the number of phases, so, for example, we can find three rings in both a three-phase and a single-phase.



SQUIRREL CAGE

WOUND

Figure 128 - Types of AC induction motors' rotors

When the rotor is immersed in the rotating magnetic field, an induced current is generated on it which in turn generates a magnetic field that opposes the first. The rotor spins at a speed similar to, but not the same as, the rotation of the magnetic field. If the rotor had the same speed as the magnetic field, its conductors would not intersect any field lines, and there would be no induced electromotive force and therefore no torque.

The difference between the two rotation speeds is called *slip* and is expressed in terms of percentage compared to the synchronism speed. The synchronism speed, being the fixed rotation speed of the magnetic field, is a data provided by the manufacturer, as is the speed of the rotor.

We can calculate a sliding speed N as the difference between n_s and n_r , and then calculate the slip s . For example, if the rotor spins at 1850 rpm and the synchronous speed is 1900 rpm, then the slip speed is $N = 50$ rpm. So, in percentage terms (with respect to n_s) we obtain a slip of 2.63%.

The actual amount of slip, however, depends on the loads applied to the motor.

Under ideal conditions, the speed of the rotor is equal to that of the magnetic field, so s would be 0 (0%).

The advantages of asynchronous motors are the high power (three-phase reaches up to 1000 kW), the ease of starting and the relatively low cost. Furthermore, as they do not have permanent magnets, they do not have problems relating to high temperatures. In fact, since there are windings on both the rotor and the stator, heat is generated by both parts. Another downside of asynchronous motors is their high inertia, which slows acceleration.

The heat from the engines can be dissipated both naturally, through openings that facilitate air circulation, and artificially. In the latter case there may be either forced ventilation or, in higher power engines, air-air or air-fluid heat exchangers.

Mono-phase induction motors

Speaking specifically about single-phase induction motors, we have seen that they cannot produce a rotating magnetic field, because there are no phase shifts that push the magnetic field in one direction or another. For starting it is necessary to apply constructional measures or use an auxiliary (starting) circuit, which provides an external force that breaks the balance and generates rotation.

We can therefore distinguish two types of single-phase asynchronous motors, which will be detailed later.

- Shaded-pole motor
- Split-pole motor.

Three-phase induction motors

A critical aspect of the asynchronous motor is the very high starting current required which reaches values 5-8 times higher than the nominal one.

The starting current can be reduced, while maintaining the supply frequency constant, in various ways:

- By inserting starting rheostats, which increase the resistance of the rotor only during the starting phase and are then excluded during steady state operation. This method is applicable when the rotor is of the wound type.
- Using double squirrel cage rotors, practically made up of two concentric cages. When starting, the current flows in the outermost cage, with high resistance and, as the machine accelerates, the current moves into the innermost cage with lower resistance.

- Using star-delta switching. If a motor is connected in delta, it can be switched to star to be started. The star connection line currents are $1/\sqrt{3}$ times those of the delta connection, therefore, the inrush current is much lower. This method can be applied if the starting load is low, because if the supply voltage is reduced the motor torque is also reduced.
To reach the nominal speed it is then necessary to switch to delta again.

AC synchronous motors

Synchronous motors are usually powered with three-phase voltage, but low-power ones can also be powered in single-phase mode.

The stator, as in the case of asynchronous motors, is equipped with the windings that generate the rotating field. The rotor is usually made up of permanent magnets but can also be found with windings.

Synchronous motors with a permanent magnet rotor are called **brushless motors** because, as there are no windings, brushes are not necessary to power them. If the rotor has windings, these are powered by a direct current.

In any case, the rotor generates a magnetic field that opposes that of the stator.

The synchronous motor is set and constructed in such a way that the speed of the rotor is equal to that of the rotating magnetic field.

Synchronous motors can only deliver torque when they have reached the synchronous speed, therefore, external intervention is required to start them and make them reach this speed. For example, they can be started without load, making them rotate thanks to another motor (asynchronous AC or DC), and only once the operating speed is reached the rotor is powered. At that point the rotor magnetic field binds to the stator and the synchronism speed is reached.

Another way to start these motors is to use a double squirrel cage with different resistance values. However, the most modern solutions consist in the use of electronic elements that increase the frequency gradually.

Permanent magnet synchronous motors can also be used as generators.

Starting a single-phase synchronous motor is more difficult than starting an asynchronous one, because it is necessary to make it reach the synchronous speed. Among the methods of starting the single-phase synchronous motor is the use of an auxiliary winding, which generates a current that is late compared to that generated by the stator and causes the starting torque. Once the synchronous speed is reached there may be a centrifugal switch that disconnects the secondary winding.

The advantages of synchronous motors are the high power/weight ratio, reliability, low inertia, and heat production only on the stator. The disadvantages are the lower power compared to the asynchronous ones (of the order of tens of kW) and the fact that, if the rotor is a permanent magnet, it risks becoming demagnetized due to the high temperature.

Methods of speed control and direction of rotation

To regulate the speed of a **synchronous motor**, it is necessary to act on the synchronism speed which, as we have seen, depends on the frequency and the number of poles. Acting on the number of poles is difficult, but the power frequency can be varied using electronic regulators. For example, if the load increases and the speed decreases to obtain a greater torque, the speed acquired by the rotor is sent as a signal to an inverter, which regulates the frequency and therefore reaches a new synchronism speed.

As for an **asynchronous motor**, even in this case the number of poles could be varied but it would be difficult. Another solution would be to reduce or increase the voltage, for example by placing adjustable impedances between the motor and the power supply. The most used solution consists in regulating the frequency, therefore acting on the synchronism speed and then on the rotor.

The direction of rotation of a motor can be reversed by acting on the phases of the stator power supply. In a three-phase, two of the three phases can be exchanged, while in a single-phase you just need to reverse the connections.



Methods of producing a rotating field: capacitor, shaded or split pole

Single-phase motors have two stator windings, placed at 90° to each other. The first is the main one (also called "running") and the second is the starting one, which can be constructed in different ways.

Split Pole

In split-pole, or "split-phase" motors, the second winding is connected in series with a centrifugal switch. This switch will disconnect the auxiliary winding when the motor speed has reached 75-80% of the synchronism speed.

The objective is to create a 90° electrical phase shift between the main and auxiliary windings, to obtain a rotating field similar to that generated by a two-phase motor.

The auxiliary winding has very high resistance. We know that if the impedance is mostly resistive, the current is practically in phase with the voltage. At the same time, in the first winding (which is practically only inductive), the current is out of phase (lags) by a very large angle with respect to the voltage. We can therefore say that the current in the two windings has the same value, but has a very large phase shift, which causes the rotation of the generated magnetic field.

This type of motors has low starting current and moderate torque. It is used for example in household appliances such as washing machines. Powers range up to half a kilowatt.

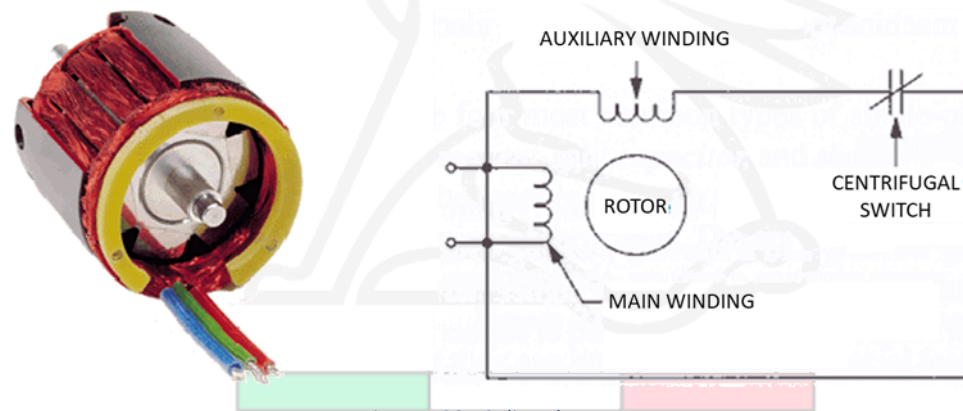


Figure 129 - Split pole motor

To create the phase shift between the two windings, a capacitor or an inductor can also be used, which is connected in series with the starting winding. In this way, the current will be out of phase with respect to the voltage and therefore, again, with respect to the current of the primary winding. The consequence, as before, is that the magnetic field becomes rotating. The condenser is then disconnected thanks to the centrifugal switch.

Use of permanent capacitor

The capacitor can also remain permanently connected. In this case the power factor increases and the costs due to the disjunction elements of the second winding decrease. This type of construction is more reliable and is, in fact, the most widespread.

Shaded Pole

In “shaded pole” motors (where “shaded” literally means “shielded” or “shadowed”) there is only one stator winding.

The pole pieces are cut into two unequal parts. A small copper ring is applied to the smallest part, which is called the "shadow band".

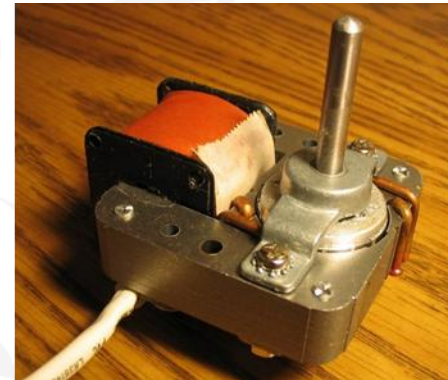
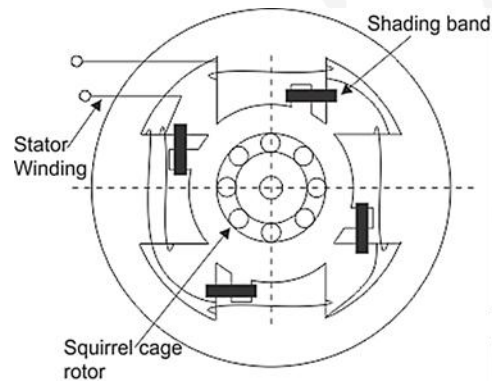


Figure 130 – Shaded-pole motor

Practically, when the stator is powered, an electromotive force is produced in the shielded part which is opposite to the supply current. For this reason, there are two magnetic fields with different directions, and therefore a rotating magnetic field is generated.

This type of motor has a low starting torque and reaches powers up to 0.05 kW. The advantages are reduced costs and simple construction. Other disadvantages are low power factor and efficiency and difficulty in reversing speeds.

